

Topological Energy Condensate Theory (TECT): 20

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I. EXPLICIT ENRICHED SHELL BASIS AND CANONICAL CORE BLOCK

Short explanation

The next nonredundant step is to stop speaking abstractly about

$$\mathcal{H}_D^{(N)}$$

and to write it explicitly.

The correct 4-dimensional Dirac-carrier candidate is the enriched opposite-valley shell space

$$\mathcal{H}_D^{(N)} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

sitting inside the truncated shell Bloch space

$$\mathcal{H}_{\text{shell}}^{(N)} = \mathcal{H}_D^{(N)} \oplus \mathcal{H}_{\text{non-enr}}^{(N)} \oplus \mathcal{H}_{\text{rest}}^{(N)}.$$

Once this basis is fixed, the projected block

$$H_{\text{core}}^{(N)}(p) := P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)}$$

has an explicit 2×2 valley-block form. Its constant term gives the possible Dirac mass, its pair-diagonal linear splitting gives the longitudinal slope, and its pair-off-diagonal linear term gives the transverse matrices W^1, W^2 .

II. EXPLICIT DEFINITION OF THE ENRICHED SHELL CARRIER

Fix the selected shell pair $\pm G_*$, and choose a shell-adapted orthonormal frame

$$\hat{n} := \frac{G_*}{|G_*|}, \quad (e_1, e_2, \hat{n}).$$

Let

$$\{\chi_1, \chi_2\}$$

be an orthonormal basis of the shell-relevant enriched internal doublet. Inside the truncated Bloch shell space, define the four basis vectors

$$|+, a\rangle := |G_*, \chi_a\rangle, \quad |-, a\rangle := |-G_*, \chi_a\rangle, \quad a = 1, 2. \quad (1)$$

Equivalently, in the folded Bragg notation $k_* = G_*/2$, one may write

$$\Psi_{+,a}^{(p)}(x) = \frac{1}{\sqrt{V_{\text{cell}}}} e^{i(k_*+p)\cdot x} \chi_a, \quad \Psi_{-,a}^{(p)}(x) = \frac{1}{\sqrt{V_{\text{cell}}}} e^{i(k_*+p-G_*)\cdot x} \chi_a. \quad (2)$$

These span the 4-dimensional enriched shell carrier

$$\boxed{\mathcal{H}_D^{(N)} = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\}. \quad (3)}$$

The corresponding orthogonal projector is

$$P_D^{(N)} = \sum_{\sigma=\pm} \sum_{a=1}^2 |\sigma, a\rangle \langle \sigma, a|. \quad (4)$$

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III. PROJECTED CORE BLOCK

In the ordered basis

$$(|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle),$$

the projected shell Hamiltonian is

$$H_{\text{core}}^{(N)}(p) = P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)} = \begin{pmatrix} E_+^{(N)}(p) & B^{(N)}(p) \\ B^{(N)}(p)^\dagger & E_-^{(N)}(p) \end{pmatrix}, \quad (5)$$

where

$$E_\pm^{(N)}(p), B^{(N)}(p) \in \text{Mat}_{2 \times 2}(\mathbb{C}).$$

We now expand around $p = 0$:

$$E_\pm^{(N)}(p) = E_\pm^{(N)}(0) + p_i V_{\pm,i}^{(N)} + O(|p|^2), \quad (6)$$

$$B^{(N)}(p) = M^{(N)} + p_i \mathcal{W}_i^{(N)} + O(|p|^2), \quad (7)$$

with

$$M^{(N)} := B^{(N)}(0).$$

IV. CANONICAL SEPARATION OF CONSTANT AND LINEAR PIECES

Subtract the irrelevant common scalar energy

$$E_*^{(N)} := \frac{1}{4} \text{Tr} H_{\text{core}}^{(N)}(0). \quad (8)$$

Then

$$H_{\text{core}}^{(N)}(p) - E_*^{(N)} \mathbf{1}_4$$

contains three logically distinct pieces.

A. Constant pair-mixing term

The constant off-diagonal block $M^{(N)}$ is the candidate Dirac mass source. Under the little-group-scalar reduction on the doubled carrier, its symmetry-allowed scalar part is

$$M_{\text{sc}}^{(N)} = m_1^{(N)} \tau^1 \otimes \mathbf{1}_2 + m_2^{(N)} \tau^2 \otimes \mathbf{1}_2. \quad (9)$$

After a constant valley-phase rotation, one may always choose a gauge in which

$$M_{\text{sc}}^{(N)} = m^{(N)} \tau^1 \otimes \mathbf{1}_2, \quad m^{(N)} := \sqrt{(m_1^{(N)})^2 + (m_2^{(N)})^2}. \quad (10)$$

Therefore the explicit constant mass parameter is

$$m^{(N)} = \left\| \Pi_{\text{scalar}} B^{(N)}(0) \right\|. \quad (11)$$

Hence the massless condition is simply

$$m^{(N)} = 0. \quad (12)$$

B. Pair-diagonal linear splitting

Define

$$\Delta E^{(N)}(p) := \frac{1}{2} (E_+^{(N)}(p) - E_-^{(N)}(p)). \quad (13)$$

Its shell-adapted scalar longitudinal coefficient is

$$v_{\parallel}^{(N)} := \frac{1}{4} \text{Tr}_{\text{int}} \left[\partial_{p_{\parallel}} (E_+^{(N)}(p) - E_-^{(N)}(p)) \right]_{p=0}. \quad (14)$$

The minimal Dirac form requires

$$\Delta E^{(N)}(p) = v_{\parallel}^{(N)} p_{\parallel} \mathbf{1}_2 + O(|p|^2), \quad v_{\parallel}^{(N)} \neq 0. \quad (15)$$

This contributes the valley-diagonal term

$$v_{\parallel}^{(N)} p_{\parallel} \tau^3 \otimes \mathbf{1}_2. \quad (16)$$

C. Transverse pair-mixing matrices

Define the two transverse linear matrices

$$W_{(N)}^1 := \partial_{p_1} B^{(N)}(p) \Big|_{p=0}, \quad W_{(N)}^2 := \partial_{p_2} B^{(N)}(p) \Big|_{p=0}. \quad (17)$$

Only their traceless Hermitian parts carry Dirac data:

$$\widetilde{W}_{(N)}^{\alpha} := \frac{1}{2} (W_{(N)}^{\alpha} + W_{(N)}^{\alpha \dagger}) - \frac{1}{2} \text{tr} \left[\frac{1}{2} (W_{(N)}^{\alpha} + W_{(N)}^{\alpha \dagger}) \right] \mathbf{1}_2. \quad (18)$$

Since every traceless Hermitian 2×2 matrix is a Pauli vector, there exist real coefficients w_a^{α} such that

$$\widetilde{W}_{(N)}^{\alpha} = w_a^{\alpha} \sigma_a. \quad (19)$$

V. LOCKED PAULI BASIS AND EXPLICIT CANONICAL FORM

Choose the internal basis so that the two transverse matrices take the locked form

$$\boxed{\widetilde{W}_{(N)}^1 = \alpha^{(N)} \sigma_1 + \beta^{(N)} \sigma_2, \quad \widetilde{W}_{(N)}^2 = -\beta^{(N)} \sigma_1 + \alpha^{(N)} \sigma_2.} \quad (20)$$

Then

$$\{\widetilde{W}_{(N)}^1, \widetilde{W}_{(N)}^2\} = 0, \quad (\widetilde{W}_{(N)}^1)^2 = (\widetilde{W}_{(N)}^2)^2 = ((\alpha^{(N)})^2 + (\beta^{(N)})^2) \mathbf{1}_2. \quad (21)$$

Hence the transverse speed is

$$v_{\perp}^{(N)} = \sqrt{(\alpha^{(N)})^2 + (\beta^{(N)})^2}. \quad (22)$$

Substituting everything into (5), the projected core block becomes

$$\boxed{H_{\text{core}}^{(N)}(p) = E_*^{(N)}(p) \mathbf{1}_4 + m^{(N)} \tau^1 \otimes \mathbf{1}_2 + v_{\parallel}^{(N)} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1 \widetilde{W}_{(N)}^1 + p_2 \widetilde{W}_{(N)}^2) + O(|p|^2).} \quad (23)$$

Using (20), this is

$$\begin{aligned} H_{\text{core}}^{(N)}(p) &= E_*^{(N)}(p) \mathbf{1}_4 + m^{(N)} \tau^1 \otimes \mathbf{1}_2 + v_{\parallel}^{(N)} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 \\ &\quad + \tau^1 \otimes [(\alpha^{(N)} p_1 - \beta^{(N)} p_2) \sigma_1 + (\beta^{(N)} p_1 + \alpha^{(N)} p_2) \sigma_2] + O(|p|^2). \end{aligned} \quad (24)$$

This is exactly the canonical target form that was missing.

VI. DIRAC-CARRIER CONDITIONS IN EXPLICIT FINITE FORM

The explicit finite conditions are now:

$$\boxed{m^{(N)} = 0,} \quad (25)$$

$$\boxed{v_{\parallel}^{(N)} \neq 0,} \quad (26)$$

$$\boxed{(\alpha^{(N)}, \beta^{(N)}) \neq (0, 0).} \quad (27)$$

Equivalently,

$$\boxed{\widetilde{W}_{(N)}^1, \widetilde{W}_{(N)}^2 \text{ are nonzero, traceless Hermitian, and linearly independent.}} \quad (28)$$

Under these conditions, define

$$\Gamma^1 := \tau^1 \otimes \sigma_1, \quad \Gamma^2 := \tau^1 \otimes \sigma_2, \quad \Gamma^3 := \tau^3 \otimes \mathbf{1}_2. \quad (29)$$

Then

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4,$$

and after a constant transverse rotation the principal symbol is

$$H_{\text{core}}^{(N), \text{pr}}(p) = v_{\perp}^{(N)} p_1' \Gamma^1 + v_{\perp}^{(N)} p_2' \Gamma^2 + v_{\parallel}^{(N)} p_{\parallel} \Gamma^3. \quad (30)$$

Thus $H_{\text{core}}^{(N)}(p)$ is a genuine anisotropic massless Dirac carrier if and only if

$$m^{(N)} = 0, \quad v_{\parallel}^{(N)} \neq 0, \quad v_{\perp}^{(N)} \neq 0.$$

VII. WHAT IS NOW CLOSED AND WHAT REMAINS OPEN

What is now explicitly closed is the exact form of the enriched shell basis and the canonical 4×4 core block:

$$\mathcal{H}_D^{(N)} = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\},$$

$$H_{\text{core}}^{(N)}(p) = E_*^{(N)} \mathbf{1}_4 + m^{(N)} \tau^1 \otimes \mathbf{1}_2 + v_{\parallel}^{(N)} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes (p_1 \widetilde{W}_{(N)}^1 + p_2 \widetilde{W}_{(N)}^2) + O(|p|^2).$$

What is still not unconditionally closed is the final microscopic evaluation of

$$m^{(N)}, \quad v_{\parallel}^{(N)}, \quad \alpha^{(N)}, \quad \beta^{(N)}$$

from the true refined locked + Class II Bloch operator of TECT. That is now a finite symbolic/numerical extraction problem, not a conceptual one.

VIII. FINAL SUMMARY

The explicit enriched shell Dirac carrier is

$$\mathcal{H}_D^{(N)} = \text{span}\{|+, 1\rangle, |+, 2\rangle, |-, 1\rangle, |-, 2\rangle\} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

With

$$P_D^{(N)} = \sum_{\sigma=\pm} \sum_{a=1}^2 |\sigma, a\rangle \langle \sigma, a|,$$

the core block is

$$H_{\text{core}}^{(N)}(p) = P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)} = \begin{pmatrix} E_+^{(N)}(p) & B^{(N)}(p) \\ B^{(N)}(p)^\dagger & E_-^{(N)}(p) \end{pmatrix}.$$

After subtracting the common scalar piece and fixing the valley phase, its canonical form is

$$H_{\text{core}}^{(N)}(p) = E_*^{(N)}(p) \mathbf{1}_4 + m^{(N)} \tau^1 \otimes \mathbf{1}_2 + v_{\parallel}^{(N)} p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + \tau^1 \otimes \left(p_1 \widetilde{W}_{(N)}^1 + p_2 \widetilde{W}_{(N)}^2 \right) + O(|p|^2).$$

In the locked Pauli basis,

$$\widetilde{W}_{(N)}^1 = \alpha^{(N)} \sigma_1 + \beta^{(N)} \sigma_2, \quad \widetilde{W}_{(N)}^2 = -\beta^{(N)} \sigma_1 + \alpha^{(N)} \sigma_2,$$

so that

$$(\widetilde{W}_{(N)}^1)^2 = (\widetilde{W}_{(N)}^2)^2 = ((\alpha^{(N)})^2 + (\beta^{(N)})^2) \mathbf{1}_2, \quad \{\widetilde{W}_{(N)}^1, \widetilde{W}_{(N)}^2\} = 0.$$

Therefore the explicit Dirac-carrier conditions are

$$m^{(N)} = 0, \quad v_{\parallel}^{(N)} \neq 0, \quad (\alpha^{(N)}, \beta^{(N)}) \neq (0, 0).$$

Equivalently,

$$\widetilde{W}_{(N)}^1, \widetilde{W}_{(N)}^2 \text{ are traceless Hermitian, nonzero, and linearly independent.}$$

Under these conditions, defining

$$\Gamma^1 = \tau^1 \otimes \sigma_1, \quad \Gamma^2 = \tau^1 \otimes \sigma_2, \quad \Gamma^3 = \tau^3 \otimes \mathbf{1}_2,$$

one obtains

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4,$$

and the principal core symbol is the anisotropic massless Dirac operator

$$H_{\text{core}}^{(N), \text{pr}}(p) = v_{\perp}^{(N)} p'_{\parallel} \Gamma^1 + v_{\perp}^{(N)} p'_2 \Gamma^2 + v_{\parallel}^{(N)} p_{\parallel} \Gamma^3, \quad v_{\perp}^{(N)} = \sqrt{(\alpha^{(N)})^2 + (\beta^{(N)})^2}.$$

Thus the remaining microscopic frontier is no longer the abstract existence of the carrier, but the finite extraction of

$$m^{(N)}, \quad v_{\parallel}^{(N)}, \quad \alpha^{(N)}, \quad \beta^{(N)}$$

from the refined locked + Class II Bloch operator.

IX. EXPLICIT ENRICHED SHELL BASIS AND CANONICAL CORE EXTRACTION

X. GOAL

The current frontier is to make the enriched shell Dirac carrier explicit. The correct task is not to reprove the abstract reduced Dirac theorem, but to:

1. fix an explicit 4-dimensional enriched shell basis,
2. write the projected block

$$H_{\text{core}}^{(N)}(p) := P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)},$$

3. separate its constant and linear pieces,
4. identify the exact finite conditions for a massless Dirac carrier.

XI. EXPLICIT ENRICHED SHELL BASIS

Fix the selected shell pair $\pm G_*$, and define the Bragg quasimomentum

$$k_* = \frac{G_*}{2}. \quad (31)$$

Choose a shell-relevant orthonormal internal doublet

$$\xi_1, \xi_2 \in \mathbb{C}_{\text{int}}^2, \quad \xi_a^\dagger \xi_b = \delta_{ab}. \quad (32)$$

In the constant-envelope branch, define

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a, \quad a = 1, 2. \quad (33)$$

Then the enriched shell-pair basis states are

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2. \quad (34)$$

We define the 4-dimensional enriched shell carrier by

$$\boxed{\mathcal{H}_D^{(N)} = \text{span}\{\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}\}.} \quad (35)$$

Equivalently,

$$\mathcal{H}_D^{(N)} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2. \quad (36)$$

Let

$$P_D^{(N)} = \sum_{\sigma=\pm} \sum_{a=1}^2 |\sigma, a\rangle \langle \sigma, a| \quad (37)$$

be the orthogonal projector onto this carrier.

XII. PROJECTED CORE BLOCK

The shell-projected Hamiltonian is

$$H_{\text{shell}}^{(N)}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}. \quad (38)$$

Its restriction to the enriched carrier is

$$\boxed{H_{\text{core}}^{(N)}(p) = P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)}}. \quad (39)$$

In the ordered basis

$$(\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}),$$

the core block has the general 2×2 valley-block form

$$H_{\text{core}}^{(N)}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}, \quad E_\pm(p), B(p) \in \text{Mat}_{2 \times 2}(\mathbb{C}). \quad (40)$$

XIII. CONSTANT AND LINEAR EXPANSION

Expand around $p = 0$:

$$E_\pm(p) = E_\pm(0) + p_i \partial_{p_i} E_\pm(0) + O(|p|^2), \quad (41)$$

$$B(p) = B(0) + p_i \partial_{p_i} B(0) + O(|p|^2). \quad (42)$$

The common scalar energy is

$$E_*(0) := \frac{1}{4} \text{Tr} H_{\text{core}}^{(N)}(0). \quad (43)$$

Subtracting the irrelevant common scalar shift, the physically meaningful constant part is

$$H_{\text{core}}^{(N)}(0) - E_*(0) \mathbf{1}_4. \quad (44)$$

XIV. CONSTANT MASS CLASSIFICATION

At the doubled shell-block level, the general little-group-scalar constant Hermitian operator is

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2. \quad (45)$$

Among these, the genuine constant Dirac-mass space is exactly

$$\boxed{\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}}. \quad (46)$$

Therefore the constant off-diagonal scalar part of $H_{\text{core}}^{(N)}(0)$ is the candidate Dirac mass.

Choose the valley phase so that this constant off-diagonal scalar part is aligned along $\tau_x \otimes \mathbf{1}_2$. Then

$$H_{\text{core}}^{(N)}(0) - E_*(0) \mathbf{1}_4 = \delta E_0 \tau_z \otimes \mathbf{1}_2 + m \tau_x \otimes \mathbf{1}_2 + (\text{non-scalar internal part}). \quad (47)$$

Here:

- δE_0 is the opposite-valley energy mismatch,
- m is the genuine constant Dirac mass.

Thus the exact massless condition is

$$\boxed{m = 0}. \quad (48)$$

If the residual valley symmetry generated by

$$Q_V = \tau_z \otimes \mathbf{1}_2 \quad (49)$$

is exact on the isolated doubled shell block, then

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}, \quad (50)$$

so $m = 0$ is symmetry-protected at shell-block level.

XV. LONGITUDINAL LINEAR PIECE

The longitudinal Dirac channel comes from the odd valley splitting of the pair-diagonal blocks. Define

$$v_{\parallel} := \frac{1}{4} \text{Tr} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0}. \quad (51)$$

Then the pair-diagonal linear part is

$$\frac{1}{2} (E_+(p) - E_-(p)) = v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2), \quad (52)$$

hence contributes

$$v_{\parallel} p_{\parallel} \tau_z \otimes \mathbf{1}_2. \quad (53)$$

The nondegeneracy condition is

$$\boxed{v_{\parallel} \neq 0.} \quad (54)$$

XVI. TRANSVERSE LINEAR PIECE

The transverse Dirac channel comes from the linear off-diagonal block $B(p)$. Define

$$W^1 := \frac{1}{2} [\partial_{p_1} B(0) + \partial_{p_1} B(0)^{\dagger}]_{\text{trl}}, \quad W^2 := \frac{1}{2} [\partial_{p_2} B(0) + \partial_{p_2} B(0)^{\dagger}]_{\text{trl}}, \quad (55)$$

where $(\cdot)_{\text{trl}}$ denotes the traceless Hermitian part on the internal 2×2 space.

After a constant valley-phase choice, the off-diagonal linear term can be written canonically as

$$\tau_x \otimes (p_1 W^1 + p_2 W^2). \quad (56)$$

Thus the full canonical core form becomes

$$\boxed{H_{\text{core}}^{(N)}(p) = E_*(p) \mathbf{1}_4 + m \tau_x \otimes \mathbf{1}_2 + v_{\parallel} p_{\parallel} \tau_z \otimes \mathbf{1}_2 + \tau_x \otimes (p_1 W^1 + p_2 W^2) + O(|p|^2).} \quad (57)$$

This is exactly the canonical target form required by the current program.

XVII. LOCKED PAULI FORM OF THE TRANSVERSE MATRICES

Now choose the internal basis so that the transverse matrices are in locked form:

$$\boxed{W^1 = \alpha \sigma_1 + \beta \sigma_2, \quad W^2 = -\beta \sigma_1 + \alpha \sigma_2.} \quad (58)$$

Then

$$(W^1)^2 = (W^2)^2 = (\alpha^2 + \beta^2) \mathbf{1}_2, \quad \{W^1, W^2\} = 0, \quad (59)$$

and the transverse speed is

$$v_{\perp} = \sqrt{\alpha^2 + \beta^2}. \quad (60)$$

Substituting (58) into (57),

$$\begin{aligned} H_{\text{core}}^{(N)}(p) &= E_*(p) \mathbf{1}_4 + m \tau_x \otimes \mathbf{1}_2 + v_{\parallel} p_{\parallel} \tau_z \otimes \mathbf{1}_2 \\ &\quad + \tau_x \otimes [(\alpha p_1 - \beta p_2) \sigma_1 + (\beta p_1 + \alpha p_2) \sigma_2] + O(|p|^2). \end{aligned} \quad (61)$$

XVIII. DIRAC-CARRIER CONDITIONS

The explicit finite Dirac-carrier conditions are therefore

$$\boxed{m = 0, \quad v_{\parallel} \neq 0, \quad W^1, W^2 \text{ traceless Hermitian, nonzero, and linearly independent.}} \quad (62)$$

In the locked basis this is equivalent to

$$\boxed{m = 0, \quad v_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (63)$$

XIX. CLIFFORD CLOSURE

Define

$$\Gamma^1 = \tau_x \otimes \sigma_1, \quad \Gamma^2 = \tau_x \otimes \sigma_2, \quad \Gamma^3 = \tau_z \otimes \mathbf{1}_2. \quad (64)$$

Then

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4. \quad (65)$$

Hence, after a transverse rotation, the principal symbol of the massless core block is

$$H_{\text{core}}^{(N),\text{pr}}(p) = v_{\perp} p'_1 \Gamma^1 + v_{\perp} p'_2 \Gamma^2 + v_{\parallel} p_{\parallel} \Gamma^3. \quad (66)$$

Therefore the explicit enriched shell carrier is a genuine anisotropic massless Dirac carrier whenever (63) holds.

XX. CONNECTION WITH THE FINITE MICROSCOPIC CARRIER AUDIT

The shell-adapted microscopic seed space is

$$V_{\min} = \text{span}\{B_{\parallel}, B_I, B_J\}. \quad (67)$$

For any microscopic transport basis $\{E_A\} \subset V_{\text{TCT}}$, decompose

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A, \quad R_A \perp V_{\min}. \quad (68)$$

Then:

- $a_A \neq 0$ feeds the longitudinal coefficient $v_{\parallel}$,
- $(b_A, c_A) \neq (0, 0)$ feeds the transverse matrices W^1, W^2 .

Hence the microscopic completion criterion is exactly the finite audit:

$$\boxed{\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).} \quad (69)$$

Thus the explicit 4-dimensional shell basis and the finite basis-overlap audit fit together perfectly: the former gives the carrier, the latter tells us whether the actual microscopic linearization populates it nontrivially.

XXI. FINAL SUMMARY

The enriched shell Dirac carrier can be fixed explicitly as

$$\mathcal{H}_D^{(N)} = \text{span}\{\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}\}, \quad \Psi_{\pm,a}(x) = e^{\pm i k_* \cdot x} u_a(x), \quad k_* = \frac{G_*}{2}.$$

In the ordered basis

$$(\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}),$$

the projected enriched shell core is

$$H_{\text{core}}^{(N)}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix},$$

with 2×2 internal blocks.

Its canonical expansion is

$$H_{\text{core}}^{(N)}(p) = E_*(p)\mathbf{1}_4 + m\tau_x \otimes \mathbf{1}_2 + v_{\parallel}p_{\parallel}\tau_z \otimes \mathbf{1}_2 + \tau_x \otimes (p_1W^1 + p_2W^2) + O(|p|^2).$$

The exact finite Dirac-carrier conditions are

$$m = 0, \quad v_{\parallel} \neq 0, \quad W^1, W^2 \text{ traceless Hermitian, nonzero, and linearly independent.}$$

Choosing the internal basis in locked Pauli form,

$$W^1 = \alpha\sigma_1 + \beta\sigma_2, \quad W^2 = -\beta\sigma_1 + \alpha\sigma_2,$$

one obtains

$$v_{\perp} = \sqrt{\alpha^2 + \beta^2},$$

and the principal core symbol becomes

$$H_{\text{core}}^{(N),\text{pr}}(p) = v_{\perp}p'_1\Gamma^1 + v_{\perp}p'_2\Gamma^2 + v_{\parallel}p_{\parallel}\Gamma^3,$$

where

$$\Gamma^1 = \tau_x \otimes \sigma_1, \quad \Gamma^2 = \tau_x \otimes \sigma_2, \quad \Gamma^3 = \tau_z \otimes \mathbf{1}_2, \quad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4.$$

Therefore the enriched shell carrier is a genuine anisotropic massless Dirac carrier if and only if

$$m = 0, \quad v_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

At the microscopic finite-audit level, the completion criterion is

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0),$$

where

$$E_A = a_AB_{\parallel} + b_AB_I + c_AB_J + R_A, \quad R_A \perp \text{span}\{B_{\parallel}, B_I, B_J\}.$$

Thus the remaining frontier is no longer conceptual: it is the explicit extraction of

$$m, \quad v_{\parallel}, \quad W^1, \quad W^2$$

from the refined locked + Class II microscopic Bloch operator.

XXII. EXPLICIT EXTRACTION OF $m, v_{\parallel}, W^1$, AND W^2 FROM THE REFINED LOCKED + CLASS II BRANCH

XXIII. STARTING POINT

We start from the refined locked microscopic branch with one heavy Class II mediator:

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k), \quad H_{XX}(k) = M_X^2\mathbf{1}_X. \quad (70)$$

Hence

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger. \quad (71)$$

Fix the selected shell pair $\pm G_*$, and set

$$k_* = \frac{G_*}{2}. \quad (72)$$

Choose the enriched shell basis

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad a = 1, 2. \quad (73)$$

Then the shell-projected operator is

$$H_{\text{shell}}^{(N)}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}, \quad (74)$$

and the enriched core block is

$$H_{\text{core}}^{(N)}(p) = P_D^{(N)} H_{\text{shell}}^{(N)}(p) P_D^{(N)}. \quad (75)$$

XXIV. VALLEY-BLOCK MATRIX ELEMENTS

In the ordered basis

$$(\Psi_{+,1}, \Psi_{+,2}, \Psi_{-,1}, \Psi_{-,2}),$$

the projected core block has the form

$$H_{\text{core}}^{(N)}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^\dagger & E_-(p) \end{pmatrix}, \quad E_\pm(p), B(p) \in \text{Mat}_{2 \times 2}(\mathbb{C}), \quad (76)$$

where

$$(E_+(p))_{ab} = \langle \Psi_{+,a}, H_{\text{eff}}(k_* + p) \Psi_{+,b} \rangle, \quad (77)$$

$$(E_-(p))_{ab} = \langle \Psi_{-,a}, H_{\text{eff}}(k_* + p) \Psi_{-,b} \rangle, \quad (78)$$

$$B_{ab}(p) = \langle \Psi_{+,a}, H_{\text{eff}}(k_* + p) \Psi_{-,b} \rangle. \quad (79)$$

Thus the complete microscopic extraction problem reduces to computing these finite matrix elements.

XXV. CONSTANT AND LINEAR DECOMPOSITION

Expand around $p = 0$:

$$E_\pm(p) = E_\pm(0) + p_i V_{\pm,i} + O(|p|^2), \quad (80)$$

$$B(p) = B(0) + p_i \mathcal{W}_i + O(|p|^2). \quad (81)$$

We write the common scalar part as

$$E_*(p) = \frac{1}{4} \text{Tr} H_{\text{core}}^{(N)}(p). \quad (82)$$

Then three quantities control the Dirac carrier:

1. the constant pair-off-diagonal block $B(0)$;
2. the odd pair-diagonal linear splitting $E_+(p) - E_-(p)$;
3. the linear pair-off-diagonal matrices $\mathcal{W}_1, \mathcal{W}_2$.

XXVI. CONSTANT MASS EXTRACTION

The constant Dirac mass is the little-group-scalar part of $B(0)$. At the doubled block level, the genuine constant Dirac mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}. \quad (83)$$

Therefore the scalar constant pair-mixing amplitude is

$$M_0 := \frac{1}{2} \text{tr}_{\text{int}} B(0), \quad (84)$$

and after a constant valley-phase choice the mass term is

$$m \tau_x \otimes \mathbf{1}_2, \quad m = |M_0|. \quad (85)$$

In the constant-envelope shell basis,

$$\boxed{M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2,} \quad (86)$$

hence

$$\boxed{m = 0 \iff \widehat{U}_{\text{eff}}(G_*) = 0.} \quad (87)$$

XXVII. LONGITUDINAL SLOPE EXTRACTION

Define the shell-adapted longitudinal direction

$$\hat{n} = \frac{G_*}{|G_*|}, \quad p_{\parallel} = \hat{n} \cdot p.$$

The longitudinal pair splitting is

$$\Delta E(p) := \frac{1}{2} (E_+(p) - E_-(p)). \quad (88)$$

Its scalar shell-relevant coefficient is

$$\boxed{v_{\parallel} = \frac{1}{4} \text{Tr}_{\text{int}} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0}.} \quad (89)$$

Equivalently,

$$E_+(p) - E_-(p) = 2v_{\parallel} p_{\parallel} \mathbf{1}_2 + O(|p|^2) \quad (90)$$

is the exact shell-level longitudinal criterion.

For the refined working branch, the total longitudinal slope decomposes as

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}.} \quad (91)$$

Thus the longitudinal carrier condition is simply

$$\boxed{v_{\parallel} \neq 0.} \quad (92)$$

XXVIII. TRANSVERSE MATRIX EXTRACTION

Fix a shell-adapted orthonormal frame

$$(e_1, e_2, \hat{n}).$$

Define the two transverse momentum coordinates

$$p_1 = e_1 \cdot p, \quad p_2 = e_2 \cdot p.$$

The transverse Dirac matrices are the traceless Hermitian parts of

$$\mathcal{W}_1 = \partial_{p_1} B(p)|_{p=0}, \quad \mathcal{W}_2 = \partial_{p_2} B(p)|_{p=0}. \quad (93)$$

Thus

$$W^1 = \left[\frac{1}{2} (\mathcal{W}_1 + \mathcal{W}_1^\dagger) \right]_{\text{trl}}, \quad W^2 = \left[\frac{1}{2} (\mathcal{W}_2 + \mathcal{W}_2^\dagger) \right]_{\text{trl}}. \quad (94)$$

In the constant-envelope reduction, these are controlled by first-shell Fourier coefficients:

$$W^\alpha = \hat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2. \quad (95)$$

Moreover, the effective transverse coefficients split into explicit microscopic sources:

$$W^\alpha = \left[\hat{T}_{\text{bare}}^\alpha(G_*) + \hat{T}_{(II)}^\alpha(G_*) + \hat{T}_{(\text{sp})}^\alpha(G_*) \right] s^\alpha. \quad (96)$$

Hence the transverse nondegeneracy condition is

$$\hat{T}_{\text{eff}}^1(G_*) \neq 0, \quad \hat{T}_{\text{eff}}^2(G_*) \neq 0. \quad (97)$$

XXIX. LOCKED PAULI FORM AND (α, β) EXTRACTION

Because W^1, W^2 are traceless Hermitian 2×2 matrices, one may choose the internal basis so that

$$W^1 = \alpha \sigma_1 + \beta \sigma_2, \quad W^2 = -\beta \sigma_1 + \alpha \sigma_2. \quad (98)$$

Then the coefficients are read off exactly by

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 W^1 + \sigma_2 W^2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 W^1 - \sigma_1 W^2). \quad (99)$$

Equivalently, in the shell-seed dictionary,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_I I_{\theta}, \quad \beta = c_J I_{\theta}. \quad (100)$$

Hence, since $I_{\theta} \neq 0$ and $I_{\theta'^3} \neq 0$, the finite basis audit

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0)$$

is equivalent to

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

XXX. CANONICAL CORE HAMILTONIAN

Combining the previous steps, the explicit enriched shell core becomes

$$\boxed{H_{\text{core}}^{(N)}(p) = E_*(p)\mathbf{1}_4 + m\tau_x \otimes \mathbf{1}_2 + v_{\parallel}p_{\parallel}\tau_z \otimes \mathbf{1}_2 + \tau_x \otimes (p_1W^1 + p_2W^2) + O(|p|^2).} \quad (101)$$

In the locked basis (98),

$$\begin{aligned} H_{\text{core}}^{(N)}(p) &= E_*(p)\mathbf{1}_4 + m\tau_x \otimes \mathbf{1}_2 + v_{\parallel}p_{\parallel}\tau_z \otimes \mathbf{1}_2 \\ &\quad + \tau_x \otimes \left[(\alpha p_1 - \beta p_2)\sigma_1 + (\beta p_1 + \alpha p_2)\sigma_2 \right] + O(|p|^2). \end{aligned} \quad (102)$$

Define

$$\Gamma^1 = \tau_x \otimes \sigma_1, \quad \Gamma^2 = \tau_x \otimes \sigma_2, \quad \Gamma^3 = \tau_z \otimes \mathbf{1}_2. \quad (103)$$

Then

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4. \quad (104)$$

Hence the massless principal symbol is

$$H_{\text{core}}^{\text{pr}}(p) = v_{\perp}p'_1\Gamma^1 + v_{\perp}p'_2\Gamma^2 + v_{\parallel}p_{\parallel}\Gamma^3, \quad v_{\perp} = \sqrt{\alpha^2 + \beta^2}. \quad (105)$$

XXXI. FINAL FINITE EXTRACTION DICTIONARY

The complete refined locked + Class II extraction problem is therefore reduced to the finite computation of the following four objects:

$$\boxed{\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.} \quad (106)$$

Equivalently:

$$\boxed{m \iff \hat{U}_{\text{eff}}(G_*), \quad W^{\alpha} \iff \hat{T}_{\text{eff}}^{\alpha}(G_*), \quad v_{\parallel} \iff \partial_{p_{\parallel}}(E_+ - E_-)|_0.} \quad (107)$$

Thus the explicit massless Dirac-carrier conditions are

$$\boxed{\hat{U}_{\text{eff}}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad \hat{T}_{\text{eff}}^1(G_*) \neq 0, \quad \hat{T}_{\text{eff}}^2(G_*) \neq 0.} \quad (108)$$

XXXII. FINAL SUMMARY

Fix the enriched shell basis

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a(x), \quad \Psi_{-,a}(x) = e^{-ik_* \cdot x} u_a(x), \quad k_* = \frac{G_*}{2}.$$

The shell core block is

$$H_{\text{core}}^{(N)}(p) = \begin{pmatrix} E_+(p) & B(p) \\ B(p)^{\dagger} & E_-(p) \end{pmatrix}.$$

The exact extraction rules are:

$$\boxed{m = \left| \frac{1}{2} \text{tr}_{\text{int}} B(0) \right|, \quad B(0) = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2,}$$

so

$$m = 0 \iff \widehat{U}_{\text{eff}}(G_*) = 0.$$

$$v_{\parallel} = \frac{1}{4} \text{Tr}_{\text{int}} [\partial_{p_{\parallel}} (E_+(p) - E_-(p))]_{p=0},$$

with

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}.$$

$$W^{\alpha} = \left[\frac{1}{2} (\partial_{p_{\alpha}} B(0) + \partial_{p_{\alpha}} B(0)^{\dagger}) \right]_{\text{trl}} = \widehat{T}_{\text{eff}}^{\alpha}(G_*) s^{\alpha}, \quad \alpha = 1, 2.$$

Choosing the internal basis in locked Pauli form,

$$W^1 = \alpha \sigma_1 + \beta \sigma_2, \quad W^2 = -\beta \sigma_1 + \alpha \sigma_2,$$

the explicit coefficients are

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 W^1 + \sigma_2 W^2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 W^1 - \sigma_1 W^2).$$

Therefore the refined locked + Class II branch gives a massless shell Dirac carrier if and only if

$$\widehat{U}_{\text{eff}}(G_*) = 0, \quad v_{\parallel} \neq 0, \quad \widehat{T}_{\text{eff}}^1(G_*) \neq 0, \quad \widehat{T}_{\text{eff}}^2(G_*) \neq 0.$$

Equivalently,

$$m = 0, \quad \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Thus the remaining microscopic frontier is the finite evaluation of

$$\widehat{U}_{\text{eff}}(G_*), \quad \widehat{T}_{\text{eff}}^1(G_*), \quad \widehat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}$$

for the actual refined locked TECT Bloch operator.

XXXIII. TERM-BY-TERM ROLE OF THE REFINED LOCKED + CLASS II BRANCH

XXXIV. GOAL

The remaining shell-side microscopic problem is to separate the refined locked + Class II contribution into:

$$\widehat{U}_{(II)}(G_*), \quad \widehat{T}_{(II)}^1(G_*), \quad \widehat{T}_{(II)}^2(G_*), \quad \delta v_{\parallel}^{(II)}. \quad (109)$$

These govern, respectively:

1. the scalar intervalley Bragg block,
2. the first transverse Dirac matrix,
3. the second transverse Dirac matrix,
4. the longitudinal shell slope.

The key point is that these four quantities do *not* play symmetric roles.

XXXV. 1. SCHUR-COMPLEMENT DECOMPOSITION OF THE CLASS II CORRECTION

Start from the refined locked branch with one heavy auxiliary Class II field:

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k), \quad H_{XX}(k) = M_X^2 \mathbf{1}_X. \quad (110)$$

Using

$$H_{\Psi X}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad (111)$$

one gets

$$\Delta_X(k) := -H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k) \quad (112)$$

$$= -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger \quad (113)$$

$$= -\frac{\alpha_X^2}{M_X^2} \mathcal{J} \mathcal{J}^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\mathcal{J} \mathcal{K}^\dagger + \mathcal{K} \mathcal{J}^\dagger) - \frac{\beta_X^2}{M_X^2} \mathcal{K} \mathcal{K}^\dagger. \quad (114)$$

Therefore the Class II shell correction naturally decomposes into

$$JJ, \quad JK, \quad KK$$

sectors.

XXXVI. 2. FOURIER-SHELL DECOMPOSITION AT $Q = G_*$

At the folded Bragg point

$$k_* = \frac{G_*}{2},$$

the shell-relevant constant-envelope extraction is controlled by the $Q = G_*$ primitive-cell Fourier coefficient.

Thus the total effective shell data are

$$M_0 = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2, \quad W^\alpha = \hat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2, \quad (115)$$

together with

$$v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}. \quad (116)$$

The exact split is

$$\hat{U}_{\text{eff}}(G_*) = \hat{U}_{\text{bare}}(G_*) + \hat{U}_{(II)}(G_*) + \hat{U}_{(\text{sp})}(G_*), \quad (117)$$

$$\hat{T}_{\text{eff}}^\alpha(G_*) = \hat{T}_{\text{bare}}^\alpha(G_*) + \hat{T}_{(II)}^\alpha(G_*) + \hat{T}_{(\text{sp})}^\alpha(G_*), \quad \alpha = 1, 2. \quad (118)$$

So the Class II problem is to identify the $Q = G_*$ identity and traceless Pauli parts of Δ_X .

XXXVII. 3. FORMAL SHELL PROJECTIONS OF THE CLASS II BLOCK

Let $(\cdot)_1$ denote the internal identity part and $(\cdot)_\alpha$ the s^α coefficient. Then formally:

$$\hat{U}_{(II)}(G_*) = \left[\widehat{\Delta_X}(G_*) \right]_1, \quad (119)$$

$$\hat{T}_{(II)}^\alpha(G_*) = \left[\widehat{\Delta_X}(G_*) \right]_\alpha, \quad \alpha = 1, 2. \quad (120)$$

Substituting (114), this gives

$$\begin{aligned}\widehat{U}_{(II)}(G_*) &= -\frac{\alpha_X^2}{M_X^2} \left[\widehat{\mathcal{J}\mathcal{J}^\dagger}(G_*) \right]_1 - \frac{\alpha_X \beta_X}{M_X^2} \left[\widehat{\mathcal{J}\mathcal{K}^\dagger + \mathcal{K}\mathcal{J}^\dagger}(G_*) \right]_1 \\ &\quad - \frac{\beta_X^2}{M_X^2} \left[\widehat{\mathcal{K}\mathcal{K}^\dagger}(G_*) \right]_1,\end{aligned}\tag{121}$$

and

$$\begin{aligned}\widehat{T}_{(II)}^\alpha(G_*) &= -\frac{\alpha_X^2}{M_X^2} \left[\widehat{\mathcal{J}\mathcal{J}^\dagger}(G_*) \right]_\alpha - \frac{\alpha_X \beta_X}{M_X^2} \left[\widehat{\mathcal{J}\mathcal{K}^\dagger + \mathcal{K}\mathcal{J}^\dagger}(G_*) \right]_\alpha \\ &\quad - \frac{\beta_X^2}{M_X^2} \left[\widehat{\mathcal{K}\mathcal{K}^\dagger}(G_*) \right]_\alpha.\end{aligned}\tag{122}$$

So, in complete generality, both JK and KK can contribute to the traceless transverse sector, while the scalar identity part controls the Bragg mass.

XXXVIII. 4. EXPLICIT TRANSVERSE CLASS II OVERLAP

Inside the folded-Bragg Class II audit isolated in the notes, the transverse coefficient compresses to the shell convolution

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q').\tag{123}$$

If the shell-relevant K -coefficient is zero-mode dominated, then

$$\widehat{K}_{i,\alpha}^A(Q') \approx \widehat{K}_{i,\alpha}^A(0) \delta_{Q',0},\tag{124}$$

and (123) reduces to

$$\widehat{T}_{(II)}^\alpha(G_*) \approx -\frac{\alpha_X \beta_X}{M_X^2} \sum_A iG_{*i} A_{G_*} j^A \widehat{K}_{i,\alpha}^A(0).\tag{125}$$

Hence the canonical first-shell form is

$$\widehat{T}_{(II)}^\alpha(G_*) = iA_{G_*}(C_\alpha \cdot G_*),\tag{126}$$

for an effective microscopic vector C_α .

Therefore the transverse witness is structurally natural:

$$\widehat{T}_{(II)}^\alpha(G_*) \neq 0 \iff A_{G_*} \neq 0, \quad C_\alpha \cdot G_* \neq 0 \quad \text{modulo accidental cancellations.}\tag{127}$$

XXXIX. 5. EXPLICIT LONGITUDINAL CLASS II COEFFICIENT

The longitudinal coefficient remains separate from the transverse shell Fourier overlaps. At the reduced level,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.\tag{128}$$

In the explicit local Class II-compatible branch, one has

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the active longitudinal route is the primitive Branch-B one:

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.\tag{129}$$

Equivalently, in the pure Class II working truncation,

$$A_1^{\text{Class II}}(p) = \frac{\nu_{\parallel} \Gamma c_{\parallel}}{2} I_{\theta'^3} p_{\parallel} \sigma_3 + \nu_{\perp} \Gamma I_{\theta} \left[c_I (p_1 \sigma_1 + p_2 \sigma_2) + c_J (-p_2 \sigma_1 + p_1 \sigma_2) \right]. \quad (130)$$

Thus the explicit Class II contribution to the shell slope is

$$\delta v_{\parallel}^{(II)} = \frac{\nu_{\parallel} \Gamma c_{\parallel}}{2} I_{\theta'^3}, \quad (131)$$

while the transverse reduced coefficients are

$$\alpha^{(II)} = \nu_{\perp} \Gamma c_I I_{\theta}, \quad \beta^{(II)} = \nu_{\perp} \Gamma c_J I_{\theta}. \quad (132)$$

XL. 6. WHY CLASS II DOES NOT SOLVE THE SCALAR BRAGG-MASS PROBLEM

Now the crucial asymmetry appears.

A naive scalar periodic-core contribution such as

$$U_{\text{quad}}(x) = u_2 \phi_0(x)^2$$

gives

$$\widehat{U}_{\text{quad}}(G_*) = 4u_2 A^2, \quad M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0 \quad (133)$$

in the equal-amplitude 12-mode branch.

By contrast, every Class II contribution carries an explicit internal Pauli generator. Hence its constant pair-mixing block lies in

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

not in the scalar identity channel

$$\tau_1 \otimes \mathbf{1}_2.$$

Since identity and traceless Pauli channels are orthogonal, the Class II sector cannot directly cancel the scalar Bragg block.

Moreover, in the natural odd/reflection parity of the shell-Dirac construction, the constant gradient-sector pair mixing vanishes at principal order:

$$\delta B_0^{\text{Class II}} = 0.$$

Therefore:

$$\boxed{\text{Class II renormalizes the derivative witness sector } (\lambda_{\parallel}, \alpha, \beta), \text{ but does not remove the scalar identity-channel obstruction } M_0.} \quad (134)$$

XLI. 7. ROUTE SELECTION

Because of (134), the plain shell-pair route has only two logically distinct exits:

$$\boxed{\text{(i) identity-channel cancellation,} \quad \text{(ii) graded } 2 + 2 \text{ bypass.}} \quad (135)$$

The exact scalar cancellation law is

$$2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) - \sum_{\eta, \zeta \in S_{\text{bi}}} \widehat{V}(\eta)(R_0)_{\eta\zeta} \overline{\widehat{V}(\zeta + G_*)} = 0. \quad (136)$$

If one does not want to rely on this cancellation, the correct mainline is the graded $2 + 2$ infrared route, where the principal operator is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad D_1(0) = 0, \quad (137)$$

so the scalar Bragg block is bypassed rather than cancelled.

XLII. 8. FINAL STRATEGIC CONCLUSION

At the strongest honest current stage:

1. the explicit Class II branch already supplies

$$\delta v_{\parallel}^{(II)}, \quad \alpha^{(II)}, \quad \beta^{(II)};$$

2. the transverse shell witness is structurally natural and reduced to finite shell convolutions;
3. the longitudinal primitive Branch-B carrier is explicitly available;
4. the true bottleneck of the plain shell route is not the witness sector but the scalar Bragg mass

$$M_0 = \hat{U}_{\text{eff}}(G_*);$$

5. therefore the strongest current mainline is:

use Class II to extract the coefficient point $(\lambda_{\parallel}, \alpha, \beta)$, but use graded $2 + 2$ for the robust massless Dirac route.

XLIII. FINAL SUMMARY

The refined locked + Class II correction is

$$\Delta_X(k) = -\frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger,$$

hence it decomposes into

$$JJ, \quad JK, \quad KK$$

channels.

At the folded Bragg shell $Q = G_*$, the shell data are

$$M_0 = \hat{U}_{\text{eff}}(G_*) \mathbf{1}_2, \quad W^\alpha = \hat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}.$$

The explicit Class II transverse overlap is

$$\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q').$$

In the minimal first-shell channel this reduces to

$$\hat{T}_{(II)}^\alpha(G_*) = i A_{G_*}(C_\alpha \cdot G_*),$$

so the transverse witness is structurally natural.

The explicit longitudinal Class II contribution is

$$\delta v_{\parallel}^{(II)} = \frac{\nu_{\parallel} \Gamma c_{\parallel}}{2} I_{\theta^3},$$

and

$$\alpha^{(II)} = \nu_{\perp} \Gamma c_I I_{\theta}, \quad \beta^{(II)} = \nu_{\perp} \Gamma c_J I_{\theta}.$$

Thus Class II genuinely supplies the derivative witness sector

$$(\lambda_{\parallel}, \alpha, \beta).$$

However, the scalar Bragg mass behaves differently. A naive scalar periodic core gives

$$\hat{U}_{\text{quad}}(G_*) = 4u_2 A^2, \quad M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0.$$

Moreover, the Class II constant block lies in the traceless internal Pauli channel and is orthogonal to the scalar identity channel. Hence:

$$\text{Class II renormalizes the derivative witness sector, but does not remove the scalar Bragg-mass obstruction } M_0.$$

Therefore the plain shell-pair route has only two exits:

$$(i) \text{ identity-channel cancellation,} \quad (ii) \text{ graded } 2 + 2 \text{ bypass.}$$

The strongest honest strategic conclusion is:

$$\text{use the refined locked + Class II branch to extract } (\lambda_{\parallel}, \alpha, \beta), \text{ but use the graded } 2 + 2 \text{ route as the robust massless Dirac main}$$

XLIV. EXACT THREE-DIRECTION PROJECTION OF THE INTERVALLEY CLASS-II TENSOR

XLV. STARTING POINT

The shell-pair intervalley channel is controlled by the single harmonic $Q = G_*$. At the current refined locked + Class II stage, the surviving microscopic tensor is

$$\mathbf{C}_{i,r}^a(G_*; 0) = \alpha_X \mathbf{J}_{i,r}^a(G_*; 0) + \beta_X \mathbf{K}_{i,r}^a(G_*; 0). \quad (138)$$

This is the unique object to be projected onto the three shell-adapted Dirac witness channels.

XLVI. SHELL-ADAPTED FRAME AND PROJECTOR TENSORS

Fix the adapted shell frame

$$(\hat{n}, e_1, e_2), \quad \hat{n} = \frac{G_*}{|G_*|},$$

and the tangent-doublet Pauli basis

$$(\Sigma_1, \Sigma_2, \Sigma_3).$$

Define the three projector tensors

$$P^{(\parallel) i, r}_a := \mathcal{N}_{\parallel} \hat{n}^i (\Sigma_3)^r_a, \quad (139)$$

$$P^{(E) i, r}_a := \mathcal{N}_E \frac{1}{2} [e_1^i (\Sigma_1)^r_a + e_2^i (\Sigma_2)^r_a], \quad (140)$$

$$P^{(O) i, r}_a := \mathcal{N}_O \frac{1}{2} [-e_2^i (\Sigma_1)^r_a + e_1^i (\Sigma_2)^r_a]. \quad (141)$$

Here $\mathcal{N}_{\parallel}, \mathcal{N}_E, \mathcal{N}_O$ are fixed normalization constants.

XLVII. EXACT COEFFICIENT EXTRACTION

The three seed coefficients are exactly

$$c_{\parallel} = P^{(\parallel) i, r}{}_a C_{i, r}^a(G_*; 0), \quad (142)$$

$$c_E = P^{(E) i, r}{}_a C_{i, r}^a(G_*; 0), \quad (143)$$

$$c_O = P^{(O) i, r}{}_a C_{i, r}^a(G_*; 0). \quad (144)$$

Substituting (138), one gets the exact linear decomposition

$$c_{\parallel} = \alpha_X c_{\parallel}^{(J)} + \beta_X c_{\parallel}^{(K)}, \quad (145)$$

$$c_E = \alpha_X c_E^{(J)} + \beta_X c_E^{(K)}, \quad (146)$$

$$c_O = \alpha_X c_O^{(J)} + \beta_X c_O^{(K)}, \quad (147)$$

where

$$c_{\parallel}^{(J)} := P^{(\parallel) i, r}{}_a J_{i, r}^a(G_*; 0), \quad c_{\parallel}^{(K)} := P^{(\parallel) i, r}{}_a K_{i, r}^a(G_*; 0), \quad (148)$$

$$c_E^{(J)} := P^{(E) i, r}{}_a J_{i, r}^a(G_*; 0), \quad c_E^{(K)} := P^{(E) i, r}{}_a K_{i, r}^a(G_*; 0), \quad (149)$$

$$c_O^{(J)} := P^{(O) i, r}{}_a J_{i, r}^a(G_*; 0), \quad c_O^{(K)} := P^{(O) i, r}{}_a K_{i, r}^a(G_*; 0). \quad (150)$$

Thus the coefficient extraction problem is reduced to six finite tensor contractions.

XLVIII. CURRENT EXPLICIT LEADING BRANCH INPUT

At the current explicit leading local level, the K -sector intervalley tensor has the shell-adapted form

$$K_{i, r}^a(G_*; 0) = \hat{n}_i k^a(\theta_*) \left[\Lambda_E^{(0)}(\Sigma_1)_r + \Lambda_O^{(0)}(\Sigma_2)_r \right] + K_{i, r}^{\text{null} a}, \quad (151)$$

with no leading longitudinal component:

$$c_{\parallel}^{(K)} = 0. \quad (152)$$

Moreover, the pure J -sector is axial-only at the shell-adapted local level, so it does not supply transverse witnesses:

$$c_E^{(J)} = 0, \quad c_O^{(J)} = 0. \quad (153)$$

Therefore the exact split (145)–(147) reduces at current explicit leading order to

$$c_{\parallel} = \alpha_X c_{\parallel}^{(J)}, \quad (154)$$

$$c_E = \beta_X c_E^{(K)}, \quad c_O = \beta_X c_O^{(K)}. \quad (155)$$

Thus the primitive longitudinal carrier and the transverse carrier are separated already at the level of explicit shell projectors.

XLIX. REDUCED DIRAC DICTIONARY

Once $(c_{\parallel}, c_E, c_O)$ are known, the reduced coefficients are

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}. \quad (156)$$

Hence

$$\lambda_{\parallel} = \frac{1}{2}(\alpha_X c_{\parallel}^{(J)} + \beta_X c_{\parallel}^{(K)})I_{\theta'^3}, \quad (157)$$

$$\alpha = (\alpha_X c_E^{(J)} + \beta_X c_E^{(K)})I_{\theta}, \quad \beta = (\alpha_X c_O^{(J)} + \beta_X c_O^{(K)})I_{\theta}. \quad (158)$$

Under the current explicit leading branch input,

$$\lambda_{\parallel} = \frac{1}{2}\alpha_X c_{\parallel}^{(J)}I_{\theta'^3}, \quad \alpha = \beta_X c_E^{(K)}I_{\theta}, \quad \beta = \beta_X c_O^{(K)}I_{\theta}. \quad (159)$$

L. IMMEDIATE YES/NO CRITERION

Therefore the shell-level nondegeneracy test is exactly

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0). \quad (160)$$

Equivalently, in the present branch,

$$c_{\parallel}^{(J)} \neq 0, \quad (c_E^{(K)}, c_O^{(K)}) \neq (0, 0), \quad (161)$$

up to accidental cancellations and higher-order corrections.

LI. HONEST CAVEAT: WHY THIS IS NOT YET THE FINAL UNCONDITIONAL TRANSVERSE CLOSURE

The projector algebra above is exact. However, the current explicit leading transverse bookkeeping should still be described honestly.

A naive local real Hermitian JK bilinear vanishes because

$$\{\Sigma_3, \Sigma_1\} = 0, \quad \{\Sigma_3, \Sigma_2\} = 0.$$

Therefore the genuine microscopic transverse frontier is not a naive undifferentiated local JK term, but the derivative-resolved shell audit:

$$\partial_{p_{\nu}} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p_{\nu}} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0}. \quad (162)$$

Hence (155) should be read as the strongest current explicit branch-level extraction structure, not yet as the unconditional final closure of the full microscopic TECT PDE.

LII. CONNECTION TO THE BLOCH CALIBRATION

The transport coefficients $(c_{\parallel}, c_E, c_O)$ feed the reduced coefficients $(\lambda_{\parallel}, \alpha, \beta)$, while the shell-core Bloch block yields the overlap-normalized coefficients $(C_{\parallel}, C_E, C_O)$. The final microscopic viability test remains

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (163)$$

Thus the present tensor-contraction step is the correct transport-side half of the final Bloch calibration.

LIII. FINAL SUMMARY

The intervalley shell-pair tensor is

$$\mathbf{C}_{i,r}^a(G_*; 0) = \alpha_X \mathbf{J}_{i,r}^a(G_*; 0) + \beta_X \mathbf{K}_{i,r}^a(G_*; 0).$$

Its three exact shell-adapted projections are

$$c_{\parallel} = \mathcal{N}_{\parallel} \hat{n}^i(\Sigma_3)^r{}_a \mathbf{C}_{i,r}^a,$$

$$c_E = \mathcal{N}_E \frac{1}{2} [e_1^i(\Sigma_1)^r{}_a + e_2^i(\Sigma_2)^r{}_a] \mathbf{C}_{i,r}^a,$$

$$c_O = \mathcal{N}_O \frac{1}{2} [-e_2^i(\Sigma_1)^r{}_a + e_1^i(\Sigma_2)^r{}_a] \mathbf{C}_{i,r}^a.$$

Therefore

$$c_{\parallel} = \alpha_X c_{\parallel}^{(J)} + \beta_X c_{\parallel}^{(K)}, \quad c_E = \alpha_X c_E^{(J)} + \beta_X c_E^{(K)}, \quad c_O = \alpha_X c_O^{(J)} + \beta_X c_O^{(K)}.$$

At the current explicit leading branch level,

$$c_{\parallel}^{(K)} = 0, \quad c_E^{(J)} = 0, \quad c_O^{(J)} = 0.$$

Hence the leading bookkeeping reduces to

$$\boxed{c_{\parallel} = \alpha_X c_{\parallel}^{(J)}, \quad c_E = \beta_X c_E^{(K)}, \quad c_O = \beta_X c_O^{(K)}}.$$

The reduced Dirac coefficients are then

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}}.$$

So the present explicit branch-level nondegeneracy test is

$$\boxed{c_{\parallel}^{(J)} \neq 0, \quad (c_E^{(K)}, c_O^{(K)}) \neq (0, 0),}$$

equivalently

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)}.$$

However, this should still be stated honestly: the exact projector algebra is closed, but the genuine microscopic transverse frontier is not a naive local JK bilinear. It is the derivative-resolved shell audit

$$\boxed{\partial_{p\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0}}.$$

Thus the current result is:

$$\boxed{\text{the exact tensor-extraction formulas are closed,}}$$

$$\boxed{\text{the leading branch-level coefficient split is explicit,}}$$

but

$$\boxed{\text{the final unconditional microscopic transverse closure still requires the derivative-resolved Bloch audit.}}$$

LIV. BRANCH-LEVEL EXPLICIT CLOSED FORM FOR $c_{\parallel}, c_E, c_O$

LV. STARTING POINT

At the current refined locked + Class II stage, the pre-Schur intervalley tensor is

$$\mathbf{C}_{i,r}^a(G_*; 0) = \alpha_X \mathbf{J}_{i,r}^a(G_*; 0) + \beta_X \mathbf{K}_{i,r}^a(G_*; 0). \quad (164)$$

However, the actual shell witness coefficients are not extracted directly from (164). They are extracted from the Schur-complement correction

$$\Delta_X = -\frac{1}{M_X^2}(\alpha_X \mathcal{J} + \beta_X \mathcal{K})(\alpha_X \mathcal{J} + \beta_X \mathcal{K})^\dagger. \quad (165)$$

Therefore the physical witness channels decompose into

$$JJ, \quad JK, \quad KK, \quad (166)$$

and their leading shell roles are different.

LVI. SHELL-ADAPTED LEADING ANSATZ

Fix the shell frame

$$(\hat{n}, e_1, e_2), \quad \hat{n} = \frac{G_*}{|G_*|},$$

and the tangent Pauli basis

$$(\Sigma_1, \Sigma_2, \Sigma_3).$$

The current explicit leading shell-adapted branch ansatz is:

$$\boxed{\mathbf{J}_{i,r}^a(G_*; 0) = \hat{n}_i j^a (\Sigma_3)_r{}^a + \mathbf{J}_{i,r}^{\text{null}a}}, \quad (167)$$

where the null part is orthogonal to the three shell witness projectors.

Likewise,

$$\boxed{\mathbf{K}_{i,r}^a(G_*; 0) = \hat{n}_i k^a \left[\Lambda_E^{(0)}(\Sigma_1)_r{}^a + \Lambda_O^{(0)}(\Sigma_2)_r{}^a \right] + \mathbf{K}_{i,r}^{\text{null}a}}. \quad (168)$$

Thus, at leading shell level, J is purely longitudinal and K is purely transverse.

LVII. PROJECTOR DEFINITIONS

Define the exact shell-adapted witness projectors

$$P^{(\parallel)}{}^{i,r}{}_a = \mathcal{N}_{\parallel} \hat{n}^i (\Sigma_3)_r{}^a, \quad (169)$$

$$P^{(E)}{}^{i,r}{}_a = \mathcal{N}_E \frac{1}{2} \left[e_1^i (\Sigma_1)_r{}^a + e_2^i (\Sigma_2)_r{}^a \right], \quad (170)$$

$$P^{(O)}{}^{i,r}{}_a = \mathcal{N}_O \frac{1}{2} \left[-e_2^i (\Sigma_1)_r{}^a + e_1^i (\Sigma_2)_r{}^a \right]. \quad (171)$$

At the pre-Schur level these define

$$c_{\parallel}^{\text{pre}}, \quad c_E^{\text{pre}}, \quad c_O^{\text{pre}}.$$

But the physical shell-core coefficients must be computed from Δ_X , not directly from $\mathbf{C}$.

LVIII. LONGITUDINAL COEFFICIENT FROM THE JJ CHANNEL

Because K is transverse-only at leading shell order, the longitudinal coefficient comes from the JJ channel. Substituting (167) into (165), the longitudinal shell scalar is proportional to

$$\hat{n}^i(\Sigma_3)^r{}_a \mathbf{J}_{i,r}^a \hat{n}^j(\Sigma_3)^s{}_b \mathbf{J}_{j,s}^b \sim j^a j^a = j^2.$$

After normalization, we obtain

$$c_{\parallel} = \frac{\alpha_X^2}{M_X^2} j^2. \quad (172)$$

This is exactly the Branch-B longitudinal coefficient. Equivalently,

$$\lambda_c = \frac{\alpha_X^2}{M_X^2} j^2. \quad (173)$$

Hence the reduced longitudinal witness is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2. \quad (174)$$

LIX. TRANSVERSE COEFFICIENTS FROM THE JK CHANNEL

Now consider the leading transverse part. Because pure J is axial-only and pure K is transverse, the first nontrivial transverse witness comes from the mixed JK channel.

Substituting (167) and (168) into the mixed part of (165), one gets a shell contraction proportional to

$$j^a k^a = j \cdot k.$$

The even transverse piece carries $\Lambda_E^{(0)}$, and the odd one carries $\Lambda_O^{(0)}$.

Therefore the exact leading branch-level coefficients are

$$c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad (175)$$

$$c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (176)$$

Hence the reduced transverse witnesses are

$$\alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad (177)$$

$$\beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (178)$$

LX. COMPLETE COEFFICIENT POINT

Collecting the results,

$$c_{\parallel} = \frac{\alpha_X^2}{M_X^2} j^2, \quad c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (179)$$

Therefore

$$\boxed{\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad \alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}}. \quad (180)$$

This is the explicit branch-level closed form of the current refined locked + Class II realization.

LXI. IMMEDIATE NONDEGENERACY TEST

The shell witness nondegeneracy conditions become

$$\boxed{j^2 \neq 0, \quad (j \cdot k) \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0)}. \quad (181)$$

Equivalently,

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)}. \quad (182)$$

LXII. ANISOTROPY RATIO

The leading anisotropy ratio is

$$\rho = \frac{|\lambda_{\parallel}|}{\sqrt{\alpha^2 + \beta^2}} = \frac{1}{2} \frac{\alpha_X}{\beta_X} \frac{I_{\theta'^3}}{I_{\theta}} \frac{j^2}{|j \cdot k|} \frac{1}{\sqrt{\Lambda_E^{(0)2} + \Lambda_O^{(0)2}}}. \quad (183)$$

Using the explicit overlap ratio

$$\frac{I_{\theta'^3}}{I_{\theta}} = \frac{3}{16} \theta_0^2 \kappa^2,$$

one gets

$$\boxed{\rho = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{j^2}{|j \cdot k|} \frac{\theta_0^2 \kappa^2}{\sqrt{\Lambda_E^{(0)2} + \Lambda_O^{(0)2}}}. \quad (184)$$

In particular, ρ is independent of the heavy mediator mass M_X .

LXIII. HONEST STATUS

The formulas (179)–(184) are the strongest current explicit leading closed form of the refined locked + Class II branch.

However, they should still be stated honestly:

1. they are branch-level explicit closure, not yet the unconditional final closure of the full microscopic TECT PDE;
2. the exact projector algebra is closed, but the full microscopic transverse frontier is still the derivative-resolved shell audit of the JK/KK sector;
3. the scalar Bragg-mass obstruction M_0 remains logically separate from the witness coefficient point $(\lambda_{\parallel}, \alpha, \beta)$.

LXIV. FINAL SUMMARY

At the current refined locked + Class II stage, the actual shell witness coefficients are extracted from the Schur-complement block

$$\Delta_X = -\frac{1}{M_X^2}(\alpha_X \mathcal{J} + \beta_X \mathcal{K})(\alpha_X \mathcal{J} + \beta_X \mathcal{K})^\dagger,$$

not directly from the pre-Schur tensor

$$\mathbb{C}^{\text{valley}} = \alpha_X \mathbf{J} + \beta_X \mathbf{K}.$$

Using the shell-adapted leading ansatz

$$\mathbf{J}_{i,r}^a(G_*; 0) = \hat{n}_i j^a (\Sigma_3)_r{}^a + \mathbf{J}_{i,r}^{\text{null}a},$$

$$\mathbf{K}_{i,r}^a(G_*; 0) = \hat{n}_i k^a \left[\Lambda_E^{(0)}(\Sigma_1)_r{}^a + \Lambda_O^{(0)}(\Sigma_2)_r{}^a \right] + \mathbf{K}_{i,r}^{\text{null}a},$$

the exact branch-level leading coefficient point becomes

$$c_{\parallel} = \frac{\alpha_X^2}{M_X^2} j^2, \quad c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Hence the reduced Dirac witness coefficients are

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad \alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Therefore the explicit branch-level nondegeneracy condition is

$$j^2 \neq 0, \quad (j \cdot k) \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0),$$

equivalently

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

The corresponding anisotropy ratio is

$$\rho = \frac{1}{2} \frac{\alpha_X}{\beta_X} \frac{I_{\theta'^3}}{I_{\theta}} \frac{j^2}{|j \cdot k|} \frac{1}{\sqrt{\Lambda_E^{(0)2} + \Lambda_O^{(0)2}}},$$

and using

$$\frac{I_{\theta'^3}}{I_{\theta}} = \frac{3}{16} \theta_0^2 \kappa^2,$$

this becomes

$$\rho = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{j^2}{|j \cdot k|} \frac{\theta_0^2 \kappa^2}{\sqrt{\Lambda_E^{(0)2} + \Lambda_O^{(0)2}}}.$$

Honest status:

$$\text{the explicit branch-level coefficient point is now closed,}$$

but

$$\text{the unconditional full microscopic transverse closure still requires the derivative-resolved shell audit,}$$

and

$$\text{the scalar Bragg-mass obstruction } M_0 \text{ remains a separate issue.}$$

LXV. FINAL BLOCH CALIBRATION: EXACT SEPARATION OF THE TRANSPORT COEFFICIENT POINT AND THE BLOCH OVERLAP LAYER

LXVI. GOAL

At the present stage, the transport-side coefficient point of the current refined locked + Class II branch is already explicit. However, the actual microscopic shell coefficients are not given directly by this transport-side point. They still require the final Bloch-frame overlap calibration.

Thus the final microscopic task is to separate exactly:

transport-side coefficients (g_c, g_E, g_O)

from

Bloch overlap constants (C_c, C_E, C_O) ,

and then reconstruct the physical reduced coefficients

$$(\lambda_{\parallel}, \alpha, \beta).$$

LXVII. TRANSPORT/BLOCH EXACT NORMALIZATION LAYER

The exact overlap-normalization dictionary is

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (185)$$

Equivalently,

$$\boxed{g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.} \quad (186)$$

This layer is exact. What remains open is only the explicit evaluation of the overlap constants

$$C_c, \quad C_E, \quad C_O.$$

LXVIII. CURRENT BRANCH-LEVEL TRANSPORT COEFFICIENT POINT

Inside the current refined locked + Class II working branch, the leading explicit coefficient point is already fixed in branch-level closed form:

$$\boxed{g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.} \quad (187)$$

Therefore the physical reduced coefficients are

$$\boxed{\lambda_{\parallel} = C_c \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2,} \quad (188)$$

$$\boxed{\alpha = C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)},} \quad (189)$$

$$\boxed{\beta = C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.} \quad (190)$$

Thus the previous branch-level formulas are not replaced; they are completed by the final Bloch overlap layer.

LXIX. PROJECTED SHELL MATRIX AUDIT

Let

$$\Pi_D$$

be the projector onto the shell-relevant doubled low-mode block

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

Let the linear-in- p projected Bloch operator be

$$\mathcal{L}_D^{(1)}(p) = \Pi_D \mathcal{L}_{\text{mic}}(p) \Pi_D \Big|_{\text{linear in } p}. \quad (191)$$

Choose low-mode bases $\{u_a^+\}$, $\{u_b^-\}$ at the opposite shell pair $\pm G_*$, and define the projected shell matrices

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle, \quad a, b = 1, 2. \quad (192)$$

Then the first-order shell symbol is

$$A_1(p) = p_\mu M^\mu. \quad (193)$$

LXX. SHELL-ADAPTED PAULI DECOMPOSITION

Fix the shell direction

$$\hat{n} = \frac{G_*}{|G_*|},$$

and an orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Define

$$M^\parallel := \hat{n}_\mu M^\mu, \quad M^1 := e_{1\mu} M^\mu, \quad M^2 := e_{2\mu} M^\mu.$$

Under the locked Pauli selection rule, the shell matrices take the form

$$\boxed{M^\parallel = \lambda_\parallel \sigma_3, \quad M^1 = \alpha \sigma_1 + \beta \sigma_2, \quad M^2 = \alpha \sigma_2 - \beta \sigma_1.} \quad (194)$$

Hence the reduced coefficients are extracted exactly by

$$\boxed{\lambda_\parallel = \frac{1}{2} \text{Tr}(\sigma_3 M^\parallel),} \quad (195)$$

$$\boxed{\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M^1 + \sigma_2 M^2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M^1 - \sigma_1 M^2).} \quad (196)$$

Equivalently, the explicit nondegeneracy test is

$$\boxed{\text{Tr}(\sigma_3 M^\parallel) \neq 0, \quad (\text{Tr}(\sigma_1 M^1), \text{Tr}(\sigma_2 M^1)) \neq (0, 0).} \quad (197)$$

LXXI. DEFINITION OF THE CALIBRATION OVERLAPS

The overlap constants are defined by comparing the projected shell matrices with the transport-side coefficient point.

Write the three unit channels:

$$(c\text{-unit}), \quad (E\text{-unit}), \quad (O\text{-unit}),$$

meaning:

1. $g_c = 1, g_E = g_O = 0;$
2. $g_E = 1, g_c = g_O = 0;$
3. $g_O = 1, g_c = g_E = 0.$

Then the Bloch overlaps are exactly the shell traces extracted from these three unit-channel calibration runs:

$$C_c = \frac{1}{2} \text{Tr}(\sigma_3 M^\parallel) \Big|_{g_c=1, g_E=g_O=0}, \quad (198)$$

$$C_E = \frac{1}{2} \text{Tr}(\sigma_1 M^1) \Big|_{g_E=1, g_c=g_O=0}, \quad (199)$$

$$C_O = \frac{1}{2} \text{Tr}(\sigma_2 M^1) \Big|_{g_O=1, g_c=g_E=0}. \quad (200)$$

Once these are known, the full coefficients follow from (185).

LXXII. FINAL MICROSCOPIC DECISION THEOREM FOR THE CURRENT BRANCH

The current working ansatz survives the final microscopic audit if and only if

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (201)$$

If (201) holds, then the current branch is dynamically sufficient to realize the required shell Dirac carrier. If instead one finds

$$C_c = 0 \quad \text{or} \quad (C_E, C_O) = (0, 0), \quad (202)$$

then this proves that the present microscopic action is structurally obstructed from being the final complete microscopic TECT PDE.

LXXIII. ANISOTROPY RATIO AFTER CALIBRATION

Before calibration, the branch-level anisotropy ratio is

$$\rho_{\text{branch}} = \frac{|g_c|}{\sqrt{g_E^2 + g_O^2}}.$$

After Bloch calibration, the physical anisotropy ratio becomes

$$\rho_{\text{phys}} = \frac{|C_c|}{\sqrt{C_E^2 g_E^2 + C_O^2 g_O^2}} |g_c|. \quad (203)$$

Equivalently, substituting (187),

$$\rho_{\text{phys}} = \frac{|C_c| \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2}{\sqrt{\left(C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)} \right)^2 + \left(C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} \right)^2}}. \quad (204)$$

Thus the transport-side ratio and the physical Bloch-calibrated ratio differ unless the shell basis is canonically normalized so that

$$C_c = C_E = C_O = 1.$$

LXXIV. HONEST STATUS

At this point the logic is sharply separated:

1. the reduced Dirac/Weyl architecture is already algebraically closed;
2. the current refined locked + Class II branch gives an explicit transport-side coefficient point (g_c, g_E, g_O) ;
3. the final microscopic arbiter is the Bloch-frame calibration (C_c, C_E, C_O) ;
4. failure of the calibration is not a minor defect but a structural obstruction of the current working microscopic ansatz.

LXXV. FINAL SUMMARY

The current refined locked + Class II branch provides an explicit transport-side coefficient point

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

However, the actual physical reduced coefficients are not these raw transport numbers. They are obtained only after the final Bloch-frame overlap calibration:

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

The Bloch overlaps are extracted from the projected 2×2 shell matrices

$$(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle, \quad A_1(p) = p_{\mu} M^{\mu}.$$

With

$$M^{\parallel} = \hat{n}_{\mu} M^{\mu}, \quad M^1 = e_{1\mu} M^{\mu}, \quad M^2 = e_{2\mu} M^{\mu},$$

the exact Pauli-trace extraction is

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}),$$

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M^1 + \sigma_2 M^2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M^1 - \sigma_1 M^2).$$

Equivalently, the overlap constants are fixed by the three unit calibration runs:

$$C_c = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}) \Big|_{g_c=1}, \quad C_E = \frac{1}{2} \text{Tr}(\sigma_1 M^1) \Big|_{g_E=1}, \quad C_O = \frac{1}{2} \text{Tr}(\sigma_2 M^1) \Big|_{g_O=1}.$$

Therefore the final microscopic viability condition for the current working ansatz is

$$\boxed{C_c \neq 0, \quad (C_E, C_O) \neq (0, 0).}$$

If this condition holds, then the current branch is dynamically sufficient and the Dirac carrier is microscopically realized. If instead one finds

$$C_c = 0 \quad \text{or} \quad (C_E, C_O) = (0, 0),$$

this rigorously proves that the present

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi, P] + \beta_X X_i^a K_i^a[P, \Psi]$$

is structurally incomplete as the final full microscopic TECT PDE.

Thus the strongest honest current status is:

$$\boxed{\text{the reduced Dirac architecture is closed,}}$$

$$\boxed{\text{the current branch-level transport coefficient point is explicit,}}$$

but

$$\boxed{\text{the final arbiter is the explicit Bloch calibration of } (C_c, C_E, C_O).}$$

LXXVI. EXACT UNIT-CHANNEL BLOCH CALIBRATION AND THE GENERAL CALIBRATION MATRIX

LXXVII. GOAL

The final microscopic frontier is the explicit evaluation of the Bloch overlap layer. The usual diagonal normalization law

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O$$

is correct only after the shell basis, Pauli basis, and unit-channel conventions are fixed so that the calibration map is diagonal.

To make the calibration protocol completely rigorous, one should first define the general linear calibration map, and only then impose the locked Pauli selection rule as the condition that diagonalizes it.

LXXVIII. TRANSPORT-SIDE COEFFICIENT VECTOR

Define the transport-side coefficient vector

$$g := \begin{pmatrix} g_c \\ g_E \\ g_O \end{pmatrix}. \tag{205}$$

For the current refined locked + Class II branch, the explicit branch-level leading point is

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \tag{206}$$

LXXIX. BLOCH-SIDE EXTRACTED COEFFICIENT VECTOR

Let

$$\Pi_D$$

be the projector onto the shell-relevant doubled low-mode block, and let

$$\{\Psi_a^+\}_{a=1}^2, \quad \{\Psi_b^-\}_{b=1}^2$$

be the orthonormal opposite-valley low-mode bases.

Project the full microscopic first-order Bloch linearization to the doubled shell block:

$$\mathcal{L}_D^{(1),\mu} = \Pi_D \mathcal{L}_{\text{mic}}^\mu \Pi_D \Big|_{\text{linear in } p}. \quad (207)$$

Define the shell matrices

$$(M^\mu)_{ab} = \langle \Psi_a^+, \mathcal{L}_D^{(1),\mu} \Psi_b^- \rangle. \quad (208)$$

With the shell-adapted frame $(\hat{n}, e_1, e_2)$, define

$$M^\parallel = \hat{n}_\mu M^\mu, \quad M^1 = e_{1\mu} M^\mu, \quad M^2 = e_{2\mu} M^\mu. \quad (209)$$

The Bloch-side extracted coefficient vector is

$$y := \begin{pmatrix} \lambda_\parallel \\ \alpha \\ \beta \end{pmatrix}, \quad (210)$$

with

$$\lambda_\parallel = \frac{1}{2} \text{Tr}(\sigma_3 M^\parallel), \quad (211)$$

$$\alpha = \frac{1}{4} \text{Tr}(\sigma_1 M^1 + \sigma_2 M^2), \quad \beta = \frac{1}{4} \text{Tr}(\sigma_2 M^1 - \sigma_1 M^2). \quad (212)$$

LXXX. GENERAL CALIBRATION MATRIX

Because the projected trace extraction is linear in the microscopic first-order operator, there exists a real 3×3 calibration matrix

$$\boxed{y = \mathbb{C} g, \quad \mathbb{C} \in M_{3 \times 3}(\mathbb{R})}. \quad (213)$$

Write

$$\mathbb{C} = \begin{pmatrix} C_{\parallel c} & C_{\parallel E} & C_{\parallel O} \\ C_{Ec} & C_{EE} & C_{EO} \\ C_{Oc} & C_{OE} & C_{OO} \end{pmatrix}. \quad (214)$$

Then

$$\lambda_\parallel = C_{\parallel c} g_c + C_{\parallel E} g_E + C_{\parallel O} g_O, \quad (215)$$

$$\alpha = C_{Ec} g_c + C_{EE} g_E + C_{EO} g_O, \quad (216)$$

$$\beta = C_{Oc} g_c + C_{OE} g_E + C_{OO} g_O. \quad (217)$$

This is the most rigorous general form of the Bloch calibration layer.

LXXXI. UNIT-CHANNEL CALIBRATION RUNS

To compute $\mathbb{C}$, one performs the three unit-channel calibration runs:

a. (c-unit) Set

$$g^{(c)} = (1, 0, 0)^T.$$

Compute the projected shell matrices

$$M_{(c)}^{\parallel}, \quad M_{(c)}^1, \quad M_{(c)}^2,$$

and define

$$\mathbb{C}_{\bullet c} = \begin{pmatrix} \frac{1}{2} \text{Tr}(\sigma_3 M_{(c)}^{\parallel}) \\ \frac{1}{4} \text{Tr}(\sigma_1 M_{(c)}^1 + \sigma_2 M_{(c)}^2) \\ \frac{1}{4} \text{Tr}(\sigma_2 M_{(c)}^1 - \sigma_1 M_{(c)}^2) \end{pmatrix}. \quad (218)$$

b. (E-unit) Set

$$g^{(E)} = (0, 1, 0)^T.$$

Compute

$$M_{(E)}^{\parallel}, \quad M_{(E)}^1, \quad M_{(E)}^2,$$

and define

$$\mathbb{C}_{\bullet E} = \begin{pmatrix} \frac{1}{2} \text{Tr}(\sigma_3 M_{(E)}^{\parallel}) \\ \frac{1}{4} \text{Tr}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2) \\ \frac{1}{4} \text{Tr}(\sigma_2 M_{(E)}^1 - \sigma_1 M_{(E)}^2) \end{pmatrix}. \quad (219)$$

c. (O-unit) Set

$$g^{(O)} = (0, 0, 1)^T.$$

Compute

$$M_{(O)}^{\parallel}, \quad M_{(O)}^1, \quad M_{(O)}^2,$$

and define

$$\mathbb{C}_{\bullet O} = \begin{pmatrix} \frac{1}{2} \text{Tr}(\sigma_3 M_{(O)}^{\parallel}) \\ \frac{1}{4} \text{Tr}(\sigma_1 M_{(O)}^1 + \sigma_2 M_{(O)}^2) \\ \frac{1}{4} \text{Tr}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2) \end{pmatrix}. \quad (220)$$

The three columns $\mathbb{C}_{\bullet c}, \mathbb{C}_{\bullet E}, \mathbb{C}_{\bullet O}$ determine the full calibration matrix $\mathbb{C}$.

LXXXII. DIAGONAL LOCKED CALIBRATION

Under the exact locked Pauli selection rule, there is no cross-channel mixing:

$$C_{\parallel E} = C_{\parallel O} = C_{Ec} = C_{EO} = C_{Oc} = C_{OE} = 0. \quad (221)$$

Then $\mathbb{C}$ becomes diagonal:

$$\boxed{\mathbb{C} = \text{diag}(C_c, C_E, C_O),} \quad (222)$$

with

$$C_c = \frac{1}{2} \text{Tr}(\sigma_3 M_{(c)}^\parallel), \quad (223)$$

$$C_E = \frac{1}{4} \text{Tr}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2), \quad (224)$$

$$C_O = \frac{1}{4} \text{Tr}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2). \quad (225)$$

In this case the calibration law reduces exactly to

$$\lambda_\parallel = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O. \quad (226)$$

LXXXIII. DECISION CRITERIA

A. Diagonal case

If the locked Pauli selection rule is exact and $\mathbb{C}$ is diagonal, the branch survives the Bloch calibration iff

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (227)$$

B. General case

If small off-diagonal mixing is present, the correct criterion is not (227) but:

$$\det \mathbb{C} \neq 0 \quad (228)$$

together with the requirement that the image of the transverse transport plane

$$\text{span}\{(0, 1, 0)^T, (0, 0, 1)^T\}$$

under $\mathbb{C}$ is genuinely two-dimensional in the shell-extracted (α, β) sector.

Equivalently, writing

$$\mathbb{C}_\perp = \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix},$$

one needs

$$C_{\parallel c} \neq 0, \quad \det \mathbb{C}_\perp \neq 0. \quad (229)$$

Thus the diagonal criterion is just the symmetry-reduced special case of the general matrix criterion.

LXXXIV. CURRENT BRANCH AFTER CALIBRATION

Substituting the current branch-level transport point (206) into the diagonal calibration law, one gets

$$\lambda_\parallel = C_c \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad (230)$$

$$\alpha = C_E I_\theta \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad (231)$$

$$\beta = C_O I_\theta \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (232)$$

Therefore the final microscopic viability of the current refined locked + Class II branch is decided entirely by the finite projected shell audit of $\mathbb{C}$.

LXXXV. HONEST STATUS

At the current strongest honest stage:

1. the reduced Dirac architecture is algebraically closed;
2. the branch-level transport coefficient point (g_c, g_E, g_O) is explicit;
3. the final remaining microscopic frontier is the explicit evaluation of the Bloch calibration matrix $\mathbb{C}$;
4. if $\mathbb{C}$ fails the required rank/nondegeneracy test, then the present working microscopic ansatz is structurally incomplete.

LXXXVI. FINAL SUMMARY

The final Bloch calibration layer should be written in its most rigorous form as a linear map

$$\begin{pmatrix} \lambda_{\parallel} \\ \alpha \\ \beta \end{pmatrix} = \mathbb{C} \begin{pmatrix} g_c \\ g_E \\ g_O \end{pmatrix}, \quad \mathbb{C} \in M_{3 \times 3}(\mathbb{R}).$$

The matrix $\mathbb{C}$ is determined by the three unit-channel calibration runs:

$$(c\text{-unit}), \quad (E\text{-unit}), \quad (O\text{-unit}),$$

by projecting the first-order Bloch linearization to the doubled shell block and evaluating the Pauli traces of

$$M^{\parallel}, \quad M^1, \quad M^2.$$

If the locked Pauli selection rule is exact, then $\mathbb{C}$ is diagonal:

$$\mathbb{C} = \text{diag}(C_c, C_E, C_O),$$

and the calibration law reduces to

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

In this diagonal case, the final microscopic viability condition is

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

In the more general case with residual cross-channel mixing, the correct criterion is:

$$C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0.$$

For the current refined locked + Class II branch, the transport-side coefficient point is already explicit:

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Therefore, after diagonal calibration,

$$\lambda_{\parallel} = C_c \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2,$$

$$\alpha = C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Thus the strongest honest current status is:

the reduced Dirac architecture is closed,

the branch-level transport coefficient point is explicit,

but

the final microscopic arbiter is the explicit Bloch calibration matrix $\mathbb{C}$.

If that calibration fails the stated nondegeneracy/rank test, then the present working microscopic ansatz is structurally incomplete.

LXXXVII. EXECUTABLE BLOCH CALIBRATION PROTOCOL FOR THE REFINED LOCKED SHELL CARRIER

LXXXVIII. GOAL

The reduced Dirac architecture is already closed. The branch-level transport coefficient point

$$(g_c, g_E, g_O)$$

is already explicit. Hence the only remaining microscopic frontier is the explicit Bloch calibration step.

The goal is to construct an executable finite protocol:

$$H_{\text{eff}}(k) \longrightarrow H_{\text{shell}}(p) \longrightarrow P_D \longrightarrow M^{\mu} \longrightarrow \mathbb{C} \longrightarrow (\lambda_{\parallel}, \alpha, \beta).$$

LXXXIX. STEP 0: FIXED INPUT DATA

Fix once and for all:

1. the selected opposite shell pair $\pm G_*$,
2. the folded shell center

$$k_* = \frac{G_*}{2},$$

3. the shell-adapted orthonormal frame

$$(\hat{n}, e_1, e_2), \quad \hat{n} = \frac{G_*}{|G_*|},$$

4. the doubled low shell block

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

with an orthonormal basis

$$\{\Psi_a^+\}_{a=1}^2, \quad \{\Psi_b^-\}_{b=1}^2.$$

XC. STEP 1: CONSTRUCT THE BLOCH EFFECTIVE OPERATOR

Start from the refined locked + Class II working branch:

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - H_{\Psi X}(k)H_{XX}(k)^{-1}H_{X\Psi}(k), \quad H_{XX}(k) = M_X^2 \mathbf{1}_X. \quad (233)$$

Equivalently,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k)) (\alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k))^\dagger. \quad (234)$$

This is the only microscopic input required for the calibration run.

XCI. STEP 2: SHELL PROJECTION

Define the shell-projected operator around the chosen shell pair:

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{eff}}(k_* + p) \Pi_{S_*}. \quad (235)$$

Here Π_{S_*} projects to the shell truncation used for the chosen opposite-valley pair.

XCII. STEP 3: LOW-SLOT SPECTRAL PROJECTOR

At $p = 0$, diagonalize the shell block

$$H_{\text{shell}}(0).$$

Choose the 4-dimensional low carrier

$$\mathcal{H}_D \subset \mathcal{H}_{\text{shell}}$$

and define its spectral projector

$$P_D = \frac{1}{2\pi i} \oint_{\Gamma_D} (z - H_{\text{shell}}(0))^{-1} dz, \quad (236)$$

where Γ_D encloses exactly the desired doubled low slot.

Then define the core shell block

$$H_{\text{core}}(p) = P_D H_{\text{shell}}(p) P_D. \quad (237)$$

XCIII. STEP 4: FIRST-ORDER BLOCH MATRICES

Expand $H_{\text{core}}(p)$ to first order in p :

$$H_{\text{core}}(p) = H_{\text{core}}(0) + p_\mu M^\mu + O(|p|^2). \quad (238)$$

Equivalently,

$$M^\mu = P_D \partial_{k_\mu} H_{\text{eff}}(k) P_D \big|_{k=k_*}. \quad (239)$$

In the chosen opposite-valley basis,

$$(M^\mu)_{ab} = \langle \Psi_a^+, M^\mu \Psi_b^- \rangle. \quad (240)$$

Now define the shell-adapted components

$$M^\parallel := \hat{n}_\mu M^\mu, \quad M^1 := e_{1\mu} M^\mu, \quad M^2 := e_{2\mu} M^\mu. \quad (241)$$

XCIV. STEP 5: DIRECT PAULI-TRACE EXTRACTION

If the locked Pauli selection rule holds exactly, then

$$M^\parallel = \lambda_\parallel \sigma_3, \quad M^1 = \alpha \sigma_1 + \beta \sigma_2, \quad M^2 = \alpha \sigma_2 - \beta \sigma_1. \quad (242)$$

Hence the reduced coefficients are extracted by

$$\boxed{\lambda_\parallel = \frac{1}{2}(\sigma_3 M^\parallel)}, \quad (243)$$

$$\boxed{\alpha = \frac{1}{4}(\sigma_1 M^1 + \sigma_2 M^2), \quad \beta = \frac{1}{4}(\sigma_2 M^1 - \sigma_1 M^2)}. \quad (244)$$

This is the direct shell-side readout.

XCV. STEP 6: GENERAL CALIBRATION MATRIX

To make the protocol fully rigorous, define

$$g = \begin{pmatrix} g_c \\ g_E \\ g_O \end{pmatrix}, \quad y = \begin{pmatrix} \lambda_\parallel \\ \alpha \\ \beta \end{pmatrix}.$$

Then there exists a real 3×3 matrix

$$\boxed{y = \mathbb{C} g.} \quad (245)$$

If the locked Pauli selection rule is exact and there is no cross-channel mixing, then

$$\boxed{\mathbb{C} = \text{diag}(C_c, C_E, C_O),} \quad (246)$$

so that

$$\boxed{\lambda_\parallel = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (247)$$

XCVI. STEP 7: THREE UNIT-CHANNEL CALIBRATION RUNS

To compute $\mathbb{C}$, perform the following three runs.

(c-unit)

Set

$$g^{(c)} = (1, 0, 0)^T.$$

Compute the corresponding first-order Bloch matrices

$$M_{(c)}^\parallel, \quad M_{(c)}^1, \quad M_{(c)}^2.$$

Then the first column of $\mathbb{C}$ is

$$\mathbb{C}_{\bullet c} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(c)}^\parallel) \\ \frac{1}{4}(\sigma_1 M_{(c)}^1 + \sigma_2 M_{(c)}^2) \\ \frac{1}{4}(\sigma_2 M_{(c)}^1 - \sigma_1 M_{(c)}^2) \end{pmatrix}. \quad (248)$$

(E-unit)

Set

$$g^{(E)} = (0, 1, 0)^T.$$

Compute

$$M_{(E)}^{\parallel}, \quad M_{(E)}^1, \quad M_{(E)}^2.$$

Then

$$\mathbb{C}_{\bullet E} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(E)}^{\parallel}) \\ \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2) \\ \frac{1}{4}(\sigma_2 M_{(E)}^1 - \sigma_1 M_{(E)}^2) \end{pmatrix}. \quad (249)$$

(O-unit)

Set

$$g^{(O)} = (0, 0, 1)^T.$$

Compute

$$M_{(O)}^{\parallel}, \quad M_{(O)}^1, \quad M_{(O)}^2.$$

Then

$$\mathbb{C}_{\bullet O} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(O)}^{\parallel}) \\ \frac{1}{4}(\sigma_1 M_{(O)}^1 + \sigma_2 M_{(O)}^2) \\ \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2) \end{pmatrix}. \quad (250)$$

Thus the full calibration matrix is

$$\mathbb{C} = (\mathbb{C}_{\bullet c} \ \mathbb{C}_{\bullet E} \ \mathbb{C}_{\bullet O}).$$

XCVII. STEP 8: DECISION CRITERIA**Diagonal locked case**

If the shell basis is already locked and no cross-channel mixing appears, then

$$\mathbb{C} = \text{diag}(C_c, C_E, C_O).$$

The branch survives iff

$$\boxed{C_c \neq 0, \quad (C_E, C_O) \neq (0, 0)}. \quad (251)$$

General case

If small cross-channel mixing is present, then the correct criterion is

$$\boxed{C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0}. \quad (252)$$

Equivalently, the longitudinal image must be nonzero and the transverse image must have rank two.

XCVIII. STEP 9: INSERT THE CURRENT BRANCH-LEVEL TRANSPORT POINT

For the present refined locked + Class II branch, the transport-side coefficient point is

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (253)$$

Therefore, after diagonal calibration,

$$\lambda_{\parallel} = C_c \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad (254)$$

$$\alpha = C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (255)$$

XCIX. STEP 10: FINAL OUTPUTS

The protocol returns exactly four microscopic verdicts:

1. whether the longitudinal shell overlap survives;
2. whether the transverse shell overlap survives;
3. whether the calibration is diagonal or mildly mixed;
4. whether the current working branch is microscopically viable.

The branch is accepted as the final microscopic carrier only if the calibration test passes. If it fails, the conclusion is structural incompleteness of the present microscopic ansatz, not merely a poor parameter choice.

C. FINAL SUMMARY

The final microscopic problem is no longer conceptual. It is the executable finite Bloch calibration protocol

$$H_{\text{eff}}(k) \longrightarrow H_{\text{shell}}(p) \longrightarrow P_D \longrightarrow M^{\mu} \longrightarrow \mathbb{C} \longrightarrow (\lambda_{\parallel}, \alpha, \beta).$$

The projected shell matrices are

$$M^{\parallel} = \hat{n}_{\mu} M^{\mu}, \quad M^1 = e_{1\mu} M^{\mu}, \quad M^2 = e_{2\mu} M^{\mu}.$$

Their exact Pauli-trace readout is

$$\lambda_{\parallel} = \frac{1}{2} (\sigma_3 M^{\parallel}),$$

$$\alpha = \frac{1}{4} (\sigma_1 M^1 + \sigma_2 M^2), \quad \beta = \frac{1}{4} (\sigma_2 M^1 - \sigma_1 M^2).$$

The most rigorous calibration law is the linear map

$$\begin{pmatrix} \lambda_{\parallel} \\ \alpha \\ \beta \end{pmatrix} = \mathbb{C} \begin{pmatrix} g_c \\ g_E \\ g_O \end{pmatrix}.$$

Under exact locked Pauli selection, this reduces to the diagonal normalization layer

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.}$$

The matrix $\mathbb{C}$ is computed by three unit-channel runs:

$$(c\text{-unit}), \quad (E\text{-unit}), \quad (O\text{-unit}),$$

with columns extracted from the corresponding shell Pauli traces.

For the current refined locked + Class II branch,

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Hence, after diagonal calibration,

$$\boxed{\lambda_{\parallel} = C_c \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2,}$$

$$\boxed{\alpha = C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Therefore the final microscopic viability criterion is:

$$\boxed{\text{diagonal locked case: } C_c \neq 0, \quad (C_E, C_O) \neq (0, 0),}$$

or, more generally,

$$\boxed{\text{mixed case: } C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0.}$$

Thus the strongest honest current status is:

$$\boxed{\text{the reduced Dirac architecture is closed,}}$$

$$\boxed{\text{the branch-level transport coefficient point is explicit,}}$$

but

$$\boxed{\text{the final microscopic arbiter is the explicit unit-channel Bloch calibration.}}$$

CI. EXPLICIT UNIT-CHANNEL IMPLEMENTATION FOR THE FINAL BLOCH CALIBRATION

CII. GOAL

The final Bloch calibration requires not only the abstract statement

$$g^{(c)} = (1, 0, 0), \quad g^{(E)} = (0, 1, 0), \quad g^{(O)} = (0, 0, 1),$$

but an explicit microscopic realization of these three unit channels as concrete first-order transport inputs.

The correct implementation is to start from the already constructed primitive shell seeds and normalize them so that their reduced witness output is exactly unit in the desired channel.

CIII. PRIMITIVE SEED OPERATORS

The explicit primitive longitudinal seed is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (256)$$

Its reduced witness is

$$\lambda_{\parallel}[V_{(\parallel)}] = \frac{1}{2} I_{\theta'^3}, \quad \alpha[V_{(\parallel)}] = 0, \quad \beta[V_{(\parallel)}] = 0. \quad (257)$$

The explicit transverse even seed is

$$V_{(\perp, E)}^\mu(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a, \quad (258)$$

with

$$\lambda_{\parallel}[V_{(\perp, E)}] = 0, \quad \alpha[V_{(\perp, E)}] = I_\theta, \quad \beta[V_{(\perp, E)}] = 0. \quad (259)$$

The explicit transverse odd seed is

$$V_{(\perp, O)}^\mu(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a, \quad (260)$$

with

$$\lambda_{\parallel}[V_{(\perp, O)}] = 0, \quad \alpha[V_{(\perp, O)}] = 0, \quad \beta[V_{(\perp, O)}] = I_\theta. \quad (261)$$

CIV. NORMALIZED UNIT-CHANNEL INPUTS

We now normalize these seeds so that the transport-side output is exactly one in the chosen channel.

(c-unit)

Define

$$V_{(c)}^\mu(\xi) := \frac{2}{I_{\theta'^3}} V_{(\parallel)}^\mu(\xi) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} n^\mu \sigma_3. \quad (262)$$

Then

$$\lambda_{\parallel}[V_{(c)}] = 1, \quad \alpha[V_{(c)}] = 0, \quad \beta[V_{(c)}] = 0. \quad (263)$$

(E-unit)

Define

$$V_{(E)}^\mu(\xi) := \frac{1}{I_\theta} V_{(\perp, E)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} E^\mu{}_a \sigma_a. \quad (264)$$

Then

$$\lambda_{\parallel}[V_{(E)}] = 0, \quad \alpha[V_{(E)}] = 1, \quad \beta[V_{(E)}] = 0. \quad (265)$$

(O-unit)

Define

$$V_{(O)}^\mu(\xi) := \frac{1}{I_\theta} V_{(\perp, O)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} J^\mu{}_a \sigma_a. \quad (266)$$

Then

$$\lambda_\parallel[V_{(O)}] = 0, \quad \alpha[V_{(O)}] = 0, \quad \beta[V_{(O)}] = 1. \quad (267)$$

Thus these three operators realize the canonical transport-side unit vectors

$$g^{(c)} = (1, 0, 0)^T, \quad g^{(E)} = (0, 1, 0)^T, \quad g^{(O)} = (0, 0, 1)^T.$$

CV. PROJECTED BLOCH IMPLEMENTATION

Let

$$\{\Psi_a^+\}_{a=1}^2, \quad \{\Psi_b^-\}_{b=1}^2$$

be the orthonormal opposite-valley Bloch basis spanning the shell-relevant doubled low block $\mathcal{H}_D$.

For any one of the three unit inputs $V_{(u)}^\mu$, with $u \in \{c, E, O\}$, define the projected first-order Bloch matrices

$$(M_{(u)}^\mu)_{ab} = \langle \Psi_a^+, V_{(u)}^\mu \Psi_b^- \rangle. \quad (268)$$

Using the shell-adapted frame $(\hat{n}, e_1, e_2)$, define

$$M_{(u)}^\parallel = \hat{n}_\mu M_{(u)}^\mu, \quad M_{(u)}^1 = e_{1\mu} M_{(u)}^\mu, \quad M_{(u)}^2 = e_{2\mu} M_{(u)}^\mu. \quad (269)$$

CVI. COLUMN-BY-COLUMN CALIBRATION MATRIX EXTRACTION

The full calibration matrix

$$\mathbb{C} = (\mathbb{C}_{\bullet c} \ \mathbb{C}_{\bullet E} \ \mathbb{C}_{\bullet O})$$

is obtained by the three columns:

(c-column)

$$\mathbb{C}_{\bullet c} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(c)}^\parallel) \\ \frac{1}{4}(\sigma_1 M_{(c)}^1 + \sigma_2 M_{(c)}^2) \\ \frac{1}{4}(\sigma_2 M_{(c)}^1 - \sigma_1 M_{(c)}^2) \end{pmatrix}. \quad (270)$$

(E-column)

$$\mathbb{C}_{\bullet E} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(E)}^\parallel) \\ \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2) \\ \frac{1}{4}(\sigma_2 M_{(E)}^1 - \sigma_1 M_{(E)}^2) \end{pmatrix}. \quad (271)$$

(O-column)

$$\mathbb{C}_{\bullet O} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(O)}^{\parallel}) \\ \frac{1}{4}(\sigma_1 M_{(O)}^1 + \sigma_2 M_{(O)}^2) \\ \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2) \end{pmatrix}. \quad (272)$$

This is the exact executable finite implementation of the Bloch calibration layer.

CVII. DIAGONAL LOCKED CASE

If the locked Pauli selection rule holds exactly, then the off-channel entries vanish and

$$\mathbb{C} = \text{diag}(C_c, C_E, C_O),$$

with

$$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}), \quad (273)$$

$$C_E = \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2), \quad (274)$$

$$C_O = \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2). \quad (275)$$

Therefore the physical coefficients are

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O. \quad (276)$$

CVIII. ACCEPTANCE CRITERIA

Diagonal locked case

The branch passes the final Bloch calibration iff

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (277)$$

General mixed case

If the calibration is not perfectly diagonal, one must use the full matrix criterion:

$$C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0. \quad (278)$$

CIX. RELATION TO THE CURRENT BRANCH-LEVEL COEFFICIENT POINT

For the present refined locked + Class II branch,

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Hence, after diagonal calibration,

$$\lambda_{\parallel} = C_c \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad (279)$$

$$\alpha = C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (280)$$

CX. HONEST STATUS

At this point, the remaining frontier is no longer the definition of the seed channels or the calibration algorithm. Both are explicit.

What remains is only the actual evaluation of the finite overlap matrices

$$M_{(c)}^{\mu}, \quad M_{(E)}^{\mu}, \quad M_{(O)}^{\mu}$$

in the true refined locked microscopic Bloch basis.

If this final projected overlap audit fails, the conclusion is not a minor normalization problem. It means that the current working microscopic ansatz is structurally incomplete as the final full TECT PDE.

CXI. FINAL SUMMARY

The correct microscopic implementation of the three unit-channel calibration runs is obtained by normalizing the already constructed primitive shell seeds.

The primitive seeds are:

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,$$

$$V_{(\perp, E)}^{\mu}(\xi) = \theta'(\xi) E^{\mu}_a \sigma_a, \quad V_{(\perp, O)}^{\mu}(\xi) = \theta'(\xi) J^{\mu}_a \sigma_a.$$

After exact unit normalization, the calibration inputs are

$$V_{(c)}^{\mu}(\xi) = \frac{2}{I_{\theta'^3}} V_{(\parallel)}^{\mu}(\xi) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} n^{\mu} \sigma_3,$$

$$V_{(E)}^{\mu}(\xi) = \frac{1}{I_{\theta}} V_{(\perp, E)}^{\mu}(\xi) = \frac{\theta'(\xi)}{I_{\theta}} E^{\mu}_a \sigma_a,$$

$$V_{(O)}^{\mu}(\xi) = \frac{1}{I_{\theta}} V_{(\perp, O)}^{\mu}(\xi) = \frac{\theta'(\xi)}{I_{\theta}} J^{\mu}_a \sigma_a.$$

These realize the canonical unit transport outputs

$$g^{(c)} = (1, 0, 0)^T, \quad g^{(E)} = (0, 1, 0)^T, \quad g^{(O)} = (0, 0, 1)^T.$$

For each $u \in \{c, E, O\}$, define the projected Bloch matrices

$$(M_{(u)}^\mu)_{ab} = \langle \Psi_a^+, V_{(u)}^\mu \Psi_b^- \rangle,$$

and their shell-adapted components

$$M_{(u)}^\parallel = \hat{n}_\mu M_{(u)}^\mu, \quad M_{(u)}^1 = e_{1\mu} M_{(u)}^\mu, \quad M_{(u)}^2 = e_{2\mu} M_{(u)}^\mu.$$

Then the calibration matrix columns are

$$\mathbb{C}_{\bullet u} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(u)}^\parallel) \\ \frac{1}{4}(\sigma_1 M_{(u)}^1 + \sigma_2 M_{(u)}^2) \\ \frac{1}{4}(\sigma_2 M_{(u)}^1 - \sigma_1 M_{(u)}^2) \end{pmatrix}, \quad u \in \{c, E, O\}.$$

Under exact locked Pauli selection, this reduces to the diagonal normalization law

$$\lambda_\parallel = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

The diagonal acceptance criterion is

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

If residual channel mixing is present, one must instead use the full matrix criterion

$$C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0.$$

Thus the seed operators and the calibration protocol are now explicit. The only remaining frontier is the actual finite overlap evaluation in the true refined locked microscopic Bloch basis.

CXII. EXACT FOLDED-BRAGG OVERLAP FORMULAS FOR THE UNIT-CHANNEL CALIBRATION

CXIII. FIXED CONVENTIONS AND MANDATORY GATE

Before any calibration, one must first fix:

1. the low-valley Pauli basis $\{\sigma_1, \sigma_2, \sigma_3\}$,
2. the shell-adapted frame $(\hat{n}, e_1, e_2)$,
3. the overlap normalization convention defining C_c, C_E, C_O ,
4. the sign convention for the lowest signed channel.

Moreover, the remote-gap condition must hold so that the low slot is spectrally isolated and the projector P_D is well-defined:

$$\Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.$$

CXIV. FOLDED BRAGG BASIS

Fix the selected opposite shell pair $\pm G_*$ and adopt the folded Bragg convention

$$\boxed{k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.} \quad (281)$$

Choose an orthonormal opposite-valley low-mode basis

$$\{u_a^+\}_{a=1}, \quad \{u_b^-\}_{b=1},$$

and define the Bloch states

$$\boxed{\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a^+(x), \quad \Psi_{-,b}(x) = e^{-ik_* \cdot x} u_b^-(x).} \quad (282)$$

Then the intervalley first-order Bloch matrices are

$$(M^\mu)_{ab} = \langle \Psi_{+,a}, V^\mu \Psi_{-,b} \rangle. \quad (283)$$

Substituting (282) gives

$$\boxed{(M^\mu)_{ab} = \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} \overline{u_a^+(x)} V^\mu(x) u_b^-(x).} \quad (284)$$

Thus every intervalley overlap is exactly the $Q = G_*$ Fourier component of the operator weighted by the envelope overlap.

CXV. NORMALIZED UNIT-CHANNEL INPUTS

The primitive seeds are

$$V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3, \quad V_{(\perp,E)}^\mu(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a, \quad V_{(\perp,O)}^\mu(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a.$$

Their correctly normalized unit-channel versions are

$$\boxed{V_{(c)}^\mu(\xi) = \frac{2}{I_{\theta'^3}} V_{(\parallel)}^\mu(\xi) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} n^\mu \sigma_3,} \quad (285)$$

$$\boxed{V_{(E)}^\mu(\xi) = \frac{1}{I_\theta} V_{(\perp,E)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} E^\mu{}_a \sigma_a,} \quad (286)$$

$$\boxed{V_{(O)}^\mu(\xi) = \frac{1}{I_\theta} V_{(\perp,O)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} J^\mu{}_a \sigma_a.} \quad (287)$$

These correspond exactly to the transport-side unit vectors

$$g^{(c)} = (1, 0, 0)^T, \quad g^{(E)} = (0, 1, 0)^T, \quad g^{(O)} = (0, 0, 1)^T.$$

CXVI. CELL-INTEGRAL OVERLAP MATRICES

For each unit input $u \in \{c, E, O\}$, define

$$\boxed{(M_{(u)}^\mu)_{ab} = \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} \overline{u_a^+(x)} V_{(u)}^\mu(x) u_b^-(x).} \quad (288)$$

Now define the shell-adapted components

$$M_{(u)}^\parallel = \hat{n}_\mu M_{(u)}^\mu, \quad M_{(u)}^1 = e_{1\mu} M_{(u)}^\mu, \quad M_{(u)}^2 = e_{2\mu} M_{(u)}^\mu. \quad (289)$$

These are the exact finite objects from which the calibration overlaps are extracted.

CXVII. EXACT INTEGRAL FORMULAS FOR THE CALIBRATION COLUMNS

The full 3×3 calibration matrix

$$\mathbb{C} = (\mathbb{C}_{\bullet c} \ \mathbb{C}_{\bullet E} \ \mathbb{C}_{\bullet O})$$

is computed column-by-column by Pauli traces.

(c-column)

$$\mathbb{C}_{\bullet c} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}) \\ \frac{1}{4}(\sigma_1 M_{(c)}^1 + \sigma_2 M_{(c)}^2) \\ \frac{1}{4}(\sigma_2 M_{(c)}^1 - \sigma_1 M_{(c)}^2) \end{pmatrix}. \quad (290)$$

(E-column)

$$\mathbb{C}_{\bullet E} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(E)}^{\parallel}) \\ \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2) \\ \frac{1}{4}(\sigma_2 M_{(E)}^1 - \sigma_1 M_{(E)}^2) \end{pmatrix}. \quad (291)$$

(O-column)

$$\mathbb{C}_{\bullet O} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(O)}^{\parallel}) \\ \frac{1}{4}(\sigma_1 M_{(O)}^1 + \sigma_2 M_{(O)}^2) \\ \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2) \end{pmatrix}. \quad (292)$$

This is the fully executable finite calibration layer.

CXVIII. DIAGONAL LOCKED CASE

If the locked Pauli selection rule holds exactly, then off-channel mixing vanishes and

$$\mathbb{C} = \text{diag}(C_c, C_E, C_O).$$

In that case,

$$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}) = \frac{1}{2I_{\theta'^3}} \left[\sigma_3 \int_{\text{cell}} d^3x \ e^{-iG_* \cdot x} U_+^\dagger(x) \theta'(\xi)^3 \sigma_3 U_-(x) \right], \quad (293)$$

where

$$U_{\pm}(x) := (u_1^{\pm}(x), u_2^{\pm}(x)).$$

Likewise,

$$C_E = \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2) = \frac{1}{4I_{\theta}} \left[\sigma_1 \int_{\text{cell}} d^3x e^{-iG_{*} \cdot x} U_{+}^{\dagger} \theta'(\xi) E^1{}_a \sigma_a U_{-} + \sigma_2 \int_{\text{cell}} d^3x e^{-iG_{*} \cdot x} U_{+}^{\dagger} \theta'(\xi) E^2{}_a \sigma_a U_{-} \right], \quad (294)$$

and

$$C_O = \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2) = \frac{1}{4I_{\theta}} \left[\sigma_2 \int_{\text{cell}} d^3x e^{-iG_{*} \cdot x} U_{+}^{\dagger} \theta'(\xi) J^1{}_a \sigma_a U_{-} - \sigma_1 \int_{\text{cell}} d^3x e^{-iG_{*} \cdot x} U_{+}^{\dagger} \theta'(\xi) J^2{}_a \sigma_a U_{-} \right]. \quad (295)$$

These are the exact folded-Bragg overlap formulas for the diagonal calibration constants.

CXIX. REDUCTION TO ONE-DIMENSIONAL RADIAL OVERLAPS

If the opposite-valley Bloch modes separate as

$$u_a^{\pm}(x) = f_{\pm}(\xi) \chi_a(x_{\perp}, \xi),$$

then all three calibration constants reduce to weighted radial overlaps of the form

$$C_c = \frac{1}{I_{\theta^3}} \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi), \quad (296)$$

$$C_E = \frac{1}{I_{\theta}} \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) \mathfrak{s}_E(\xi), \quad C_O = \frac{1}{I_{\theta}} \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) \mathfrak{s}_O(\xi), \quad (297)$$

where the profile functions $\mathfrak{s}_c, \mathfrak{s}_E, \mathfrak{s}_O$ are the corresponding internal/tangential Bloch overlaps.

Thus C_c, C_E, C_O measure the mismatch between the canonical seed normalization and the actual Bloch low-mode frame.

CXX. FINAL ACCEPTANCE CRITERION

In the diagonal locked case, the final calibration test is

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (298)$$

In the general mixed case, one must use the full matrix criterion:

$$C_{\parallel c} \neq 0, \quad \det \begin{pmatrix} C_{EE} & C_{EO} \\ C_{OE} & C_{OO} \end{pmatrix} \neq 0. \quad (299)$$

CXXI. CONNECTION TO THE CURRENT BRANCH-LEVEL COEFFICIENT POINT

For the present refined locked + Class II branch, the transport-side point is already explicit:

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Hence, after diagonal calibration,

$$\lambda_{\parallel} = C_c \frac{1}{2} I_{\theta^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad (300)$$

$$\alpha = C_E I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = C_O I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (301)$$

CXXII. HONEST STATUS

At this point the seed operators, the unit normalization, the folded-Bragg matrix elements, and the calibration traces are all explicit.

What remains is only the actual finite evaluation of the overlap integrals

$$M_{(c)}^\mu, \quad M_{(E)}^\mu, \quad M_{(O)}^\mu$$

in the true refined locked microscopic Bloch basis.

If this final overlap audit fails, the conclusion is structural incompleteness of the current working microscopic ansatz, not merely an unfortunate normalization choice.

CXXIII. FINAL SUMMARY

Using the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad \Psi_{+,a}(x) = e^{ik_* \cdot x} u_a^+(x), \quad \Psi_{-,b}(x) = e^{-ik_* \cdot x} u_b^-(x),$$

the intervalley projected Bloch matrix element of any first-order transport input V^μ is exactly

$$(M^\mu)_{ab} = \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} \overline{u_a^+(x)} V^\mu(x) u_b^-(x).$$

The correctly normalized unit-channel inputs are

$$V_{(c)}^\mu(\xi) = \frac{\theta'(\xi)^3}{I_{\theta^3}} n^\mu \sigma_3,$$

$$V_{(E)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} E^\mu{}_a \sigma_a, \quad V_{(O)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} J^\mu{}_a \sigma_a.$$

For each $u \in \{c, E, O\}$, define

$$M_{(u)}^\parallel = \hat{n}_\mu M_{(u)}^\mu, \quad M_{(u)}^1 = e_{1\mu} M_{(u)}^\mu, \quad M_{(u)}^2 = e_{2\mu} M_{(u)}^\mu.$$

Then the calibration matrix columns are exactly

$$\mathbf{C}_{\bullet u} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(u)}^\parallel) \\ \frac{1}{4}(\sigma_1 M_{(u)}^1 + \sigma_2 M_{(u)}^2) \\ \frac{1}{4}(\sigma_2 M_{(u)}^1 - \sigma_1 M_{(u)}^2) \end{pmatrix}, \quad u \in \{c, E, O\}.$$

If the locked Pauli selection rule is exact, this reduces to the diagonal overlap-normalization law

$$\lambda_\parallel = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

The exact folded-Bragg overlap formulas are then

$$C_c = \frac{1}{2I_{\theta^3}} \left[\sigma_3 \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} U_+^\dagger(x) \theta'(\xi)^3 \sigma_3 U_-(x) \right],$$

$$C_E = \frac{1}{4I_\theta} \left[\sigma_1 \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} U_+^\dagger \theta'(\xi) E^1{}_a \sigma_a U_- + \sigma_2 \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} U_+^\dagger \theta'(\xi) E^2{}_a \sigma_a U_- \right],$$

$$C_O = \frac{1}{4I_\theta} \left[\sigma_2 \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_+^\dagger \theta'(\xi) J_a^1 \sigma_a U_- - \sigma_1 \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_+^\dagger \theta'(\xi) J_a^2 \sigma_a U_- \right].$$

Thus C_c, C_E, C_O are not abstract constants; they are exact normalized $Q = G_*$ Bloch-overlap integrals in the true low-mode frame.

The diagonal acceptance criterion is

$$C_c \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

More generally, in the mixed case one must use the full calibration matrix rank condition.

Therefore the strongest honest current status is:

$$\text{the seed channels and the calibration protocol are explicit,}$$

but

$$\text{the final microscopic frontier is still the actual finite evaluation of these folded-Bragg overlap integrals.}$$

CXXIV. FACTORIZATION OF THE FOLDED-BRAGG CALIBRATION INTEGRALS

CXXV. EXACT STARTING POINT

Fix the folded Bragg basis

$$\Psi_{+,a}(x) = e^{ik_* \cdot x} u_a^+(x), \quad \Psi_{-,b}(x) = e^{-ik_* \cdot x} u_b^-(x), \quad k_* = \frac{G_*}{2}. \quad (302)$$

For any unit-channel input $V_{(u)}^\mu$, $u \in \{c, E, O\}$, the projected intervalley matrix is exactly

$$(M_{(u)}^\mu)_{ab} = \int_{\text{cell}} d^3x e^{-iG_* \cdot x} \overline{u_a^+(x)} V_{(u)}^\mu(x) u_b^-(x). \quad (303)$$

The shell-adapted matrices are

$$M_{(u)}^\parallel = \hat{n}_\mu M_{(u)}^\mu, \quad M_{(u)}^1 = e_{1\mu} M_{(u)}^\mu, \quad M_{(u)}^2 = e_{2\mu} M_{(u)}^\mu. \quad (304)$$

The overlap-calibration columns are then

$$\mathbb{C}_{\bullet u} = \begin{pmatrix} \frac{1}{2}(\sigma_3 M_{(u)}^\parallel) \\ \frac{1}{4}(\sigma_1 M_{(u)}^1 + \sigma_2 M_{(u)}^2) \\ \frac{1}{4}(\sigma_2 M_{(u)}^1 - \sigma_1 M_{(u)}^2) \end{pmatrix}. \quad (305)$$

In the exact locked diagonal case,

$$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^\parallel), \quad C_E = \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2), \quad C_O = \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2). \quad (306)$$

CXXVI. NORMALIZED UNIT-CHANNEL SEEDS

The correctly normalized calibration inputs are

$$V_{(c)}^\mu(\xi) = \frac{\theta'(\xi)^3}{I_{\theta^3}} n^\mu \sigma_3, \quad (307)$$

$$V_{(E)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} E_a^\mu \sigma_a, \quad V_{(O)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} J_a^\mu \sigma_a. \quad (308)$$

CXXVII. SEPARABLE SHELL-ADAPTED LOW-MODE ANSATZ

To reduce the 3D cell overlap to a more transparent form, assume the shell-adapted low modes admit the decomposition

$$u_a^\pm(x) = f_\pm(\xi) \chi_a^\pm(x_\perp, \xi), \quad (309)$$

where

$$x = (x_\perp, \xi), \quad \xi := \hat{n} \cdot x, \quad x_\perp \perp \hat{n}.$$

This is not an additional conceptual theorem; it is the natural factorized implementation of the defect-adapted shell basis.

Substituting (309) into (303), we obtain

$$(M_{(u)}^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) \mathcal{I}_{ab;(u)}^\mu(\xi), \quad (310)$$

with

$$\boxed{\mathcal{I}_{ab;(u)}^\mu(\xi) := \int_{\text{cell}_\perp(\xi)} d^2 x_\perp e^{-iG_* \cdot x} \overline{\chi_a^+(x_\perp, \xi)} V_{(u)}^\mu(x_\perp, \xi) \chi_b^-(x_\perp, \xi).} \quad (311)$$

Thus the full overlap is already reduced to a one-dimensional weighted integral over ξ , with a transverse primitive-cell overlap kernel $\mathcal{I}_{ab;(u)}^\mu(\xi)$.

CXXVIII. PRIMITIVE-CELL FOURIER / INTERNAL SPLIT

For the unit seeds (307)–(308), the ξ -dependence factorizes explicitly:

$$V_{(c)}^\mu(x) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} \mathcal{S}_{(c)}^\mu(x_\perp, \xi), \quad \mathcal{S}_{(c)}^\mu := n^\mu \sigma_3, \quad (312)$$

$$V_{(E)}^\mu(x) = \frac{\theta'(\xi)}{I_\theta} \mathcal{S}_{(E)}^\mu(x_\perp, \xi), \quad \mathcal{S}_{(E)}^\mu := E^\mu{}_a \sigma_a, \quad (313)$$

$$V_{(O)}^\mu(x) = \frac{\theta'(\xi)}{I_\theta} \mathcal{S}_{(O)}^\mu(x_\perp, \xi), \quad \mathcal{S}_{(O)}^\mu := J^\mu{}_a \sigma_a. \quad (314)$$

Hence

$$(M_{(c)}^\mu)_{ab} = \frac{1}{I_{\theta'^3}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{F}_{ab;(c)}^\mu(\xi), \quad (315)$$

$$(M_{(E)}^\mu)_{ab} = \frac{1}{I_\theta} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{F}_{ab;(E)}^\mu(\xi), \quad (316)$$

$$(M_{(O)}^\mu)_{ab} = \frac{1}{I_\theta} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{F}_{ab;(O)}^\mu(\xi), \quad (317)$$

where

$$\boxed{\mathfrak{F}_{ab;(u)}^\mu(\xi) := \int_{\text{cell}_\perp(\xi)} d^2 x_\perp e^{-iG_* \cdot x} \overline{\chi_a^+(x_\perp, \xi)} \mathcal{S}_{(u)}^\mu(x_\perp, \xi) \chi_b^-(x_\perp, \xi).} \quad (318)$$

This is the precise factorization:

radial weight $\times$ primitive-cell Fourier overlap $\times$ internal matrix overlap.

CXXIX. COMPRESSED SCALAR OVERLAP FUNCTIONS

Define the shell-traced scalar kernels

$$\mathfrak{s}_c(\xi) := \frac{1}{2}(\sigma_3 \hat{n}_\mu \mathfrak{F}_{(c)}^\mu(\xi)), \quad (319)$$

$$\mathfrak{s}_E(\xi) := \frac{1}{4}(\sigma_1 e_{1\mu} \mathfrak{F}_{(E)}^\mu(\xi) + \sigma_2 e_{2\mu} \mathfrak{F}_{(E)}^\mu(\xi)), \quad (320)$$

$$\mathfrak{s}_O(\xi) := \frac{1}{4}(\sigma_2 e_{1\mu} \mathfrak{F}_{(O)}^\mu(\xi) - \sigma_1 e_{2\mu} \mathfrak{F}_{(O)}^\mu(\xi)). \quad (321)$$

Then the diagonal calibration constants reduce to

$$C_c = \frac{1}{I_{\theta^3}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi), \quad (322)$$

$$C_E = \frac{1}{I_\theta} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_E(\xi), \quad (323)$$

$$C_O = \frac{1}{I_\theta} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_O(\xi). \quad (324)$$

These are the exact one-dimensional factorized forms under the separable shell-adapted basis ansatz.

CXXX. FURTHER REDUCTION IN THE CONSTANT-ENVELOPE LIMIT

If the transverse envelopes are ξ -independent and one uses the constant-envelope shell basis

$$u_a(x) = \frac{1}{\sqrt{V_{\text{cell}}}} \xi_a,$$

then the projected shell overlaps reduce to primitive-cell Fourier coefficients. In particular, first-derivative coefficient fields supported on the first BCC shell satisfy

$$\widehat{\mathcal{T}}_j(G_*) = i \left(\sum_m C_{jm} G_{*,m} \right) A_{G_*}, \quad (325)$$

and

$$W^\alpha = \widehat{\mathcal{T}}^\alpha(G_*) s^\alpha. \quad (326)$$

So in the simplest constant-envelope reduction,

$$\mathfrak{s}_c, \mathfrak{s}_E, \mathfrak{s}_O$$

collapse to fixed primitive-cell Fourier amplitudes times constant internal traces.

CXXXI. VANISHING MECHANISMS

The factorized formulas (322)–(324) show that a calibration constant can vanish for three logically distinct reasons.

(i) **Radial cancellation**

Even if the microscopic seed exists, the weighted defect overlap may vanish:

$$\int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^m \mathfrak{s}(\xi) = 0. \quad (327)$$

(ii) **Primitive-cell Fourier cancellation**

The $Q = G_*$ primitive-cell Fourier overlap may vanish:

$$\mathfrak{F}_{ab;(u)}^\mu(\xi) \equiv 0 \iff \text{the } G_*\text{-harmonic does not couple to the chosen Bloch pair.} \quad (328)$$

In the first-shell derivative case this can happen when the coefficient tensor annihilates G_* :

$$\sum_m C_{jm} G_{*,m} = 0.$$

(iii) **Internal Pauli orthogonality**

The internal matrix overlap may vanish after the shell Pauli trace:

$$(\sigma_3 \hat{n}_\mu \mathfrak{F}_{(c)}^\mu) = 0, \quad (\sigma_1 e_{1\mu} \mathfrak{F}_{(E)}^\mu + \sigma_2 e_{2\mu} \mathfrak{F}_{(E)}^\mu) = 0, \quad (\sigma_2 e_{1\mu} \mathfrak{F}_{(O)}^\mu - \sigma_1 e_{2\mu} \mathfrak{F}_{(O)}^\mu) = 0. \quad (329)$$

Therefore the final microscopic calibration problem is completely reduced to checking whether each desired seed survives these three filters.

CXXXII. CONNECTION TO THE BRANCH-LEVEL COEFFICIENT POINT

For the present refined locked + Class II branch, the transport-side point is already explicit:

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2, \quad g_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad g_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}.$$

Hence the physical reduced coefficients are

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,} \quad (330)$$

with C_c, C_E, C_O given by (322)–(324).

CXXXIII. HONEST STATUS

The exact folded-Bragg overlap problem is now reduced as far as one can reduce it analytically without inserting an explicit Bloch wave basis.

What remains open is no longer conceptual: it is the direct evaluation of the scalar kernel functions

$$\mathfrak{s}_c(\xi), \quad \mathfrak{s}_E(\xi), \quad \mathfrak{s}_O(\xi)$$

for the actual refined locked microscopic Bloch frame.

CXXXIV. FINAL SUMMARY

The exact folded-Bragg calibration starts from

$$(M_{(u)}^\mu)_{ab} = \int_{\text{cell}} d^3x e^{-iG_* \cdot x} \overline{u_a^+(x)} V_{(u)}^\mu(x) u_b^-(x), \quad u \in \{c, E, O\}.$$

With the normalized unit-channel inputs

$$V_{(c)}^\mu = \frac{\theta'(\xi)^3}{I_{\theta^3}} n^\mu \sigma_3, \quad V_{(E)}^\mu = \frac{\theta'(\xi)}{I_\theta} E^\mu{}_a \sigma_a, \quad V_{(O)}^\mu = \frac{\theta'(\xi)}{I_\theta} J^\mu{}_a \sigma_a,$$

the diagonal calibration constants are extracted by Pauli traces of the shell-adapted components.

Under the separable shell-adapted Bloch basis

$$u_a^\pm(x) = f_\pm(\xi) \chi_a^\pm(x_\perp, \xi),$$

the calibration integrals factorize into

$$\text{radial overlap} \times \text{primitive-cell Fourier overlap} \times \text{internal Pauli overlap}.$$

More precisely,

$$C_c = \frac{1}{I_{\theta^3}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi),$$

$$C_E = \frac{1}{I_\theta} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_E(\xi), \quad C_O = \frac{1}{I_\theta} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_O(\xi).$$

Thus a calibration constant can vanish for exactly three distinct reasons:

$$\boxed{\text{(i) radial cancellation,} \quad \text{(ii) primitive-cell Fourier cancellation,} \quad \text{(iii) internal Pauli orthogonality.}}$$

In the first-shell derivative case, the Fourier obstruction is explicit:

$$\hat{\mathcal{T}}_j(G_*) = i \left(\sum_m C_{jm} G_{*,m} \right) A_{G_*},$$

so vanishing occurs if the coefficient tensor annihilates G_* .

Therefore the final microscopic calibration problem has been reduced as far as possible without inserting the actual refined locked Bloch wavefunctions: it is the explicit evaluation of the scalar overlap kernels

$$\mathfrak{s}_c(\xi), \quad \mathfrak{s}_E(\xi), \quad \mathfrak{s}_O(\xi).$$

The physical reduced coefficients remain

$$\boxed{\lambda_{||} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O,}$$

with the branch-level transport point already explicit.

CXXXV. PRACTICAL NONVANISHING CRITERIA FOR THE FINAL BLOCH CALIBRATION

CXXXVI. SHORT EXPLANATION

The calibration constants C_c, C_E, C_O were reduced to folded-Bragg overlap integrals. The next task is not to rewrite those integrals, but to turn them into actual executable nonvanishing criteria.

The key point is that each calibration constant can fail only through three filters:

1. radial cancellation,
2. primitive-cell Fourier cancellation at $Q = G_*$,
3. internal Pauli orthogonality.

Thus the final carrier audit becomes a sign/support/overlap problem.

CXXXVII. STARTING POINT

The exact reduced extraction formulas are

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2), \quad (331)$$

together with the exact overlap-normalization layer

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O. \quad (332)$$

Inside the explicit seed space, the reduced symbol is nondegenerate if and only if

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).} \quad (333)$$

Thus the final Bloch calibration problem is exactly whether the explicit seed channels survive in the true Bloch low-mode frame.

CXXXVIII. FACTORIZED CALIBRATION FORM

Under the shell-adapted separable Bloch basis,

$$u_a^{\pm}(x) = f_{\pm}(\xi) \chi_a^{\pm}(x_{\perp}, \xi),$$

the diagonal calibration constants reduce to

$$C_c = \frac{1}{I_{\theta^3}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi), \quad (334)$$

$$C_E = \frac{1}{I_{\theta}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_E(\xi), \quad C_O = \frac{1}{I_{\theta}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_O(\xi). \quad (335)$$

Here:

- $f_+ f_-$ is the radial/envelope overlap,
- $\mathfrak{s}_c, \mathfrak{s}_E, \mathfrak{s}_O$ are the primitive-cell Fourier plus internal Pauli overlap kernels,
- the normalization denominators I_{θ^3}, I_{θ} are nonzero by construction.

CXXXIX. LONGITUDINAL CRITERION FOR $C_c \neq 0$

A. Necessary condition

A necessary condition is that the weighted kernel is not identically zero:

$$f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi) \not\equiv 0. \quad (336)$$

Equivalently, at least one longitudinal microscopic overlap must survive. At the tensor-basis level this is the executable criterion

$$\boxed{\exists A : \ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j \neq 0.} \quad (337)$$

At the graded overlap level this is the same condition written as

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.} \quad (338)$$

B. Simple sufficient condition

A clean sufficient condition is:

$$\boxed{f_+(\xi)f_-(\xi)\theta'(\xi)^3\mathfrak{s}_c(\xi) \text{ has a definite sign on a set of positive measure.}} \quad (339)$$

Indeed, under (339), the integral (334) cannot cancel to zero.

Thus the longitudinal overlap survives whenever there is

1. nontrivial radial overlap,
2. nonvanishing $Q = G_*$ harmonic coupling,
3. nonzero σ_3 -projected internal trace,

with no sign-changing cancellation strong enough to kill the integral.

CXL. TRANSVERSE CRITERION FOR $(C_E, C_O) \neq (0, 0)$

A. Necessary condition

A necessary condition is that at least one transverse weighted kernel is not identically zero:

$$f_+(\xi)f_-(\xi)\theta'(\xi)\mathfrak{s}_E(\xi) \neq 0 \quad \text{or} \quad f_+(\xi)f_-(\xi)\theta'(\xi)\mathfrak{s}_O(\xi) \neq 0. \quad (340)$$

At the corrected microscopic level, this requires a nontrivial derivative-resolved shell source:

$$\boxed{\partial_{p_\nu}\widehat{\Sigma}_{JK}(G_*;p)|_{p=0} \neq 0 \quad \text{or} \quad \partial_{p_\nu}\widehat{\Sigma}_{KK}(G_*;p)|_{p=0} \neq 0.} \quad (341)$$

B. Minimal sufficient mechanisms

The minimal nonvanishing mechanisms for the transverse sector are exactly:

$$\boxed{\text{(a) derivative on the shell phase,}} \quad (342)$$

$$(\partial_{p_\nu}\widehat{\mathcal{J}})(G_*;0) \neq 0 \quad \text{or} \quad (\partial_{p_\nu}\widehat{\mathcal{K}})(G_*;0) \neq 0;$$

$$\boxed{\text{(b) nontrivial internal-index contraction,}} \quad (343)$$

meaning that the mediator/internal contraction distinguishes the Σ_1, Σ_2 channels rather than collapsing to a trivial scalar;

$$\boxed{\text{(c) nontrivial complex intervalley phase data,}} \quad (344)$$

so that the $Q = G_*$ harmonic does not reduce to the naive vanishing anticommutator structure;

$$\boxed{\text{(d) direct } KK \text{ transverse contribution.}} \quad (345)$$

If at least one of these mechanisms is present and the final Pauli projection is nonzero, then $(C_E, C_O) \neq (0, 0)$ can survive.

C. Minimal first-shell sufficient test

In the explicit folded-Bragg Class II reduction, the full transverse audit compresses to two scalar Fourier overlaps:

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (346)$$

If (346) holds, then the explicit Class II branch realizes

$$\boxed{(\alpha, \beta) \neq (0, 0).} \quad (347)$$

Thus, at the practical branch level, (346) is the strongest usable explicit sufficient test for the transverse carrier.

CXLI. THREE-FILTER DECOMPOSITION OF FAILURE

The factorized formulas show that failure of a calibration constant can occur only through one of three mechanisms:

a. (i) radial cancellation

$$\int d\xi f_+ f_- \theta'^m \mathfrak{s}(\xi) = 0$$

by sign-changing or oscillatory cancellation in the defect/radial overlap;

b. (ii) primitive-cell Fourier cancellation the $Q = G_*$ harmonic does not couple to the chosen Bloch pair;

c. (iii) internal Pauli orthogonality the desired σ_3 , σ_1 , or σ_2 component vanishes after projection.

So the calibration audit is exactly a three-filter survival test.

CXLII. PRACTICAL FINAL ACCEPTANCE CRITERIA

A. Diagonal locked case

If the shell basis is locked and the overlap matrix is diagonal, the current branch passes iff

$$\boxed{C_c \neq 0, \quad (C_E, C_O) \neq (0, 0).} \quad (348)$$

B. Practical sufficient version

A convenient sufficient form is:

$$\boxed{f_+ f_- \theta'(\xi)^3 \mathfrak{s}_c(\xi) \text{ has definite nonzero sign on positive measure,}} \quad (349)$$

and either

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0} \quad (350)$$

or

$$\boxed{A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (351)$$

Under these conditions, the explicit branch survives the final Bloch calibration.

CXLIII. HONEST STATUS

At the strongest honest current stage:

1. the reduced witness architecture is closed;

2. the branch-level transport coefficient point is explicit;
3. the calibration problem is reduced to finite overlap survival tests;
4. the longitudinal side is governed by the basis-overlap / graded-overlap criterion;
5. the transverse side is governed by the derivative-resolved JK/KK shell audit, with the first-shell two-overlap test as the practical explicit sufficient criterion.

What remains open is not conceptual structure, but the explicit evaluation of the actual overlap kernels in the true refined locked Bloch frame.

CXLIV. FINAL SUMMARY

The exact final Bloch calibration problem has now been reduced to explicit nonvanishing criteria. The reduced witness extraction is

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2),$$

with the exact normalization layer

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

In the explicit seed space, the symbol is nondegenerate if and only if

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).}$$

Under the factorized shell-adapted Bloch basis,

$$C_c = \frac{1}{I_{\theta'^3}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi),$$

$$C_E = \frac{1}{I_{\theta}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_E(\xi), \quad C_O = \frac{1}{I_{\theta}} \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \mathfrak{s}_O(\xi).$$

Therefore each calibration constant can fail only through one of three filters:

$$\boxed{\text{radial cancellation,} \quad \text{primitive-cell Fourier cancellation,} \quad \text{internal Pauli orthogonality.}}$$

For the longitudinal channel, an executable criterion is

$$\boxed{\exists A : E_{A,ij}^3 \hat{n}_i \hat{n}_j \neq 0,}$$

equivalently

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.}$$

For the transverse channel, the honest microscopic frontier is not the naive local JK term, but the derivative-resolved shell audit

$$\boxed{\partial_{p\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0}.}$$

The minimal transverse nonvanishing mechanisms are:

$$\boxed{\text{(a) derivative on the shell phase,} \quad \text{(b) nontrivial internal contraction,} \quad \text{(c) nontrivial complex intervalley phase,} \quad \text{(d) direct } K \text{ term}}$$

In the explicit folded-Bragg Class II branch, this compresses to the practical sufficient test

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0,}$$

which implies

$$\boxed{(\alpha, \beta) \neq (0, 0).}$$

Hence the sharp practical acceptance criterion for the current branch is:

$$\boxed{C_c \neq 0}$$

via a nonvanishing longitudinal overlap, and

$$\boxed{(C_E, C_O) \neq (0, 0)}$$

via the derivative-resolved transverse shell audit, or in the minimal explicit reduction, via the two first-shell Fourier overlap tests.

Thus the remaining frontier is no longer conceptual. It is the explicit evaluation of the overlap kernels and their sign/support structure in the true refined locked Bloch frame.

CXLV. FINAL DECISION TREE FOR THE REFINED LOCKED + CLASS II WORKING BRANCH

CXLVI. GOAL

We now convert the current refined locked + Class II program into a final pass/fail decision tree. The purpose is to separate:

1. support gates that must hold before the shell Dirac carrier is meaningful,
2. actual carrier-survival gates,
3. mass-protection gates,
4. the downstream one-defect/one-mode realization criterion.

The logic is sequential:

$$\text{Gate 1} \implies \text{Gate 2} \implies \text{Gate 3} \implies \text{Gate 4},$$

and only after all four pass does one proceed to the protected-mode criterion.

CXLVII. GATE 1: REMOTE-GAP SUPPORT THEOREM

Criterion

The first mandatory gate is

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.} \tag{352}$$

The current explicit benchmark side is

$$\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}, \tag{353}$$

while the residual side is bounded by

$$\eta_{R,\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}, \tag{354}$$

with

$$\eta_{R,\rho}^{\text{tr}} \leq 2 \sum_{(ab) \in \mathfrak{P}_6} \left[\mathcal{D}_{ab,\rho} + \frac{\alpha_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{JJ} + \frac{|\alpha_X \beta_X|}{M_X^2} \mathcal{M}_{ab,\rho}^{JK} + \frac{\beta_X^2}{M_X^2} \mathcal{M}_{ab,\rho}^{KK} \right]. \tag{355}$$

Meaning

If Gate 1 passes, the enriched low slot is spectrally isolated from the remote sector. This is a support theorem, not the main frontier.

If Gate 1 fails, then spectral isolation is not established, and all later shell-carrier statements are unsupported.

Current honest status

The benchmark-side frontier has been reduced to one finite eigenvalue problem:

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}.$$

So Gate 1 is operationally finite.

CXLVIII. GATE 2: LONGITUDINAL SURVIVAL

Criterion

The shell-extracted longitudinal coefficient is

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}) = C_c g_c. \quad (356)$$

Therefore the carrier requires

$$C_c \neq 0 \quad \text{and} \quad g_c \neq 0. \quad (357)$$

Inside the explicit Class II transport truncation,

$$g_c \propto c_{\parallel},$$

and in the current refined working branch

$$g_c = \frac{\alpha_X^2}{M_X^2} j^2.$$

At the basis-overlap level, a practical executable criterion is

$$\exists A : E_{A,ij}^3 \hat{n}_i \hat{n}_j \neq 0. \quad (358)$$

Equivalently, at the graded/radial level,

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0. \quad (359)$$

Sufficient form

A clean sufficient condition is:

$$f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi) \text{ has definite nonzero sign on a set of positive measure.} \quad (360)$$

Meaning

If Gate 2 fails, then $\lambda_{\parallel} = 0$, so the branch does not realize the required longitudinal Dirac carrier. This is not a minor normalization issue; it kills the shell Dirac architecture.

CXLIX. GATE 3: TRANSVERSE SURVIVAL

Criterion

The shell-extracted transverse coefficients are

$$\alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2) = C_E g_E, \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2) = C_O g_O. \quad (361)$$

Hence the carrier requires

$$(C_E, C_O) \neq (0, 0) \quad \text{and} \quad (g_E, g_O) \neq (0, 0). \quad (362)$$

Honest microscopic frontier

A naive local real Hermitian JK bilinear vanishes because

$$\{\Sigma_3, \Sigma_1\} = \{\Sigma_3, \Sigma_2\} = 0.$$

Therefore the real microscopic frontier is

$$\partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0} \neq 0 \quad \text{or} \quad \partial_{p_\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0} \neq 0. \quad (363)$$

Minimal explicit sufficient mechanisms

The minimal nonvanishing mechanisms are:

1. derivative on the shell phase,
2. nontrivial internal-index contraction,
3. nontrivial complex intervalley phase data,
4. direct KK transverse contribution.

In the folded-Bragg first-shell reduction, this compresses to the explicit sufficient test

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0. \quad (364)$$

Meaning

If Gate 3 fails, then the branch does not realize a genuine two-direction transverse Dirac witness. This does not necessarily refute the reduced architecture itself, but it shows that the current microscopic branch is insufficient.

CL. GATE 4: MASS PROTECTION

Criterion

For the doubled Dirac block,

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i),$$

the exact constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}. \quad (365)$$

Let

$$A_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}$$

be the residual-symmetry allowed constant algebra. Mass protection requires

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}. \quad (366)$$

Sufficient mechanism

It is sufficient to prove exact residual valley symmetry

$$U(1)_V \quad \text{or already} \quad \mathbb{Z}_2^V. \quad (367)$$

With

$$Q_V = \tau_z \otimes \mathbf{1}_2,$$

the two mass generators rotate nontrivially under $U(1)_V$, and are odd already under $\mathbb{Z}_2^V$. Therefore either symmetry excludes the mass space.

Meaning

If Gate 4 fails, then even if Gates 1–3 pass, the doubled Dirac block is not protected against a constant symmetry-allowed mass. So the branch is not an acceptable robust massless Dirac realization.

CLI. POST-ACCEPTANCE GATE 5: ONE-DEFECT / ONE-PROTECTED-MODE REALIZATION

Only after Gates 1–4 pass does it make sense to test the downstream protected-mode criterion. In the currently cleanest operational form, the sufficient criterion is

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}. \quad (368)$$

This is not part of branch acceptance itself. It is the next physical realization step once the branch has already been accepted as a valid microscopic carrier.

CLII. FINAL DECISION TREE

The final decision tree is:

a. Gate 1. Check (352). If false:

Reject as unsupported: spectral isolation not established.

If true, continue.

b. Gate 2. Check (357), preferably via (358) or (359). If false:

Reject: no longitudinal Dirac witness.

If true, continue.

c. *Gate 3.* Check (362), with the honest frontier (405) and practical explicit test (364). If false:

Reject current branch as transversely insufficient.

If true, continue.

d. *Gate 4.* Check (366), preferably by proving (367). If false:

Reject as unprotected: constant Dirac mass may survive.

If true:

Accept the current refined locked + Class II branch as a viable microscopic massless Dirac carrier.

Then, and only then, proceed to Gate 5 (368).

CLIII. FINAL SUMMARY

The refined locked + Class II working branch is accepted or rejected by the following final decision tree.

a. *Gate 1: remote-gap support.* Require

$$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.$$

If this fails, spectral isolation is not established, so all later shell-carrier statements are unsupported.

b. *Gate 2: longitudinal survival.* Require

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}) = C_c g_c \neq 0.$$

Practical executable criteria are:

$$\exists A : E_{A,ij}^3 \hat{n}_i \hat{n}_j \neq 0,$$

or equivalently

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.$$

c. *Gate 3: transverse survival.* Require

$$(\alpha, \beta) \neq (0, 0), \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

The honest microscopic frontier is

$$\partial_{p\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0} \neq 0 \quad \text{or} \quad \partial_{p\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0} \neq 0.$$

In the explicit folded-Bragg reduction, a practical sufficient test is

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.$$

d. *Gate 4: mass protection.* Let

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Acceptance requires

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

A sufficient mechanism is exact residual valley symmetry

$$U(1)_V \quad \text{or} \quad \mathbb{Z}_2^V.$$

e. Acceptance verdict. If Gates 1–4 all pass, then

the current refined locked + Class II branch is accepted as a viable microscopic massless Dirac carrier.

If any gate fails, the branch is rejected for the corresponding reason: unsupported, longitudinally insufficient, transversely insufficient, or unprotected.

f. Post-acceptance Gate 5. Only after acceptance should one test the downstream one-defect/one-protected-mode realization criterion:

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Thus the final frontier is no longer conceptual. It is the explicit finite evaluation of Gates 1–4 in the true refined locked microscopic Bloch frame.

CLIV. FINAL MASTER CHECKLIST FOR THE REFINED LOCKED + CLASS II WORKING BRANCH

CLV. PURPOSE

The purpose of this checklist is not to introduce new structure, but to classify every required gate in the current working branch into exactly one of the following three statuses:

closed, finite computation left, structurally open.

Here:

- **closed** means the theorem-level structure is complete and no additional conceptual input is needed;
- **finite computation left** means the criterion is already exact, but the concrete branch values still have to be evaluated;
- **structurally open** means an honest nontrivial conceptual caveat remains, so one must not overstate closure.

CLVI. MASTER STATUS TABLE

Gate	Question	Status	Why
G1	$\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho} ?$	finite computation left	remote-gap theorem is structurally closed, but the sign has not yet been verified on the concrete
G2	$C_c \neq 0, \lambda_{\parallel} \neq 0 ?$	finite computation left	the exact executable criterion is already closed via basis-overlap or graded-overlap audit
G3	$(C_E, C_O) \neq (0, 0), (\alpha, \beta) \neq (0, 0) ?$	finite computation left (working branch) + structurally open caveat	the explicit folded-Bragg practical test is finite, but the full unconditional microscopic closure still honestly rests on the derivative-resolved JK/KK
G4	$A_{\text{res}} \cap \mathcal{M}_D = \{0\} ?$	finite computation left	mass protection is reduced to a finite residual-symmetry Class II is only compatible, not itself a proof
G5	one defect $\leftrightarrow$ one protected mode?	finite computation left	the Callias/radial criterion is structurally closed but it is post-acceptance and still requires branch verification

CLVII. GATE-BY-GATE EXACT STATUS

A. Gate 1: remote-gap support

The full support criterion is already fixed:

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}.} \quad (369)$$

Moreover, the remote-gap theorem and the benchmark/residual decomposition are already structurally closed, and the same-shell non-enriched part has already been reduced to a finite Hermitian eigenvalue problem. Therefore Gate 1 is not conceptually open. It is a finite explicit branch computation.

a. Status:

$$\boxed{\text{finite computation left.}}$$

B. Gate 2: longitudinal survival

The exact low-valley readout is

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}) = C_c g_c. \quad (370)$$

The remaining microscopic test is already sharply reduced to the finite overlap question

$$\boxed{\exists A : E_{A,ij}^3 \hat{n}_i \hat{n}_j \neq 0.} \quad (371)$$

Equivalently,

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.} \quad (372)$$

Thus the longitudinal frontier is no longer conceptual.

a. Status:

$$\boxed{\text{finite computation left.}}$$

C. Gate 3: transverse survival

At the honest microscopic level, the true frontier is

$$\boxed{\partial_{p_{\nu}} \hat{\Sigma}_{JK}(G_*; p)|_{p=0} \neq 0 \quad \text{or} \quad \partial_{p_{\nu}} \hat{\Sigma}_{KK}(G_*; p)|_{p=0} \neq 0.} \quad (373)$$

However, inside the explicit folded-Bragg Class II working branch, this has already been compressed to the practical finite sufficient test

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (374)$$

Therefore:

- as a *working-branch acceptance problem*, Gate 3 is finite;
- as an *unconditional microscopic theorem*, Gate 3 still carries an honest structural caveat.

a. Status:

$$\boxed{\text{finite computation left on the working branch,}}$$

but also

$$\boxed{\text{structurally open as a fully unconditional transverse closure statement.}}$$

D. Gate 4: mass protection

The exact constant mass space is already known:

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}. \quad (375)$$

The exact acceptance criterion is already fixed:

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}}. \quad (376)$$

A sufficient mechanism is also already fixed:

$$\boxed{U(1)_V \quad \text{or} \quad \mathbb{Z}_2^V}. \quad (377)$$

Thus Gate 4 is not conceptually open. It is a finite residual-symmetry exclusion test. However, the important honest caveat is: Class II is mass-protection compatible, but this is not by itself a proof of mass protection.

a. Status:

$$\boxed{\text{finite computation left.}}$$

E. Gate 5: one-defect / one-mode

The downstream radial/Callias structure is already fixed. The practical sufficient criterion is already explicit; for example, in the most practical case,

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}}. \quad (378)$$

The profile-level theorem is already structurally closed under the stated assumptions. What remains is the explicit branch verification of the quantities

$$a_0, b_0, c_0, \mu_\infty, \kappa_*, \xi_c, \xi_E, \xi_O,$$

and the competitor inequalities.

a. Status:

$$\boxed{\text{finite computation left.}}$$

CLVIII. GLOBAL HONEST VERDICT

The current working branch is *not* blocked by any new conceptual obstruction at Gates 1, 2, 4, or 5. These have all been reduced to explicit finite verification problems.

The only place where one must still speak with extra caution is Gate 3: the practical branch-level test is finite and usable, but the final fully unconditional transverse closure still carries the derivative-resolved JK/KK caveat.

Therefore the strongest honest classification is:

$$\boxed{\text{G1: finite computation left}} \quad (379)$$

$$\boxed{\text{G2: finite computation left}} \quad (380)$$

$$\boxed{\text{G3: finite computation left on the branch, with structural caveat}} \quad (381)$$

$$\boxed{\text{G4: finite computation left}} \quad (382)$$

$$\boxed{\text{G5: finite computation left after branch acceptance}} \quad (383)$$

CLIX. PRIORITY ORDER FOR THE NEXT CALCULATION

The correct next order is:

1. Gate 2: longitudinal survival $C_c \neq 0$;
2. Gate 3: transverse survival $(C_E, C_O) \neq (0, 0)$;
3. Gate 4: mass protection;
4. Gate 1: remote-gap verification in the same parameter regime;
5. Gate 5: one-defect / one-mode realization.

The reason is simple:

- Gates 2 and 3 decide whether there is any shell Dirac carrier at all;
- Gate 4 decides whether that carrier is robustly massless;
- Gate 1 is a support theorem that must ultimately hold in the same regime, but it is no longer the main conceptual frontier;
- Gate 5 only makes sense after acceptance of the carrier branch.

CLX. FINAL SUMMARY

The refined locked + Class II working branch is now best organized by the following master checklist.

a. Gate 1: remote-gap support

$$\Delta_{\text{bench}, \rho}^{\text{fin}} > \eta_{R, \rho}.$$

Status:

finite computation left.

Reason: the theorem-level structure is already closed, but the concrete branch sign has not yet been verified.

b. Gate 2: longitudinal survival

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}) = C_c g_c \neq 0.$$

Executable criteria:

$$\exists A : E_{A, ij}^3 \hat{n}_i \hat{n}_j \neq 0, \quad J_{\parallel}^{(3)} \neq 0.$$

Status:

finite computation left.

c. Gate 3: transverse survival

$$(\alpha, \beta) \neq (0, 0), \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

Honest microscopic frontier:

$$\partial_{p\nu} \hat{\Sigma}_{JK}(G_*; p)|_{p=0} \neq 0 \quad \text{or} \quad \partial_{p\nu} \hat{\Sigma}_{KK}(G_*; p)|_{p=0} \neq 0.$$

Practical branch-level sufficient test:

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.$$

Status:

finite computation left on the working branch, with a structural caveat for unconditional closure.

d. *Gate 4: mass protection*

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}, \quad \mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.$$

Sufficient mechanism:

$$U(1)_V \quad \text{or} \quad \mathbb{Z}_2^V.$$

Status:

finite computation left.

e. *Gate 5: one-defect / one-mode* This is a post-acceptance downstream criterion, not a branch-acceptance gate. In the most practical case it reduces to

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.$$

Status:

finite computation left after branch acceptance.

Therefore the strongest honest current global verdict is:

G1: finite computation left,

G2: finite computation left,

G3: finite computation left on the branch, but still with an honest structural caveat,

G4: finite computation left,

G5: finite computation left after acceptance.

Hence the next actual priority order is:

Gate 2 $\longrightarrow$ Gate 3 $\longrightarrow$ Gate 4 $\longrightarrow$ Gate 1 $\longrightarrow$ Gate 5.

This is because Gates 2 and 3 decide whether the shell Dirac carrier exists at all, Gate 4 decides whether it is protected, Gate 1 is the support theorem to be verified in the same regime, and Gate 5 is meaningful only after acceptance.

CLXI. GATE 2 CHECKLIST: ACTUAL LONGITUDINAL SURVIVAL AUDIT

CLXII. PURPOSE

Gate 2 asks whether the current refined locked + Class II working branch truly carries a nonzero longitudinal primitive shell channel.

The exact question is:

$$\lambda_{\parallel} \neq 0 ?$$

At the current stage this is no longer a conceptual PDE question. It is a finite microscopic audit that can be organized into four nested levels:

1. basis-level overlap audit,
2. shell-seed coefficient audit,
3. graded/radial overlap audit,
4. final Bloch calibration audit.

CLXIII. EXACT LONGITUDINAL TARGET

The canonical primitive longitudinal shell target is

$$B_{\parallel} : \quad N_{ij}^3 = \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = N_{ij}^2 = 0. \quad (384)$$

Equivalently, the shell covector is

$$T_{ija}^{(\parallel)} = \hat{n}_i \hat{n}_j \delta_{a3}. \quad (385)$$

The primitive transport operator representative is

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi)(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3. \quad (386)$$

Its reduced witness is

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0. \quad (387)$$

Therefore the longitudinal question is exactly whether the full microscopic theory generates a nonzero coefficient in front of this primitive odd singlet channel.

CLXIV. LEVEL A: BASIS-LEVEL OVERLAP AUDIT

Let

$$V_{\text{TECT}} \subset \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the actual microscopic transport space extracted from the full refined locked linearization.

Choose any basis

$$\{E_A\}_{A=1}^d$$

of V_{TECT} . Define the longitudinal overlaps

$$\ell_{\parallel A} := \langle T^{(\parallel)}, E_A \rangle = E_{A,ij}^3 \hat{n}_i \hat{n}_j. \quad (388)$$

Then the exact basis-level longitudinal criterion is

$$\exists A : \ell_{\parallel A} \neq 0. \quad (389)$$

This is already a complete finite linear-algebra test.

a. Checklist A.

1. Fix the shell pair G_* and the adapted frame $(e_1, e_2, \hat{n})$.
2. Extract the actual microscopic transport basis $\{E_A\}$.
3. Compute

$$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j$$

for every basis element.

4. If all $\ell_{\parallel A} = 0$, Gate 2 fails immediately.
5. If at least one $\ell_{\parallel A} \neq 0$, proceed.

CLXV. LEVEL B: SHELL-SEED COEFFICIENT AUDIT

Decompose the actual induced shell tensor onto the explicit seed space:

$$P_{\text{seed}} N_{\text{mic}} = c_{\parallel}^{(\text{mic})} B_{\parallel} + c_I^{(\text{mic})} B_I + c_J^{(\text{mic})} B_J. \quad (390)$$

The basis-overlap audit is equivalent to the nonvanishing of the longitudinal seed coefficient:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0.} \quad (391)$$

Hence

$$\boxed{c_{\parallel}^{(\text{mic})} \neq 0 \implies \lambda_{\parallel} \neq 0} \quad (392)$$

provided the overlap normalization is nonzero.

a. Checklist B.

1. Project the actual shell transport tensor onto $V_{\text{seed}} = \text{span}\{B_{\parallel}, B_I, B_J\}$.
2. Read off $c_{\parallel}^{(\text{mic})}$.
3. If $c_{\parallel}^{(\text{mic})} = 0$, Gate 2 fails.
4. If $c_{\parallel}^{(\text{mic})} \neq 0$, proceed.

CLXVI. LEVEL C: GRADED/RADIAL OVERLAP AUDIT

At the projected defect-profile level, the same condition is encoded in the σ_3 -component of the traceless quadratic response:

$$s_3(\xi).$$

The exact graded overlap is

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi).} \quad (393)$$

The graded criterion is

$$\boxed{J_{\parallel}^{(3)} \neq 0.} \quad (394)$$

A simple sufficient condition is:

$$\boxed{f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi) \text{ has definite nonzero sign on a set of positive measure.}} \quad (395)$$

a. Checklist C.

1. Construct the low-mode radial profiles $f_{\pm}(\xi)$.
2. Extract the σ_3 -projected kernel $s_3(\xi)$ or $\mathfrak{s}_c(\xi)$.
3. Evaluate

$$J_{\parallel}^{(3)} = \int d\xi f_+ f_- \theta'(\xi) s_3(\xi).$$

4. If this vanishes by sign cancellation, Gate 2 fails.
5. If it is nonzero, proceed.

CLXVII. LEVEL D: FINAL BLOCH CALIBRATION

Even after the longitudinal seed survives, the actual physical coefficient still carries the Bloch overlap factor:

$$\boxed{\lambda_{\parallel} = C_c g_c.} \quad (396)$$

In the current working branch, the transport-side coefficient g_c is already explicit. So the remaining final Gate-2 condition is

$$\boxed{C_c \neq 0.} \quad (397)$$

Using the normalized longitudinal unit input,

$$V_{(c)}^{\mu}(\xi) = \frac{\theta'(\xi)^3}{I_{\theta^3}} n^{\mu} \sigma_3,$$

the calibration overlap is

$$\boxed{C_c = \frac{1}{2I_{\theta^3}} \left[\sigma_3 \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_+^{\dagger}(x) \theta'(\xi)^3 \sigma_3 U_-(x) \right].} \quad (398)$$

Thus Gate 2 finally passes iff

$$\boxed{c_{\parallel}^{(\text{mic})} \neq 0 \quad \text{and} \quad C_c \neq 0.} \quad (399)$$

a. Checklist D.

1. Build the normalized c -unit input $V_{(c)}^{\mu}$.
2. Compute the folded-Bragg overlap matrix $M_{(c)}^{\parallel}$.
3. Evaluate

$$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}).$$

4. If $C_c = 0$, Gate 2 fails at the calibration layer.
5. If $C_c \neq 0$, Gate 2 passes completely.

CLXVIII. FINAL GATE-2 DECISION TREE

The complete Gate-2 decision tree is:

a. Step 1. Check whether

$$\exists A : \ell_{\parallel A} \neq 0.$$

If not, reject: no longitudinal microscopic seed exists.

b. Step 2. Check whether

$$c_{\parallel}^{(\text{mic})} \neq 0.$$

If not, reject: the full shell projection kills the primitive longitudinal channel.

c. Step 3. Check whether

$$J_{\parallel}^{(3)} \neq 0.$$

If not, reject: the defect/graded overlap cancels the longitudinal seed.

d. Step 4. Check whether

$$C_c \neq 0.$$

If not, reject: the Bloch-frame overlap kills the surviving longitudinal seed.

e. Final verdict. Only if all four steps pass do we conclude

$$\boxed{\lambda_{\parallel} \neq 0.} \tag{400}$$

CLXIX. STATUS

At the present stage, every ingredient of this decision tree is already structurally fixed. Therefore Gate 2 is not conceptually open. It is an exact finite executable audit.

What remains is only the actual evaluation of

$$\ell_{\parallel A}, \quad c_{\parallel}^{(\text{mic})}, \quad J_{\parallel}^{(3)}, \quad C_c$$

for the true refined locked microscopic transport basis.

CLXX. FINAL SUMMARY

Gate 2 asks whether the primitive longitudinal shell channel truly survives in the full refined locked microscopic transport space.

The canonical longitudinal target is

$$B_{\parallel} : \quad N_{ij}^3 = \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = N_{ij}^2 = 0,$$

equivalently

$$T_{ija}^{(\parallel)} = \hat{n}_i \hat{n}_j \delta_{a3}.$$

The exact basis-level overlap audit is

$$\boxed{\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j, \quad \exists A : \ell_{\parallel A} \neq 0.}$$

This is equivalent to the shell-seed survival condition

$$\boxed{c_{\parallel}^{(\text{mic})} \neq 0.}$$

At the graded/radial level, the same survival question becomes

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.}$$

A simple sufficient condition is that

$$f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi)$$

has definite nonzero sign on a set of positive measure.

Finally, the actual physical longitudinal coefficient carries the Bloch overlap factor:

$$\boxed{\lambda_{\parallel} = C_c g_c,}$$

so the final calibration condition is

$$\boxed{C_c \neq 0.}$$

Therefore the complete Gate-2 pass condition is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.}$$

Only when all of these hold can one conclude

$$\boxed{\lambda_{\parallel} \neq 0.}$$

Thus Gate 2 is not conceptually open. It is an exact finite executable audit of:

$$\ell_{\parallel A}, \quad c_{\parallel}^{(\text{mic})}, \quad J_{\parallel}^{(3)}, \quad C_c.$$

CLXXI. GATE 3 CHECKLIST: ACTUAL TRANSVERSE SURVIVAL AUDIT

CLXXII. PURPOSE

Gate 3 asks whether the current refined locked + Class II working branch truly carries nonzero transverse shell witnesses.

The exact question is:

$$\boxed{(\alpha, \beta) \neq (0, 0) ?}$$

At the present stage this should be organized into four nested layers:

1. shell-adapted witness projection level,
2. derivative-resolved microscopic shell audit,
3. folded-Bragg first-shell compressed test,
4. final Bloch calibration audit.

CLXXIII. EXACT TRANSVERSE TARGET

The exact shell-adapted transverse witness projections are

$$\boxed{c_E \propto \frac{1}{2} [\text{Tr}(\Sigma_1 \mathcal{S}_1) + \text{Tr}(\Sigma_2 \mathcal{S}_2)],} \tag{401}$$

$$\boxed{c_O \propto \frac{1}{2} [\text{Tr}(\Sigma_2 \mathcal{S}_1) - \text{Tr}(\Sigma_1 \mathcal{S}_2)].} \tag{402}$$

The reduced witness coefficients are then

$$\boxed{\alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}, \quad I_{\theta} \neq 0.} \tag{403}$$

Thus the transverse question is exactly whether the full microscopic theory generates a nonzero coefficient in at least one of these two shell-adapted transverse channels.

CLXXIV. LEVEL A: HONEST MICROSCOPIC FRONTIER

The first fact that must be fixed is negative:

$$\boxed{\text{the naive local real Hermitian } JK \text{ bilinear vanishes.}} \quad (404)$$

Indeed, in the minimal shell-adapted form,

$$J \sim j \Sigma_3, \quad K \sim k_E \Sigma_1 + k_O \Sigma_2,$$

so the simplest Hermitian bilinear reduces to

$$\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = k_E \{\Sigma_3, \Sigma_1\} + k_O \{\Sigma_3, \Sigma_2\} = 0.$$

Therefore the genuine microscopic transverse frontier is not the undifferentiated local JK term, but the derivative-resolved shell audit

$$\boxed{\partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p)|_{p=0}, \quad \partial_{p_\nu} \widehat{\Sigma}_{KK}(G_*; p)|_{p=0}.} \quad (405)$$

a. Checklist A.

1. Do *not* use the naive statement $JK \Rightarrow$ transverse automatically.
2. Construct the shell-derivative blocks

$$\partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_\nu} \widehat{\Sigma}_{KK}(G_*; p)|_0.$$

3. Pauli-decompose their shell matrices.
4. If both derivative blocks have zero Σ_1, Σ_2 projections, Gate 3 fails immediately.

CLXXV. LEVEL B: EXACT WITNESS DECOMPOSITION

After Pauli decomposition,

$$c_E = c_E^{(JK)} + c_E^{(KK)}, \quad c_O = c_O^{(JK)} + c_O^{(KK)}, \quad (406)$$

since pure JJ is axial only and does not contribute to the transverse witness pair.

Hence the exact shell-level transverse criterion is

$$\boxed{(c_E, c_O) \neq (0, 0)}. \quad (407)$$

Equivalently,

$$\boxed{(\alpha, \beta) \neq (0, 0)}. \quad (408)$$

a. Checklist B.

1. Compute the shell-derivative Pauli coefficients in the JK and KK sectors.
2. Form

$$c_E = c_E^{(JK)} + c_E^{(KK)}, \quad c_O = c_O^{(JK)} + c_O^{(KK)}.$$

3. If both vanish, Gate 3 fails.
4. If at least one is nonzero, proceed.

CLXXVI. LEVEL C: MINIMAL NONVANISHING MECHANISMS

The transverse sector can survive only if at least one of the following is nontrivial:

$$\boxed{\text{(a) derivative on the shell phase,}} \quad (409)$$

$$(\partial_{p_\nu} \hat{\mathcal{J}})(G_*; 0) \neq 0 \quad \text{or} \quad (\partial_{p_\nu} \hat{\mathcal{K}})(G_*; 0) \neq 0;$$

$$\boxed{\text{(b) nontrivial internal-index contraction,}} \quad (410)$$

meaning that the mediator/internal contraction distinguishes the Σ_1, Σ_2 channels;

$$\boxed{\text{(c) nontrivial complex intervalley phase data,}} \quad (411)$$

so that the $Q = G_*$ harmonic does not collapse to a pure anticommutator;

$$\boxed{\text{(d) direct } KK \text{ transverse piece.}} \quad (412)$$

These are not optional interpretations; they are the minimal ways to bypass the naive local anticommutation obstruction.

a. Checklist C.

1. Check whether the shell derivative acts nontrivially on the G_* -harmonic.
2. Check whether the internal contraction over mediator labels is genuinely anisotropic in the Σ_1, Σ_2 plane.
3. Check whether intervalley phase data are complex/nontrivial.
4. Check whether the KK block already contains a direct transverse Pauli component.
5. If none of these mechanisms is present, Gate 3 fails.

CLXXVII. LEVEL D: FOLDED-BRAGG COMPRESSED TEST

Inside the explicit folded-Bragg Class II branch, the derivative bookkeeping compresses to the exact Fourier convolution

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q').} \quad (413)$$

In the minimal first-shell-dominant channel, this reduces to

$$\hat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*}(C_\alpha \cdot G_*), \quad \alpha = 1, 2. \quad (414)$$

Therefore the practical explicit sufficient test becomes

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (415)$$

Under (415), the explicit Class II branch realizes

$$\boxed{(\alpha, \beta) \neq (0, 0).} \quad (416)$$

a. Checklist D.

1. Compute the first-shell Fourier amplitude A_{G_*} .
2. Compute the projected vectors C_1, C_2 .
3. Evaluate

$$A_{G_*}(C_1 \cdot G_*), \quad A_{G_*}(C_2 \cdot G_*).$$

4. If both vanish, Gate 3 fails on the working branch.
5. If at least one is nonzero, Gate 3 survives on the working branch.

CLXXVIII. LEVEL E: FINAL BLOCH CALIBRATION

Even after the transport-side transverse seed survives, the actual physical coefficients still carry the Bloch overlap factors:

$$\boxed{\alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (417)$$

Therefore the final calibration condition is

$$\boxed{(C_E, C_O) \neq (0, 0).} \quad (418)$$

Using the normalized E - and O -unit inputs,

$$V_{(E)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} E^\mu{}_a \sigma_a, \quad V_{(O)}^\mu(\xi) = \frac{\theta'(\xi)}{I_\theta} J^\mu{}_a \sigma_a,$$

the exact calibration overlaps are

$$\boxed{C_E = \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2),} \quad (419)$$

$$\boxed{C_O = \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2).} \quad (420)$$

Thus Gate 3 completely passes iff

$$\boxed{(c_E, c_O) \neq (0, 0) \quad \text{and} \quad (C_E, C_O) \neq (0, 0).} \quad (421)$$

a. Checklist E.

1. Build the normalized E - and O -unit inputs.
2. Compute the folded-Bragg overlap matrices $M_{(E)}^1, M_{(E)}^2$ and $M_{(O)}^1, M_{(O)}^2$.
3. Evaluate

$$C_E = \frac{1}{4}(\sigma_1 M_{(E)}^1 + \sigma_2 M_{(E)}^2), \quad C_O = \frac{1}{4}(\sigma_2 M_{(O)}^1 - \sigma_1 M_{(O)}^2).$$

4. If both vanish, Gate 3 fails at the calibration layer.
5. If at least one is nonzero, Gate 3 passes completely.

CLXXIX. FINAL GATE-3 DECISION TREE

The complete Gate-3 decision tree is:

- a. *Step 1.* Reject the naive undifferentiated local JK criterion.
- b. *Step 2.* Compute the derivative-resolved shell blocks

$$\partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_\nu} \widehat{\Sigma}_{KK}(G_*; p)|_0.$$

If both have zero transverse Pauli projections, reject.

- c. *Step 3.* Compute the shell witness pair

$$(c_E, c_O).$$

If both vanish, reject.

- d. *Step 4.* On the explicit folded-Bragg branch, evaluate

$$A_{G_*}(C_1 \cdot G_*), \quad A_{G_*}(C_2 \cdot G_*).$$

If both vanish, reject on the working branch.

- e. *Step 5.* Compute the Bloch calibration overlaps

$$(C_E, C_O).$$

If both vanish, reject at the calibration layer.

- f. *Final verdict.* Only if all previous steps pass do we conclude

$$\boxed{(\alpha, \beta) \neq (0, 0)}. \tag{422}$$

CLXXX. STATUS

At the present stage, Gate 3 has two distinct logical statuses:

1. as a working-branch acceptance problem, it is already reduced to finite explicit computations;
2. as a fully unconditional microscopic theorem, it still carries the honest derivative-resolved JK/KK caveat.

Therefore Gate 3 is:

$$\boxed{\text{finite computation left on the working branch, with a structural caveat for unconditional closure.}}$$

CLXXXI. FINAL SUMMARY

Gate 3 asks whether the transverse witness pair truly survives:

$$\boxed{(\alpha, \beta) \neq (0, 0)}.$$

The exact shell-level transverse coefficients are

$$c_E = c_E^{(JK)} + c_E^{(KK)}, \quad c_O = c_O^{(JK)} + c_O^{(KK)},$$

and the reduced dictionary is

$$\boxed{\alpha = c_E I_\theta, \quad \beta = c_O I_\theta.}$$

However, the naive local real Hermitian JK bilinear vanishes:

$$\boxed{\text{naive local } JK = 0 \quad \text{by} \quad \{\Sigma_3, \Sigma_1\} = \{\Sigma_3, \Sigma_2\} = 0.}$$

Therefore the honest microscopic frontier is

$$\boxed{\partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_\nu} \widehat{\Sigma}_{KK}(G_*; p)|_0.}$$

The minimal nonvanishing mechanisms are:

(a) derivative on the shell phase, (b) nontrivial internal contraction, (c) nontrivial complex intervalley phase, (d) direct K

Inside the explicit folded-Bragg Class II branch, this compresses to the finite practical test

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.$$

After transport-side survival, the final Bloch calibration layer is

$$\alpha = C_E g_E, \quad \beta = C_O g_O,$$

so the last acceptance condition is

$$(C_E, C_O) \neq (0, 0).$$

Therefore the complete Gate-3 pass condition is

$$(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0),$$

with the first-shell explicit sufficient version

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.$$

Thus Gate 3 is:

$$\text{finite computation left on the explicit working branch,}$$

but also

$$\text{still carrying the honest derivative-resolved } JK/KK \text{ caveat as an unconditional microscopic statement.}$$

CLXXXII. GATE 4 CHECKLIST: ACTUAL MASS-PROTECTION AUDIT

CLXXXIII. PURPOSE

Gate 4 asks whether the current refined locked + Class II working branch is protected against all symmetry-allowed constant Dirac masses.

The exact question is:

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\} ?$$

At the present stage this is no longer a conceptual emergence problem. It is a finite residual-symmetry audit on the isolated doubled shell block.

CLXXXIV. EXACT MASS-SPACE CLASSIFICATION

Once the isolated doubled shell carrier exists, the local Dirac block has the form

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i). \tag{423}$$

The full little-group-scalar constant Hermitian operator space is

$$K = a_0 \mathbf{1}_4 + a_3 \tau_z \otimes \mathbf{1}_2 + m_1 \tau_x \otimes \mathbf{1}_2 + m_2 \tau_y \otimes \mathbf{1}_2. \tag{424}$$

Among these, the genuine constant Dirac-mass space is exactly

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}. \quad (425)$$

Therefore the exact mass-protection problem is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}, \quad (426)$$

where

$$A_{\text{res}} = \{ K \in \text{Herm}(4) : U(g) K U(g)^{-1} = K \ \forall g \in G_{\text{res}} \}. \quad (427)$$

CLXXXV. LEVEL A: REJECT INSUFFICIENT SHORTCUTS

Two common shortcuts are *not* sufficient:

$$\text{little-group scalarity only} \implies \text{not enough}, \quad (428)$$

because both mass generators remain allowed;

$$\text{opposite-valley matching only} \implies \text{not enough}. \quad (429)$$

Opposite-valley matching may fix doubling,

$$h_-(p) = -h_+(-p),$$

but it does not remove

$$\tau_x \otimes \mathbf{1}_2, \quad \tau_y \otimes \mathbf{1}_2.$$

a. Checklist A.

1. Do *not* count little-group scalarity as a proof of mass protection.
2. Do *not* count opposite-valley matching as a proof of mass protection.
3. Start only from the exact criterion (426).

CLXXXVI. LEVEL B: SUFFICIENT RESIDUAL SYMMETRY MECHANISMS

A sufficient mechanism is exact residual valley $U(1)_V$. Define

$$Q_V := \tau_z \otimes \mathbf{1}_2, \quad U_\theta := e^{i\theta Q_V/2}. \quad (430)$$

Since

$$[Q_V, H_D(p)] = 0, \quad (431)$$

the doubled kinetic block is compatible with residual valley symmetry.

Now

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos \theta (\tau_x \otimes \mathbf{1}_2) + \sin \theta (\tau_y \otimes \mathbf{1}_2), \quad (432)$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin \theta (\tau_x \otimes \mathbf{1}_2) + \cos \theta (\tau_y \otimes \mathbf{1}_2). \quad (433)$$

Hence no nonzero element of $\mathcal{M}_D$ is invariant under all θ , so

$$\boxed{A_{U(1)_V} \cap \mathcal{M}_D = \{0\}}. \quad (434)$$

The weaker remnant

$$\boxed{S_V := \tau_z \otimes \mathbf{1}_2, \quad S_V^2 = \mathbf{1}_4} \quad (435)$$

already suffices, because

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \quad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \quad (436)$$

Therefore

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M}_D = \{0\}}. \quad (437)$$

a. Checklist B.

1. Determine whether the isolated doubled shell block preserves exact Q_V .
2. If exact $U(1)_V$ survives, Gate 4 passes immediately.
3. If only the discrete remnant S_V survives, Gate 4 still passes.
4. If neither survives, continue to Level C.

CLXXXVII. LEVEL C: EXPLICIT RESIDUAL-ALGEBRA AUDIT

If exact $U(1)_V$ or $\mathbb{Z}_2^V$ is not manifest, one must compute the full residual constant operator algebra

$$A_{\text{res}}$$

directly on the isolated doubled block.

The exact audit is:

1. list all residual symmetries of the corrected 4×4 shell block;
2. compute the constant Hermitian operators commuting with all of them;
3. intersect the resulting algebra with $\mathcal{M}_D$.

If

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

Gate 4 passes. If not, Gate 4 fails.

a. Checklist C.

1. Extract the exact residual group G_{res} of the corrected isolated shell block.
2. Compute

$$A_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K, \forall g \in G_{\text{res}}\}.$$

3. Compute the intersection

$$A_{\text{res}} \cap \mathcal{M}_D.$$

4. If the intersection is nonzero, Gate 4 fails.
5. If it is zero, Gate 4 passes.

CLXXXVIII. LEVEL D: REMOTE-SECTOR COMPATIBILITY AUDIT

Mass protection must survive the remote-sector elimination as well. So it is not enough that the principal kinetic block preserve the residual symmetry; the full corrected microscopic constant piece must preserve it too.

Operationally, one must verify that the constant corrected block and the remote self-energy preserve the same residual generator. In the simplest theorem-level form, this is the requirement that the full corrected constant sector remain Q_V -invariant.

Thus Gate 4 is not only a bare-shell test. It is a test of the corrected isolated block after Schur-complement elimination.

a. Checklist D.

1. Compute the constant corrected shell block δH_{const} .
2. Compute the zero-momentum self-energy $\Sigma(0,0)$ from remote-sector elimination.
3. Check whether the same residual symmetry generator is preserved by both.
4. If the corrected constant sector breaks the protecting symmetry, Gate 4 fails.
5. If the protecting symmetry survives the correction, Gate 4 passes.

CLXXXIX. LEVEL E: ROLE OF CLASS II

The canonical minimal Class II correction is gradient-linear. Therefore it renormalizes derivative shell coefficients rather than directly inserting an independent constant Dirac mass.

Hence:

$$\boxed{\text{Class II is compatible with mass protection, but is not by itself a proof of it.}} \quad (438)$$

a. Checklist E.

1. Do use Class II compatibility as a consistency check.
2. Do *not* count Class II compatibility as a substitute for the residual-symmetry audit.

CXC. FINAL GATE-4 DECISION TREE

The complete Gate-4 decision tree is:

- a. *Step 1.* Construct the isolated doubled shell block and identify the exact mass space $\mathcal{M}_D$.
- b. *Step 2.* Test whether exact residual $U(1)_V$ survives. If yes, accept Gate 4 immediately.
- c. *Step 3.* If not, test whether exact residual $\mathbb{Z}_2^V$ survives. If yes, accept Gate 4 immediately.
- d. *Step 4.* If neither symmetry is manifest, compute the residual constant algebra A_{res} . If

$$A_{\text{res}} \cap \mathcal{M}_D \neq \{0\},$$

reject: a symmetry-allowed constant Dirac mass remains.

e. *Step 5.* Check that the same protecting symmetry survives the corrected constant block after remote-sector elimination. If it fails at this level, reject.

f. *Final verdict.* Only if all previous steps pass do we conclude

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}}, \quad (439)$$

hence the branch is genuinely mass-protected.

CXCI. HONEST STATUS

At the present stage, Gate 4 is theorem-level closed. The only remaining task is the explicit microscopic symmetry audit of the actual corrected refined locked shell block.

Therefore Gate 4 is:

$$\boxed{\text{finite computation left.}}$$

CXCII. FINAL SUMMARY

Gate 4 asks whether the refined locked + Class II branch is protected against all constant Dirac masses. For the isolated doubled shell block

$$H_D(p) = \sum_{i=1}^3 v_i p_i (\tau_z \otimes \sigma_i),$$

the exact constant Dirac-mass space is

$$\mathcal{M}_D = \text{span}_{\mathbb{R}} \{ \tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2 \}.$$

Hence the exact Gate-4 criterion is

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\},$$

where

$$A_{\text{res}} = \{ K \in \text{Herm}(4) : U(g) K U(g)^{-1} = K \ \forall g \in G_{\text{res}} \}.$$

Two shortcuts are insufficient:

$$\text{little-group scalarity only} \not\Rightarrow \text{mass protection},$$

$$\text{opposite-valley matching only} \not\Rightarrow \text{mass protection}.$$

A sufficient mechanism is exact residual valley symmetry:

$$Q_V = \tau_z \otimes \mathbf{1}_2, \quad U_\theta = e^{i\theta Q_V/2}.$$

Under exact $U(1)_V$, the two mass generators rotate as a charged doublet, so

$$A_{U(1)_V} \cap \mathcal{M}_D = \{0\}.$$

The weaker remnant

$$S_V = \tau_z \otimes \mathbf{1}_2$$

already suffices, because both mass generators are odd under it:

$$A_{\mathbb{Z}_2^V} \cap \mathcal{M}_D = \{0\}.$$

Therefore the complete Gate-4 checklist is:

1. identify the isolated doubled shell block and the exact mass space $\mathcal{M}_D$;
2. test whether exact residual $U(1)_V$ survives;
3. if not, test whether exact residual $\mathbb{Z}_2^V$ survives;
4. if neither is manifest, compute the full residual constant algebra A_{res} ;
5. check whether the protecting symmetry survives the corrected constant block after remote-sector elimination.

Only if

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}$$

holds for the corrected isolated shell block does Gate 4 pass.

Class II remains only

$$\text{compatible with mass protection, but not by itself a proof of it.}$$

Thus Gate 4 is not conceptually open. It is a finite residual-symmetry audit on the actual corrected refined locked shell block.

CXCIII. INTEGRATED EXECUTION TABLE FOR GATES 2–4

CXCIV. PURPOSE

We now combine the three actual branch-acceptance gates into one operational checklist. The goal is to make the current refined locked + Class II program executable in the following sense:

$$\text{input} \longrightarrow \text{compute} \longrightarrow \text{pass/fail} \longrightarrow \text{failure meaning.}$$

The three gates are:

$$\text{Gate 2: longitudinal survival,} \quad \text{Gate 3: transverse survival,} \quad \text{Gate 4: mass protection.}$$

CXCV. FIXED EXTRACTION LAYER

The shell-extracted reduced coefficients are read from the projected low-valley matrices by

$$\boxed{\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel})}, \quad (440)$$

$$\boxed{\alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2)}. \quad (441)$$

The exact normalization layer is

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O}. \quad (442)$$

Therefore Gates 2 and 3 each have two nested layers:

1. transport-side survival (g -side),
2. Bloch-frame calibration survival (C -side).

CXCVI. INTEGRATED GATE 2–4 TABLE

Gate	Input	Output
G2	microscopic transport basis $\{E_A\}$, shell target $T^{(\parallel)}$, low-mode profiles $f_{\pm}$	$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j$
G3	derivative-resolved shell blocks $\partial_{p_{\nu}} \hat{\Sigma}_{JK} _0, \partial_{p_{\nu}} \hat{\Sigma}_{KK} _0$, first-shell data A_{G_*}, C_1, C_2	$c_E^{(JK)}, c_O^{(JK)}, c_E^{(KK)}, c_O^{(KK)}, (c_E, c_O)$
G4	isolated doubled shell block, residual symmetry generators, corrected constant block	$\mathcal{M}_D = \text{span}\{\tau_x \otimes \mathbf{1}_2,$

CXCVII. GATE 2: EXPLICIT EXECUTION ORDER

The exact basis-level longitudinal criterion is

$$\boxed{\exists A : \ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j \neq 0.} \quad (443)$$

This is equivalent to the shell-seed condition

$$\boxed{c_{\parallel}^{(\text{mic})} \neq 0.} \quad (444)$$

At the graded/radial level the same survival question is

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.} \quad (445)$$

Finally, the Bloch-frame calibration layer requires

$$\boxed{C_c \neq 0.} \quad (446)$$

Hence Gate 2 passes iff

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.} \quad (447)$$

CXCVIII. GATE 3: EXPLICIT EXECUTION ORDER

The honest microscopic frontier is not the naive local JK term, but the derivative-resolved audit

$$\boxed{\partial_{p_\nu} \widehat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_\nu} \widehat{\Sigma}_{KK}(G_*; p)|_0.} \quad (448)$$

The shell witness pair is

$$\boxed{c_E = c_E^{(JK)} + c_E^{(KK)}, \quad c_O = c_O^{(JK)} + c_O^{(KK)}. \quad (449)}$$

Hence the exact shell-side pass condition is

$$\boxed{(c_E, c_O) \neq (0, 0).} \quad (450)$$

Inside the explicit folded-Bragg Class II branch, the practical compressed test is

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (451)$$

Finally, the Bloch-frame calibration layer requires

$$\boxed{(C_E, C_O) \neq (0, 0).} \quad (452)$$

Thus Gate 3 passes on the working branch iff

$$\boxed{(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0),} \quad (453)$$

with the first-shell overlap test as the practical sufficient criterion.

CXCIX. GATE 4: EXPLICIT EXECUTION ORDER

The exact constant Dirac-mass space is

$$\boxed{\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.} \quad (454)$$

The exact mass-protection condition is

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (455)$$

Exact residual valley $U(1)_V$ is sufficient:

$$Q_V = \tau_z \otimes \mathbf{1}_2, \quad U_\theta = e^{i\theta Q_V/2}, \quad A_{U(1)_V} \cap \mathcal{M}_D = \{0\}. \quad (456)$$

Even exact residual $\mathbb{Z}_2^V$ is already sufficient:

$$S_V = \tau_z \otimes \mathbf{1}_2, \quad A_{\mathbb{Z}_2^V} \cap \mathcal{M}_D = \{0\}. \quad (457)$$

Hence Gate 4 passes iff either:

1. exact residual $U(1)_V$ survives,
2. or exact residual $\mathbb{Z}_2^V$ survives,
3. or, more generally, explicit computation shows $A_{\text{res}} \cap \mathcal{M}_D = \{0\}$.

CC. INTEGRATED ACCEPTANCE LOGIC

The branch is accepted at the carrier level iff Gates 2, 3, and 4 all pass:

$$\boxed{\text{G2 passes} \wedge \text{G3 passes} \wedge \text{G4 passes.}} \quad (458)$$

Equivalently:

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (459)$$

Only after this carrier-level acceptance does it become meaningful to proceed to the remote-gap support check in the same parameter regime and then to the one-defect/one-mode criterion.

CCI. FINAL SUMMARY

The carrier-level branch acceptance problem is completely reduced to the integrated Gate 2–4 audit. The fixed shell-extraction layer is

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{4}(\sigma_1 M_1 + \sigma_2 M_2), \quad \beta = \frac{1}{4}(\sigma_2 M_1 - \sigma_1 M_2),$$

with the exact normalization law

$$\lambda_{\parallel} = C_c g_c, \quad \alpha = C_E g_E, \quad \beta = C_O g_O.$$

a. Gate 2 (longitudinal survival). Compute

$$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j, \quad c_{\parallel}^{(\text{mic})}, \quad J_{\parallel}^{(3)}, \quad C_c.$$

Pass iff

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.}$$

b. Gate 3 (transverse survival). Do not use the naive local JK shortcut. Instead compute

$$\partial_{p_{\nu}} \hat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_{\nu}} \hat{\Sigma}_{KK}(G_*; p)|_0, \quad (c_E, c_O), \quad (C_E, C_O).$$

On the explicit folded-Bragg working branch, use the practical compressed test

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.$$

Pass iff

$$\boxed{(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0).}$$

c. Gate 4 (mass protection). Compute

$$\mathcal{M}_D = \text{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}, \quad A_{\text{res}}, \quad A_{\text{res}} \cap \mathcal{M}_D.$$

Pass iff

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.}$$

Exact residual valley $U(1)_V$ or already $\mathbb{Z}_2^V$ is sufficient. Class II is compatible only, not itself a proof.

d. Integrated carrier acceptance. The current refined locked + Class II branch is accepted as a viable microscopic massless Dirac carrier iff

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad A_{\text{res}} \cap \mathcal{M}_D = \{0\}.}$$

Thus the remaining frontier is no longer conceptual. It is the explicit finite evaluation of the Gate 2–4 checklist on the actual corrected microscopic Bloch basis.

CCII. GATE 2 ACTUAL INPUT TABLE: EXECUTABLE LONGITUDINAL SURVIVAL AUDIT

CCIII. GOAL

The actual remaining microscopic longitudinal question is not conceptual. It is the explicit finite audit of whether the primitive longitudinal shell seed survives through all four layers:

$$\ell_{\parallel A} \longrightarrow c_{\parallel}^{(\text{mic})} \longrightarrow J_{\parallel}^{(3)} \longrightarrow C_c.$$

The final Gate-2 objective is

$$\boxed{\lambda_{\parallel} \neq 0.} \quad (460)$$

CCIV. PRIMITIVE LONGITUDINAL TARGET

The canonical primitive longitudinal seed target is

$$\boxed{B_{\parallel} : \quad N_{ij}^3 = \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = N_{ij}^2 = 0.} \quad (461)$$

Equivalently, the shell covector is

$$\boxed{T_{ija}^{(\parallel)} = \hat{n}_i \hat{n}_j \delta_{a3}.} \quad (462)$$

Inside the explicit seed space,

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3},} \quad (463)$$

with $I_{\theta^3} \neq 0$. After Bloch calibration,

$$\boxed{\lambda_{\parallel} = C_c g_c.} \quad (464)$$

Thus Gate 2 requires both a nonzero transport-side longitudinal seed and a nonzero Bloch-frame calibration overlap.

CCV. LEVEL A: BASIS-LEVEL AUDIT

Let

$$V_{\text{TECT}} \subset \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the actual microscopic transport space, and let

$$\{E_A\}_{A=1}^d$$

be any basis.

Define the longitudinal overlap coefficients

$$\boxed{\ell_{\parallel A} := \langle T^{(\parallel)}, E_A \rangle = E_{A,ij}^3 \hat{n}_i \hat{n}_j.} \quad (465)$$

Then the exact basis-level criterion is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0.} \quad (466)$$

a. Input.

$$G_*, \quad \hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\}, \quad \{E_A\}.$$

b. *Compute.*

$$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j \quad \text{for all } A.$$

c. *Pass.* At least one $\ell_{\parallel A} \neq 0$.

d. *Fail means.* The microscopic transport space contains no admissible longitudinal seed direction.

CCVI. LEVEL B: SHELL-SEED PROJECTION AUDIT

Project the actual induced shell tensor onto the explicit seed space:

$$P_{\text{seed}} N_{\text{mic}} = c_{\parallel}^{(\text{mic})} B_{\parallel} + c_I^{(\text{mic})} B_I + c_J^{(\text{mic})} B_J. \quad (467)$$

This is equivalent to the basis test:

$$\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0. \quad (468)$$

a. *Input.*

$$N_{\text{mic}}, \quad \{B_{\parallel}, B_I, B_J\}.$$

b. *Compute.* The seed coefficient $c_{\parallel}^{(\text{mic})}$.

c. *Pass.*

$$c_{\parallel}^{(\text{mic})} \neq 0.$$

d. *Fail means.* The full shell projection annihilates the primitive longitudinal channel.

CCVII. LEVEL C: GRADED/RADIAL OVERLAP AUDIT

At the projected defect-profile level, the same survival condition becomes the σ_3 -graded overlap

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi). \quad (469)$$

a. *Input.*

$$f_+(\xi), \quad f_-(\xi), \quad \theta'(\xi), \quad s_3(\xi).$$

b. *Compute.* The scalar integral $J_{\parallel}^{(3)}$.

c. *Pass.*

$$J_{\parallel}^{(3)} \neq 0.$$

d. *Fail means.* The defect/envelope overlap cancels the surviving longitudinal seed.

A simple sufficient condition is:

$$f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi) \text{ has definite nonzero sign on a set of positive measure.} \quad (470)$$

CCVIII. LEVEL D: FINAL BLOCH CALIBRATION AUDIT

Even after the longitudinal seed survives on the transport side, the physical coefficient still depends on the Bloch-frame overlap

$$C_c = \frac{1}{2} (\sigma_3 M_{(c)}^{\parallel}). \quad (471)$$

Equivalently, with the normalized c -unit input,

$$C_c = \frac{1}{2I_{\theta^3}} \left[\sigma_3 \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_+^{\dagger}(x) \theta'(\xi)^3 \sigma_3 U_-(x) \right]. \quad (472)$$

a. *Input.*

$$V_{(c)}^\mu(\xi) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} n^\mu \sigma_3, \quad \Psi_a^\pm, \quad M_{(c)}^\parallel.$$

b. *Compute.*

$$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^\parallel).$$

c. *Pass.*

$$C_c \neq 0.$$

d. *Fail means.* The longitudinal seed survives transport-side projection but dies in the actual Bloch frame.

CCIX. INTEGRATED GATE-2 PASS CONDITION

Putting all four layers together, Gate 2 passes iff

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.} \quad (473)$$

Only then do we conclude

$$\boxed{\lambda_{\parallel} \neq 0.} \quad (474)$$

CCX. ACTUAL INPUT TABLE

Layer	Input	Compute	Pass condition	Failure meaning
A: basis audit	$G_*, \hat{n}, \{E_A\}$	$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j$	$\exists A : \ell_{\parallel A} \neq 0$	no longitudinal basis direction
B: seed projection	$N_{\text{mic}}, \{B_{\parallel}, B_I, B_J\}$	$c_{\parallel}^{(\text{mic})}$	$c_{\parallel}^{(\text{mic})} \neq 0$	shell projection kills $B_{\parallel}$
C: graded overlap	$f_{\pm}, \theta', s_3$	$J_{\parallel}^{(3)} = \int d\xi f_+ f_- \theta' s_3$	$J_{\parallel}^{(3)} \neq 0$	radial/defect cancellation
D: Bloch calibration	$V_{(c)}^\mu, \Psi_a^\pm, M_{(c)}^\parallel$	$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^\parallel)$	$C_c \neq 0$	Bloch-frame overlap kills longitudinal seed

CCXI. HONEST STATUS

Every layer of Gate 2 is already structurally fixed. Therefore Gate 2 is not conceptually open. It is an exact finite executable audit of the four quantities

$$\ell_{\parallel A}, \quad c_{\parallel}^{(\text{mic})}, \quad J_{\parallel}^{(3)}, \quad C_c.$$

CCXII. FINAL SUMMARY

Gate 2 is the longitudinal survival audit. Its purpose is to decide whether the primitive longitudinal shell seed

$$B_{\parallel} : \quad N_{ij}^3 = \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = N_{ij}^2 = 0$$

truly survives in the full refined locked microscopic realization.

The exact audit has four layers:

$$\ell_{\parallel A} \longrightarrow c_{\parallel}^{(\text{mic})} \longrightarrow J_{\parallel}^{(3)} \longrightarrow C_c.$$

The basis-level criterion is

$$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j, \quad \exists A : \ell_{\parallel A} \neq 0.$$

This is equivalent to the shell-seed condition

$$c_{\parallel}^{(\text{mic})} \neq 0.$$

At the graded/radial level the same survival question is

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.$$

Finally, after Bloch calibration,

$$\lambda_{\parallel} = C_c g_c, \quad C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}),$$

so the final calibration condition is

$$C_c \neq 0.$$

Therefore the complete Gate-2 pass condition is

$$\exists A : \ell_{\parallel A} \neq 0 \iff c_{\parallel}^{(\text{mic})} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.$$

Only then can one conclude

$$\lambda_{\parallel} \neq 0.$$

Thus Gate 2 is not conceptually open. It is an exact finite executable audit of

$$\ell_{\parallel A}, \quad c_{\parallel}^{(\text{mic})}, \quad J_{\parallel}^{(3)}, \quad C_c.$$

CCXIII. GATE 2 ACTUAL BRANCH INPUT SHEET

CCXIV. SHORT EXPLANATION

At the current strongest honest stage, the transverse carrier is already structurally realized, and the only remaining carrier-level microscopic question is the survival of the primitive longitudinal seed. This is no longer conceptual. It is a finite executable audit of the longitudinal branch data.

CCXV. 1. FIXED SHELL DATA

The Gate-2 audit begins by fixing:

$$G_* \neq 0, \quad \hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal.} \quad (475)$$

The canonical primitive longitudinal target is

$$B_{\parallel} : \quad N_{ij}^3 = \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = N_{ij}^2 = 0. \quad (476)$$

Equivalently, the shell covector is

$$T_{ija}^{(\parallel)} = \hat{n}_i \hat{n}_j \delta_{a3}. \quad (477)$$

CCXVI. 2. BASIS-LEVEL INPUT

Let

$$V_{\text{TECT}} \subset \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the actual microscopic transport space and let

$$\{E_A\}_{A=1}^d$$

be any basis.

Write

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A, \quad R_A \perp \text{span}\{B_{\parallel}, B_I, B_J\}. \quad (478)$$

Then the exact longitudinal overlap is

$$\ell_{\parallel A} = \langle T^{(\parallel)}, E_A \rangle = E_{A,ij}^3 \hat{n}_i \hat{n}_j = \mu_{\parallel} a_A, \quad \mu_{\parallel} \neq 0. \quad (479)$$

Thus the first executable Gate-2 criterion is simply

$$\exists A : a_A \neq 0 \iff \exists A : \ell_{\parallel A} \neq 0. \quad (480)$$

a. Actual input list:

$$\{E_{A,ij}^a\}_{A=1}^d, \quad \hat{n}_i, \quad \ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j.$$

CCXVII. 3. PRIMITIVE LONGITUDINAL SEED INPUT

Inside the explicit seed construction, the primitive longitudinal operator is

$$V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3. \quad (481)$$

Its intrinsic seed-space witness is

$$\lambda_{\parallel}^{\text{seed}} = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (482)$$

Hence, provided

$$\theta_0 \kappa \neq 0,$$

the primitive longitudinal seed itself is nonzero inside the explicit seed representative.

a. Actual input list:

$$\theta_0, \quad \kappa, \quad I_{\theta'^3}.$$

CCXVIII. 4. SHELL-SEED / MEDIATED COEFFICIENT INPUT

There are two practically relevant longitudinal coefficient forms.

(a) direct shell-seed form

Inside the basic seed dictionary,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}. \quad (483)$$

(b) current mediated 2 + 2 working form

Inside the current mediated Class II completion,

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}. \quad (484)$$

This is the strongest current practical working formula for the longitudinal coefficient.

a. Actual input list:

$$c_{\parallel}, \quad \lambda_c, \quad I_{\theta}, \quad I_{\theta'^3}, \quad \mu_X, \nu_X, m, M_X.$$

b. Immediate pass condition:

$$I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0. \quad (485)$$

CCXIX. 5. GRADED/RADIAL OVERLAP INPUT

At the defect-profile level, the same survival question becomes

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi). \quad (486)$$

A simple sufficient condition is:

$$f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi) \text{ has definite nonzero sign on a set of positive measure.} \quad (487)$$

a. Actual input list:

$$f_+(\xi), \quad f_-(\xi), \quad \theta'(\xi), \quad s_3(\xi) \text{ or } \mathfrak{s}_c(\xi).$$

CCXX. 6. FINAL BLOCH CALIBRATION INPUT

After transport-side survival, the physical longitudinal coefficient is

$$\lambda_{\parallel} = C_c g_c. \quad (488)$$

Using the normalized c -unit input,

$$V_{(c)}^{\mu}(\xi) = \frac{\theta'(\xi)^3}{I_{\theta'^3}} n^{\mu} \sigma_3,$$

the Bloch overlap is

$$C_c = \frac{1}{2} (\sigma_3 M_{(c)}^{\parallel}). \quad (489)$$

Equivalently,

$$C_c = \frac{1}{2 I_{\theta'^3}} \left[\sigma_3 \int_{\text{cell}} d^3 x e^{-i G_{*} \cdot x} U_{+}^{\dagger}(x) \theta'(\xi)^3 \sigma_3 U_{-}(x) \right]. \quad (490)$$

a. Actual input list:

$$V_{(c)}^{\mu}, \quad \Psi_a^{\pm}, \quad M_{(c)}^{\parallel}, \quad U_{\pm}(x).$$

CCXXI. 7. FINAL ACTUAL INPUT TABLE

Layer	Actual branch inputs	Compute	Pass test
A: basis audit	$\{E_{A,ij}^a\}, \hat{n}_i$	$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j = \mu_{\parallel} a_A$	$\exists A : a_A \neq 0$
B: seed coefficient	$c_{\parallel}, \lambda_c, I_{\theta'^3}$	$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3} \text{ or } \lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}$	$I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0$
C: graded overlap	$f_{\pm}, \theta', s_3$	$J_{\parallel}^{(3)} = \int d\xi f_+ f_- \theta' s_3$	$J_{\parallel}^{(3)} \neq 0$
D: Bloch calibration	$V_{(c)}^{\mu}, \Psi_a^{\pm}, M_{(c)}^{\parallel}$	$C_c = \frac{1}{2} (\sigma_3 M_{(c)}^{\parallel})$	$C_c \neq 0$

CCXXII. FINAL GATE-2 VERDICT

Therefore the current refined locked + Class II working branch passes Gate 2 iff

$$\boxed{\exists A : a_A \neq 0, \quad I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.} \quad (491)$$

Only then can one conclude

$$\boxed{\lambda_{\parallel} \neq 0.} \quad (492)$$

CCXXIII. FINAL SUMMARY

The honest remaining carrier frontier is the survival of the primitive longitudinal seed. This is already reduced to a finite executable audit.

The basis-level criterion is

$$\boxed{\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j = \mu_{\parallel} a_A, \quad \exists A : a_A \neq 0.}$$

The explicit primitive longitudinal seed is

$$\boxed{V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}} n^{\mu} = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.}$$

Its seed-level overlap is already explicit:

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.}$$

In the strongest current working branch, the longitudinal coefficient is

$$\boxed{\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}.}$$

At the graded/radial level one must also check

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.}$$

Finally, after Bloch calibration,

$$\boxed{\lambda_{\parallel} = C_c g_c, \quad C_c = \frac{1}{2} (\sigma_3 M_{(c)}^{\parallel}),}$$

so the final calibration condition is

$$\boxed{C_c \neq 0.}$$

Therefore the actual Gate-2 pass condition for the current refined locked + Class II branch is

$$\boxed{\exists A : a_A \neq 0, \quad I_\theta \Gamma_X c_\parallel + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad J_\parallel^{(3)} \neq 0, \quad C_c \neq 0.}$$

Only then can one conclude

$$\boxed{\lambda_\parallel \neq 0.}$$

CCXXIV. GATE 2 STATUS SHEET IN ACTUAL WORKING-BRANCH VARIABLES

CCXXV. GOAL

The longitudinal survival gate is no longer conceptually open. The only question is which variables are already explicit, which have explicit formulas but no inserted branch value yet, and whether any object remains completely undefined.

CCXXVI. 1. PRIMITIVE LONGITUDINAL SEED

The primitive longitudinal seed is already explicit:

$$\boxed{V_{(\parallel)}^\mu(\xi) = \theta'(\xi) (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3.} \quad (493)$$

This belongs to the category:

$$\boxed{\text{already explicit.}}$$

CCXXVII. 2. SEED OVERLAP INTEGRAL

The intrinsic seed overlap is also already explicit:

$$\boxed{\lambda_\parallel^{\text{seed}} = \frac{1}{2} I_{\theta'^3}, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.} \quad (494)$$

Thus $I_{\theta'^3}$ also belongs to:

$$\boxed{\text{already explicit.}}$$

CCXXVIII. 3. BASIS-LEVEL CARRIER COEFFICIENT

For any actual microscopic basis element E_A , one has

$$\boxed{E_A = a_A B_\parallel + b_A B_I + c_A B_J + R_A, \quad R_A \perp \text{span}\{B_\parallel, B_I, B_J\}.} \quad (495)$$

and

$$\boxed{\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j = \mu_\parallel a_A, \quad \mu_\parallel \neq 0.} \quad (496)$$

Therefore

$$\boxed{\exists A : a_A \neq 0 \iff \exists A : \ell_{\parallel A} \neq 0.} \quad (497)$$

The variables a_A and $\ell_{\parallel A}$ belong to:

$$\boxed{\text{formula explicit but value not inserted.}}$$

CCXXIX. 4. SHELL-SEED LONGITUDINAL COEFFICIENT

The transport-side longitudinal seed coefficient is

$$c_{\parallel}^{(\text{mic})} \neq 0 \quad (498)$$

as the sharpened carrier-survival question.

This belongs to:

$$\text{formula explicit but value not inserted.}$$

CCXXX. 5. WORKING MEDIATED COEFFICIENT POINT

In the strongest current working branch,

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}. \quad (499)$$

Thus the actual branch-level longitudinal pass test is

$$I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0. \quad (500)$$

Here:

$$\Gamma_X, \quad c_{\parallel}, \quad \lambda_c$$

belong to

$$\text{formula explicit but value not inserted.}$$

CCXXXI. 6. GRADED/RADIAL OVERLAP

At the projected defect-profile level, the longitudinal survival question is

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi). \quad (501)$$

This also belongs to:

$$\text{formula explicit but value not inserted.}$$

A sufficient sign test is already explicit:

$$f_+(\xi) f_-(\xi) \theta'(\xi)^3 \mathfrak{s}_c(\xi) \text{ has definite nonzero sign on a set of positive measure.} \quad (502)$$

CCXXXII. 7. BLOCH CALIBRATION OVERLAP

After transport-side survival, the physical coefficient is

$$\lambda_{\parallel} = C_c g_c, \quad C_c = \frac{1}{2} (\sigma_3 M_{(c)}^{\parallel}). \quad (503)$$

Equivalently,

$$C_c = \frac{1}{2I_{\theta'^3}} \left[\sigma_3 \int_{\text{cell}} d^3x e^{-iG_* \cdot x} U_+^{\dagger}(x) \theta'(\xi)^3 \sigma_3 U_-(x) \right]. \quad (504)$$

Thus C_c belongs to:

$$\text{formula explicit but value not inserted.}$$

CCXXXIII. 8. ACTUAL STATUS TABLE

Object	Status	Reason
$V_{(\parallel)}^\mu(\xi) = \frac{1}{2}\theta'(\xi)^3 n^\mu \sigma_3$	already explicit	primitive longitudinal seed is fixed explicitly
$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3$	already explicit	overlap integral already evaluated
$a_A, \ell_{\parallel A} = \mu_{\parallel} a_A$	formula explicit but value not inserted	requires the actual microscopic basis $\{E_A\}$
$c_{\parallel}^{(\text{mic})}$	formula explicit but value not inserted	requires the actual seed projection of N_{mic}
$\Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}$	formula explicit but value not inserted	requires actual background prefactors
λ_c	formula explicit but value not inserted	requires actual primitive longitudinal amplitude extr
$J_{\parallel}^{(3)} = \int d\xi f_+ f_- \theta' s_3$	formula explicit but value not inserted	requires $f_{\pm}, s_3$ in the actual defect background
$C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel})$	formula explicit but value not inserted	requires the actual Bloch-frame overlap
completely pending object	none	Gate 2 has no remaining undefined structure; only explicit

CCXXXIV. 9. FINAL ACTUAL GATE-2 PASS CRITERION

Therefore the current refined locked + Class II branch passes Gate 2 iff

$$\boxed{\exists A : a_A \neq 0, \quad I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.} \quad (505)$$

This is no longer conceptually open. It is a pure finite insertion problem.

CCXXXV. FINAL SUMMARY

The primitive longitudinal seed and its intrinsic overlap are already explicit:

$$\boxed{V_{(\parallel)}^\mu(\xi) = \frac{1}{2}\theta'(\xi)^3 n^\mu \sigma_3,}$$

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3.}$$

The basis-level longitudinal audit is

$$\boxed{\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j = \mu_{\parallel} a_A, \quad \exists A : a_A \neq 0.}$$

The strongest current working longitudinal coefficient formula is

$$\boxed{\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}.}$$

The graded/radial survival test is

$$\boxed{J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.}$$

After Bloch calibration, the physical longitudinal coefficient is

$$\lambda_{\parallel} = C_c g_c, \quad C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}).$$

Therefore the actual Gate-2 pass condition for the current refined locked + Class II branch is

$$\exists A : a_A \neq 0, \quad I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.$$

Status classification:

$$V_{(\parallel)}^{\mu}, I_{\theta'^3} \text{ are already explicit;}$$

$$a_A, c_{\parallel}, \Gamma_X, \lambda_c, J_{\parallel}^{(3)}, C_c \text{ have explicit formulas but no inserted branch values yet;}$$

$$\text{there is no completely pending new object in Gate 2.}$$

CCXXXVI. FINAL SUMMARY

The primitive longitudinal seed is already explicit:

$$V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2} \theta'(\xi)^3 n^{\mu} \sigma_3.$$

Its intrinsic overlap is also already explicit:

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

For the actual microscopic transport basis,

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A, \quad R_A \perp \text{span}\{B_{\parallel}, B_I, B_J\},$$

the longitudinal overlap is

$$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j = \mu_{\parallel} a_A.$$

Thus the first carrier question is exactly

$$\exists A : a_A \neq 0.$$

The sharpened longitudinal frontier is

$$c_{\parallel}^{(\text{mic})} \neq 0?$$

and in the strongest current working branch the practical coefficient formula is

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}.$$

At the graded/radial level one must also check

$$J_{\parallel}^{(3)} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) s_3(\xi) \neq 0.$$

Finally, after Bloch calibration,

$$\lambda_{\parallel} = C_c g_c, \quad C_c = \frac{1}{2}(\sigma_3 M_{(c)}^{\parallel}),$$

so the final calibration condition is

$$C_c \neq 0.$$

Status classification:

$$V_{(\parallel)}^{\mu}, I_{\theta'^3} \text{ are already explicit;}$$

$$a_A, \ell_{\parallel A}, c_{\parallel}, \Gamma_X, \lambda_c, J_{\parallel}^{(3)}, C_c \text{ have explicit formulas but no inserted branch values yet;}$$

$$\text{there is no completely pending new structure in Gate 2.}$$

Therefore the actual Gate-2 pass condition is

$$\exists A : a_A \neq 0, \quad I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad J_{\parallel}^{(3)} \neq 0, \quad C_c \neq 0.$$

CCXXXVII. GATE 2-A: ACTUAL CANDIDATE BLOCKS FOR THE PRIMITIVE LONGITUDINAL SEED

CCXXXVIII. SHORT EXPLANATION

The honest remaining carrier question is only the survival of the primitive longitudinal seed. This is already reduced to a finite microscopic audit:

$$\exists A : \ell_{\parallel A} \neq 0.$$

At the blockwise Hessian level, the strongest current statement is sharper:

$$OO \text{ is the natural primitive longitudinal block for } \ell_0^{(3)}.$$

Therefore the correct nonredundant next step is to start the longitudinal audit from the OO block, and only if this fails, extend to the full 9-block sum.

CCXXXIX. 1. STARTING POINT: FULL BLOCK HESSIAN

Let the full background fluctuation field be decomposed as

$$\varphi = (a, \nu, \zeta),$$

where

$$a \sim T, \quad \nu \sim O, \quad \zeta \sim D.$$

Then the full background Hessian has the exact block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}. \quad (506)$$

Each block expands as

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots. \quad (507)$$

Only the first-order transport term $\mathcal{V}_{AB}^\mu$ contributes to the primitive longitudinal audit.

CCXL. 2. BLOCKWISE LONGITUDINAL COEFFICIENTS

For each block $AB \in \{T, O, D\}^2$, reduce the invariant shell transport part to

$$\mathcal{M}_{AB,\text{inv}}^\mu(\theta) = h_{0;AB}(\theta) n^\mu \mathbf{1}_2 + h_{\parallel;AB}(\theta) n^\mu \sigma_3 + h_{E;AB}(\theta) E^\mu{}_a \sigma_a + h_{O;AB}(\theta) J^\mu{}_a \sigma_a. \quad (508)$$

Then the longitudinal block coefficient is extracted by the exact Pauli trace

$$h_{\parallel;AB}(\theta) = \frac{1}{2} [\sigma_3 n_\mu \mathcal{M}_{AB,\text{inv}}^\mu(\theta)]. \quad (509)$$

The exact longitudinal monomial coefficients are

$$\ell_{0;AB}^{(1)} = \lim_{\theta \rightarrow 0} \frac{h_{\parallel;AB}(\theta)}{\theta}, \quad \ell_{0;AB}^{(3)} = \lim_{\theta \rightarrow 0} \frac{h_{\parallel;AB}(\theta)}{\theta^3} \quad (\text{if } \ell_{0;AB}^{(1)} = 0). \quad (510)$$

Summing over all blocks gives

$$\ell_0^{(1)} = \sum_{A,B} \ell_{0;AB}^{(1)}, \quad \ell_0^{(3)} = \sum_{A,B} \ell_{0;AB}^{(3)}. \quad (511)$$

The exact microscopic longitudinal survival criterion is therefore

$$\ell_0^{(1)} \neq 0 \quad \text{or} \quad \ell_0^{(3)} \neq 0 \implies c_{\parallel}^{(\text{mic})} \neq 0. \quad (512)$$

CCXLI. 3. BRANCH A / BRANCH B INTERPRETATION

Identify

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \frac{\lambda_c}{2} = \ell_0^{(3)}. \quad (513)$$

Hence the longitudinal audit splits into two mutually exclusive practical routes:

a. Branch A:

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)} \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (514)$$

b. Branch B:

$$\Lambda_{\parallel}^{(0)} = 0, \quad \ell_0^{(3)} = \frac{\lambda_c}{2} \neq 0 \implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta^3} \neq 0. \quad (515)$$

Thus the actual Gate-2 bottleneck is:

$$\boxed{\text{Does the microscopic branch realize Branch A or Branch B?}} \quad (516)$$

CCXLII. 4. WHY OO IS THE FIRST BLOCK TO AUDIT

At the current sharpest blockwise stage, the natural carrier assignment is

$$\boxed{OO \text{ is the natural primitive longitudinal block for } \ell_0^{(3)}}. \quad (517)$$

The orientational diagonal block has the exact form

$$\boxed{\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y)}, \quad (518)$$

where the last term is the background-induced first-order transport operator.

This makes OO the natural first candidate for carrying the primitive odd singlet $\theta'(\xi)^3 \sigma_3$, i.e. the $\ell_0^{(3)} / \lambda_c$ channel. Therefore the correct nonredundant audit order is:

$$\boxed{\text{first compute } \ell_{0;OO}^{(1)}, \ell_{0;OO}^{(3)}}. \quad (519)$$

CCXLIII. 5. MINIMAL CANDIDATE LIST FOR GATE 2-A

The actual candidate list is:

a. Primary candidate:

$$\boxed{OO \text{ for } \ell_0^{(3)} / \text{Branch B} / \lambda_c.}$$

b. Secondary candidate: all remaining blocks only if the primary candidate fails:

$$\boxed{TT, TO, TD, OT, OD, DT, DO, DD.}$$

Operationally, one computes the blockwise traces

$$\boxed{\ell_{0;AB}^{(1)} = \lim_{\theta \rightarrow 0} \frac{\frac{1}{2}[\sigma_3 n_\mu \mathcal{M}_{AB,\text{inv}}^\mu]}{\theta}, \quad \ell_{0;AB}^{(3)} = \lim_{\theta' \rightarrow 0} \frac{\frac{1}{2}[\sigma_3 n_\mu \mathcal{M}_{AB,\text{inv}}^\mu]}{\theta'^3}}. \quad (520)$$

Then the Gate-2 blockwise decision rule is:

1. compute $(\ell_{0;OO}^{(1)}, \ell_{0;OO}^{(3)})$;
2. if either is nonzero, Gate 2-A passes immediately;
3. if both vanish, compute the remaining eight blocks and test

$$\sum_{A,B} \ell_{0;AB}^{(1)} \neq 0 \quad \text{or} \quad \sum_{A,B} \ell_{0;AB}^{(3)} \neq 0.$$

CCXLIV. 6. RELATION TO THE BASIS-OVERLAP AUDIT

The blockwise audit and the basis-overlap audit are equivalent formulations.

The basis-overlap statement is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0 \iff \exists A : a_A \neq 0.} \quad (521)$$

The blockwise statement is

$$\boxed{\ell_0^{(1)} \neq 0 \quad \text{or} \quad \ell_0^{(3)} \neq 0.} \quad (522)$$

So Gate 2-A can be executed in either language:

- basis language: find one microscopic basis vector with $a_A \neq 0$;
- block language: find one nonzero longitudinal block coefficient.

CCXLV. 7. FINAL PRACTICAL GATE-2-A EXECUTION SHEET

Step	Input	Compute	Pass test
1	$\mathcal{H}_{OO}, \mathcal{T}_{OO}, \mathcal{M}_{OO,\text{inv}}^\mu$	$\ell_{0;OO}^{(1)}, \ell_{0;OO}^{(3)}$	$\ell_{0;OO}^{(1)} \neq 0$ or $\ell_{0;OO}^{(3)} \neq 0$
2	$\mathcal{H}_{AB}, \mathcal{M}_{AB,\text{inv}}^\mu, AB \in \{T, O, D\}^2$	$\ell_{0;AB}^{(1)}, \ell_{0;AB}^{(3)}$	$\sum_{A,B} \ell_{0;AB}^{(1)} \neq 0$ or $\sum_{A,B} \ell_{0;AB}^{(3)} \neq 0$
3	$\{E_A\}$	$\ell_{\parallel A} = E_{A,ij}^3 \hat{n}_i \hat{n}_j$	$\exists A : \ell_{\parallel A} \neq 0$

CCXLVI. HONEST STATUS

The only remaining longitudinal carrier question is exactly this finite blockwise / basis-overlap audit. No new conceptual structure is needed.

At the current strongest explicit stage, the OO block is the natural first carrier to test. If it already yields nonzero $\ell_{0;OO}^{(3)}$, then the primitive longitudinal channel is microscopically realized and Gate 2 is essentially open only at the overlap/calibration layer.

CCXLVII. FINAL SUMMARY

The honest remaining carrier question is only the survival of the primitive longitudinal seed. This is reduced to a finite longitudinal audit:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0}$$

or equivalently

$$\boxed{\ell_0^{(1)} \neq 0 \quad \text{or} \quad \ell_0^{(3)} \neq 0.}$$

The blockwise Hessian audit uses

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix},$$

with first-order transport blocks extracted from

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)} - i\mathcal{V}_{AB}^\mu \partial_{y^\mu} + \dots$$

For each block,

$$\boxed{\ell_{0;AB}^{(1)} = \lim_{\theta \rightarrow 0} \frac{\frac{1}{2}[\sigma_3 n_\mu \mathcal{M}_{AB,\text{inv}}^\mu]}{\theta}, \quad \ell_{0;AB}^{(3)} = \lim_{\theta' \rightarrow 0} \frac{\frac{1}{2}[\sigma_3 n_\mu \mathcal{M}_{AB,\text{inv}}^\mu]}{\theta'^3}.$$

The sharpest current blockwise statement is

$$\boxed{OO \text{ is the natural primitive longitudinal block for } \ell_0^{(3)}.$$

Hence the correct nonredundant Gate-2-A procedure is:

$$\boxed{\text{first compute } \ell_{0;OO}^{(1)}, \ell_{0;OO}^{(3)}.$$

If either is nonzero, the longitudinal primitive channel is microscopically realized.

If both vanish, extend to all 9 blocks and test

$$\sum_{A,B} \ell_{0;AB}^{(1)} \neq 0 \quad \text{or} \quad \sum_{A,B} \ell_{0;AB}^{(3)} \neq 0.$$

This is equivalent to the basis-overlap audit

$$\exists A : a_A \neq 0.$$

Thus Gate 2-A is no longer conceptually open. The immediate next executable task is the explicit *OO*-block longitudinal audit.

CCXLVIII. TERM-BY-TERM LONGITUDINAL AUDIT OF THE *OO* BLOCK

CCXLIX. GOAL

We now perform the actual Gate-2-A audit on the *OO* block. The purpose is to determine, term by term, which pieces of

$$\mathcal{H}_{OO} = -Z_{O,k} \partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y)$$

can contribute to the longitudinal primitive coefficients

$$\ell_{0;OO}^{(1)}, \quad \ell_{0;OO}^{(3)}.$$

The answer will show that in the current explicit library the only genuine leading longitudinal carrier is the primitive Branch-B term inside $\mathcal{T}_{OO}$.

CCL. 1. BLOCKWISE EXTRACTION PRINCIPLE

For each block *AB*, the background Hessian is expanded as

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + O(\partial_y^2). \quad (523)$$

Only the first-order transport operator $\mathcal{V}_{AB}^\mu$ can contribute to the local shell-carrier audit. Therefore the longitudinal *OO*-coefficient is extracted from the invariant tangent reduction

$$\mathcal{M}_{OO,\text{inv}}^\mu(\xi) = h_{0;OO}(\xi) n^\mu \mathbf{1}_2 + h_{\parallel;OO}(\xi) n^\mu \Sigma_3 + h_{E;OO}(\xi) E^\mu{}_a \Sigma_a + h_{O;OO}(\xi) J^\mu{}_a \Sigma_a. \quad (524)$$

The exact longitudinal trace is

$$h_{\parallel;OO}(\xi) = \frac{1}{2 Q_B} (\Sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu(\xi)). \quad (525)$$

Then the two blockwise longitudinal coefficients are

$$\ell_{0;OO}^{(1)} = \lim_{\theta \rightarrow 0} \frac{h_{\parallel;OO}(\xi)}{\theta'(\xi)}, \quad \ell_{0;OO}^{(3)} = \lim_{\theta \rightarrow 0} \frac{h_{\parallel;OO}(\xi) - \ell_{0;OO}^{(1)} \theta'(\xi)}{\theta'(\xi)^3}. \quad (526)$$

Equivalently, if

$$h_{\parallel;OO}(\xi) = \ell_{0;OO}^{(1)} \theta'(\xi) + \ell_{0;OO}^{(3)} \theta'(\xi)^3 + \dots, \quad (527)$$

then $\ell_{0;OO}^{(1)}$ is the Branch-A coefficient and $\ell_{0;OO}^{(3)}$ is the primitive Branch-B coefficient.

CCLI. 2. THE ACTUAL OO BLOCK

The background-dependent orientational block is

$$\boxed{\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y).} \quad (528)$$

We now audit each term separately.

CCLII. 3. KINETIC TERM $-Z_{O,k}\partial^2$

The isotropic kinetic term contributes only a second-order derivative operator. Hence it does not directly produce any first-order worldvolume transport tensor of the form

$$n^\mu \Sigma_3 \partial_{y^\mu}.$$

Therefore:

$$\boxed{-Z_{O,k}\partial^2 \text{ contributes to } \mathcal{H}_{OO}^{(0)} \text{ and higher-derivative pieces, but not directly to } \ell_{0;OO}^{(1)} \text{ or } \ell_{0;OO}^{(3)}.} \quad (529)$$

CCLIII. 4. POTENTIAL TERM $\mathcal{M}_{OO}^{\text{pot}}(\xi)$

The background-dependent potential term is zeroth order in worldvolume derivatives. So it may shift the local mass matrix, but it cannot generate a transport operator carrying one derivative index and one longitudinal shell-tangent Σ_3 structure.

Therefore:

$$\boxed{\mathcal{M}_{OO}^{\text{pot}}(\xi) \text{ cannot directly contribute to } \ell_{0;OO}^{(1)} \text{ or } \ell_{0;OO}^{(3)}.} \quad (530)$$

CCLIV. 5. GRADIENT-ANISOTROPY TERM $Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}$

This is the only nontrivial diagonal term that could, in principle, generate a first-order longitudinal structure. So we must be honest here.

At the operator level, a Branch-A longitudinal contribution would require

$$\boxed{\Lambda_{\parallel}^{(0)} \theta'(\xi) n^\mu \Sigma_3.} \quad (531)$$

But the current no-go theorem for the solved lowest-order complete realization states

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (532)$$

Equivalently, the current complete LO action class contains no additional linear longitudinal tangent carrier of the form $\theta'(\xi) n^\mu \Sigma_3$. Thus Branch A is absent and Branch B is forced. Therefore:

$$\boxed{Y_k \mathcal{K}_{OO}^{\nabla\text{ani}} \text{ does not produce a nonzero } \ell_{0;OO}^{(1)} \text{ in the current solved LO class.}} \quad (533)$$

This is the precise sense in which the diagonal anisotropy term does not rescue Branch A inside the present library.

CCLV. 6. TRANSPORT TERM $\mathcal{T}_{OO}$

This is the only term that is structurally designed to contribute to the first-order transport operator. Hence all genuine leading longitudinal information must sit inside $\mathcal{T}_{OO}$.

The exact tangent first-order ansatz is

$$V^\mu(\xi) = a_0(\xi) n^\mu \Pi_Q + a_3(\xi) n^\mu \Sigma_3 + b_E(\xi) E^\mu{}_a \Sigma_a + b_O(\xi) J^\mu{}_a \Sigma_a + V_{\text{null}}^\mu(\xi). \quad (534)$$

The branch-sensitive longitudinal coefficient is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi). \quad (535)$$

Using (532), this reduces in the current solved lowest-order realization to

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi), \quad \Lambda_{\parallel}^{(0)} = 0. \quad (536)$$

Therefore:

$$\ell_{0;OO}^{(1)} = 0, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}. \quad (537)$$

This is exactly the first closed blockwise longitudinal coefficient point inside the current explicit operator library.

CCLVI. 7. FINAL TERM-BY-TERM VERDICT FOR OO

Collecting the previous sections:

$$-Z_{O,k} \partial^2 \Rightarrow \text{no direct contribution to } \ell_{0;OO}^{(1)}, \ell_{0;OO}^{(3)}, \quad (538)$$

$$\mathcal{M}_{OO}^{\text{pot}} \Rightarrow \text{no direct contribution to } \ell_{0;OO}^{(1)}, \ell_{0;OO}^{(3)}, \quad (539)$$

$$Y_k \mathcal{K}_{OO}^{\nabla \text{ani}} \Rightarrow \ell_{0;OO}^{(1)} = 0 \text{ in the current LO no-go class,} \quad (540)$$

$$\mathcal{T}_{OO} \Rightarrow \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} \text{ is the leading genuine longitudinal primitive carrier.} \quad (541)$$

Hence the complete current blockwise result is

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}. \quad (542)$$

CCLVII. 8. CONSEQUENCE FOR GATE 2-A

Thus Gate 2-A for the current working library is reduced to one finite question:

$$\lambda_{c;OO} \neq 0 ? \quad (543)$$

If yes, then Branch B is microscopically realized already at the OO -block level, and the longitudinal carrier survives at transport level:

$$\lambda_{\parallel} = I_{\theta^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta^3} \neq 0. \quad (544)$$

Only if $\lambda_{c;OO} = 0$ should one continue to the remaining eight blocks and ask whether a hidden Branch-A carrier or subleading longitudinal channel appears elsewhere.

CCLVIII. FINAL SUMMARY

The actual orientational diagonal block is

$$\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y).$$

The blockwise longitudinal coefficients are defined by

$$h_{\parallel;OO}(\xi) = \frac{1}{2} \frac{1}{Q_B} (\Sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu(\xi)),$$

and

$$\ell_{0;OO}^{(1)}, \quad \ell_{0;OO}^{(3)}$$

are the coefficients of $\theta'(\xi)$ and $\theta'(\xi)^3$ in its shell expansion.

Term by term:

$$-Z_{O,k}\partial^2 \text{ gives no direct first-order longitudinal transport contribution,}$$

$$\mathcal{M}_{OO}^{\text{pot}}(\xi) \text{ gives no direct first-order longitudinal transport contribution,}$$

$$Y_k \mathcal{K}_{OO}^{\nabla \text{ani}} \text{ does not produce a nonzero Branch-A coefficient in the current solved LO class,}$$

because

$$\Lambda_{\parallel}^{(0)} = 0.$$

Thus the only genuine leading longitudinal carrier inside the OO block is the first-order transport term

$$\mathcal{T}_{OO}.$$

Its tangent coefficient has the branch-sensitive form

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).$$

In the current solved lowest-order realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

so

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 + \dots$$

Therefore the first closed blockwise longitudinal point is

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.$$

Hence Gate 2-A in the current working library is reduced to the single finite question

$$\lambda_{c;OO} \neq 0 ?$$

If yes, then Branch B is already microscopically realized at the OO -block level and

$$\lambda_{\parallel} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3} \neq 0.$$

CCLIX. ACTUAL EXTRACTION FORMULA FOR $\lambda_{c;OO}$

CCLX. SHORT EXPLANATION

The Gate-2-A problem is now reduced to extracting the primitive longitudinal coefficient from the first-order tangent transport block.

The load-bearing projected coefficient is

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)), \quad V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q.$$

In the current solved lowest-order class, the linear Branch-A term vanishes,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the only principal longitudinal source is the primitive commutator-square channel, i.e. the $\theta'(\xi)^3 \Sigma_3$ term.

CCLXI. 1. EXACT PROJECTED FIRST-ORDER TRANSPORT BLOCK

Start from the exact term-by-term split of the first-order worldvolume Hessian:

$$\boxed{\mathbb{H}_1^\mu(\xi) = \mathbb{H}_{1,\Psi}^\mu + \mathbb{H}_{1,P}^\mu + \mathbb{H}_{1,\sigma}^\mu + \mathbb{H}_{1,X}^\mu + \mathbb{H}_{1,J}^\mu + \mathbb{H}_{1,K}^\mu + \mathbb{H}_{1,EO}^\mu.} \quad (545)$$

Projecting to the tangent Q_B -bundle gives

$$\boxed{V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q.} \quad (546)$$

The longitudinal coefficient is extracted by

$$\boxed{a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)).} \quad (547)$$

CCLXII. 2. BRANCH-SENSITIVE DECOMPOSITION OF $a_3(\xi)$

The exact branch-sensitive parameterization is

$$\boxed{a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).} \quad (548)$$

The complete microscopic decision tree is therefore:

$$\boxed{\text{first test } \Lambda_{\parallel}^{(0)} \neq 0; \quad \text{if it vanishes, test } \lambda_c \neq 0.} \quad (549)$$

CCLXIII. 3. PRIMITIVE COMMUTATOR-SQUARE SEED

Define

$$\boxed{\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}.} } \quad (550)$$

For the explicit defect representative,

$$\boxed{S_\xi = \frac{1}{2} \theta'(\xi)^2 \Sigma_3.} \quad (551)$$

Hence the primitive longitudinal transport block is

$$V_{\parallel, \text{prim}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3. \quad (552)$$

Therefore

$$a_3^{(\text{prim})}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3. \quad (553)$$

CCLXIV. 4. WHY BRANCH A VANISHES IN THE CURRENT SOLVED LOCAL CLASS

Inside the current local isotropic scalar library, the first nonzero tangent Σ_3 -object is quadratic in the defect gradient:

$$(\Pi_Q(\partial_\xi P)^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3. \quad (554)$$

Thus every local higher-order correction built from the same algebra generates only the odd tower

$$\theta'(\xi)^3 \Sigma_3, \quad \theta'(\xi)^5 \Sigma_3, \quad \theta'(\xi)^7 \Sigma_3, \quad \dots \quad (555)$$

in the first-order transport kernel.

Therefore no linear Branch-A term appears:

$$\Lambda_{\parallel}^{(0)} = 0 \quad (556)$$

within the current solved local class.

So the principal longitudinal coefficient reduces to Branch B:

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.} \quad (557)$$

CCLXV. 5. ACTUAL EXTRACTION FORMULA FOR $\lambda_{c;OO}$

Now restrict to the OO -candidate transport contribution. Define

$$V_{OO}^\mu(\xi) := \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q. \quad (558)$$

Then its longitudinal Pauli trace is

$$a_{3;OO}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V_{OO}^\mu(\xi)). \quad (559)$$

If the current blockwise interpretation is correct and OO is indeed the natural primitive longitudinal carrier, then

$$a_{3;OO}(\xi) = \frac{\lambda_{c;OO}}{2} \theta'(\xi)^3 + O(\theta'(\xi)^5). \quad (560)$$

Therefore the exact extraction formula is

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}. \quad (561)$$

Equivalently,

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} = \lim_{\theta \rightarrow 0} \frac{\frac{1}{2} Q_B (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}. \quad (562)$$

CCLXVI. 6. PROJECTION TO THE LOW BLOCH MODES

Let

$$u_a^\pm(\xi) = f_\pm(\xi) e_a(\xi)$$

be the opposite-valley low modes. The projected shell matrix is

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (563)$$

Using (552), define

$$I_3 := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (564)$$

Then the longitudinal reduced coefficient is

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3. \quad (565)$$

So if OO indeed carries the primitive branch,

$$\lambda_\parallel = \frac{\lambda_{c;OO}}{2} I_3. \quad (566)$$

CCLXVII. 7. HONEST BLOCK-LABEL CAVEAT

There is a notation drift across the current notes:

- the blockwise audit summary states that OO is the natural primitive longitudinal block;
- some explicit primitive formulas were written in a document whose local section/file labeling still carries a TT -style notation.

Therefore the fully honest statement is:

$$\text{the algebraic principal longitudinal source is unambiguously the primitive } \theta'(\xi)^3 n^\mu \Sigma_3 \text{ channel,} \quad (567)$$

and

$$\text{the current blockwise working interpretation assigns this channel to the } OO \text{ candidate.} \quad (568)$$

CCLXVIII. 8. FINAL GATE-2-A REDUCTION

Hence the actual Gate-2-A problem is reduced to the single finite extraction:

$$\lambda_{c;OO} \neq 0 ? \quad (569)$$

If yes, then the primitive Branch-B longitudinal channel is microscopically realized already at the transport level, and Gate 2-A is closed up to the already-separated graded/Bloch overlap checks.

CCLXIX. FINAL SUMMARY

The actual longitudinal transport coefficient is extracted from

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)), \quad V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q.$$

Its branch-sensitive expansion is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).$$

Inside the current solved local isotropic scalar class,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the principal longitudinal source is forced to Branch B:

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.}$$

The primitive commutator-square seed is

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3,$$

hence

$$V_{\parallel, \text{prim}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3.$$

Under the current blockwise working interpretation, the natural primitive longitudinal candidate is the OO block. Therefore the actual extraction formula is

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3},$$

equivalently

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} = \lim_{\theta \rightarrow 0} \frac{\frac{1}{2} Q_B (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}.$$

Projection to the low Bloch modes gives

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3.$$

Hence, if the OO candidate is confirmed,

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3.$$

So the Gate-2-A problem is now reduced to the single finite extraction:

$$\lambda_{c;OO} \neq 0 ?$$

CCLXX. SUBTERM DECOMPOSITION OF $\Pi_Q \mathbb{H}_{1,OO}^\mu \Pi_Q$ AND ACTUAL EXTRACTION OF $\lambda_{c;OO}$

CCLXXI. GOAL

The Gate-2-A problem is to determine which microscopic subterm of the first-order tangent transport block actually contributes to the longitudinal primitive coefficient in the OO candidate block.

The full first-order worldvolume Hessian splits as

$$\mathbb{H}_1^\mu(\xi) = \mathbb{H}_{1,\Psi}^\mu + \mathbb{H}_{1,P}^\mu + \mathbb{H}_{1,\sigma}^\mu + \mathbb{H}_{1,X}^\mu + \mathbb{H}_{1,J}^\mu + \mathbb{H}_{1,K}^\mu + \mathbb{H}_{1,EO}^\mu. \quad (570)$$

Projecting to the tangent Q_B -bundle gives

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q. \quad (571)$$

The longitudinal Pauli trace is

$$a_3(\xi) = \frac{1}{2} \frac{1}{Q_B} (\Sigma_3 n_\mu V^\mu(\xi)). \quad (572)$$

The branch-sensitive parameterization is

$$a_3(\xi) = \Lambda_{||}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi). \quad (573)$$

CCLXXII. PRIMITIVE LONGITUDINAL SOURCE

Define the commutator seed

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (574)$$

For the explicit defect representative,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3. \quad (575)$$

Therefore the primitive longitudinal transport block is

$$V_{||,\text{prim}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3. \quad (576)$$

Hence

$$a_3^{(\text{prim})}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3. \quad (577)$$

CCLXXIII. WHY BRANCH A IS ABSENT IN THE SOLVED LO CLASS

The solved lowest-order class satisfies the no-go result

$$\Lambda_{||}^{(0)} = 0, \quad \lambda_c \neq 0. \quad (578)$$

Equivalently, the current local isotropic scalar library contains no linear longitudinal tangent carrier of the form

$$\theta'(\xi) n^\mu \Sigma_3.$$

Thus the principal longitudinal coefficient reduces to

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.} \quad (579)$$

CCLXXIV. K-SECTOR CONTRIBUTION

The direct tangent-block contribution from the K -sector has the form

$$V_K^\mu = \theta'(\xi) \left[\kappa_E(\xi) E^\mu{}_a \Sigma_a + \kappa_O(\xi) J^\mu{}_a \Sigma_a \right]. \quad (580)$$

Therefore

$$a_3^{(K)} = 0, \quad b_E^{(K)} = \kappa_E(\xi) \theta'(\xi), \quad b_O^{(K)} = \kappa_O(\xi) \theta'(\xi). \quad (581)$$

So the K -sector is purely transverse at principal order and cannot contribute to λ_c .

CCLXXV. EO-SLOT CONTRIBUTION

The explicit extra E/O -slot contribution is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (582)$$

Hence

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (583)$$

So the EO -slot is also purely transverse and cannot contribute to λ_c .

CCLXXVI. CANONICAL SOURCE TABLE

At principal order, the canonical source table is therefore:

source	canonical form	a_3 -contribution
primitive longitudinal	$\frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3$	$\frac{\lambda_c}{2} \theta'(\xi)^3$
K -sector	$\theta'(\xi) [\kappa_E E^\mu{}_a \Sigma_a + \kappa_O J^\mu{}_a \Sigma_a]$	0
EO -slot	$\theta'(\xi) \cos \theta(\xi) [\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a]$	0
irreducible/forbidden	V_{irr}^μ	0

(584)

Thus the longitudinal principal source is unique.

CCLXXVII. OO-CANDIDATE EXTRACTION

Under the current blockwise working interpretation, the natural primitive longitudinal candidate is the OO block. Define

$$V_{OO}^\mu(\xi) := \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q. \quad (585)$$

Then

$$a_{3;OO}(\xi) = \frac{1}{2_{QB}} (\Sigma_3 n_\mu V_{OO}^\mu(\xi)). \quad (586)$$

If the current blockwise assignment is correct, then at principal order

$$a_{3;OO}(\xi) = \frac{\lambda_{c;OO}}{2} \theta'(\xi)^3 + O(\theta'(\xi)^5). \quad (587)$$

Therefore the actual extraction formula is

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B(\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}. \quad (588)$$

Equivalently,

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} = \lim_{\theta \rightarrow 0} \frac{\frac{1}{2} Q_B(\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}. \quad (589)$$

CCLXXVIII. PROJECTION TO THE LOW BLOCH MODES

Projecting to the opposite-valley low modes gives

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (590)$$

Define

$$I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (591)$$

Then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3. \quad (592)$$

Under the OO -candidate interpretation,

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3. \quad (593)$$

CCLXXIX. HONEST NOTATION CAVEAT

There is a notation drift across the current notes: some primitive formulas were written in a local notation that resembles a TT -style label, while the blockwise audit summary assigns the primitive longitudinal carrier to OO .

Therefore the fully honest statement is:

$$\text{the algebraic principal longitudinal source is unambiguously the primitive } \theta'(\xi)^3 n^\mu \Sigma_3 \text{ channel,} \quad (594)$$

and

$$\text{the current blockwise working interpretation assigns this channel to the } OO \text{ candidate.} \quad (595)$$

CCLXXX. FINAL GATE-2-A REDUCTION

Hence the actual Gate-2-A problem is reduced to the single finite extraction:

$$\lambda_{c;OO} \neq 0 ? \quad (596)$$

If yes, then the primitive Branch-B longitudinal channel is microscopically realized already at the transport level, and Gate 2-A is closed up to the already-separated graded/Bloch overlap checks.

CCLXXXI. FINAL SUMMARY

The actual first-order tangent transport block is

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q,$$

and the longitudinal coefficient is extracted by

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)).$$

Its exact branch-sensitive form is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).$$

In the solved local lowest-order class,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so the principal longitudinal source is forced to Branch B:

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.}$$

The primitive commutator-square seed is

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3,$$

hence

$$V_{\parallel, \text{prim}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3.$$

At principal order:

$$a_3^{(K)} = 0, \quad a_3^{(EO)} = 0,$$

so the K -sector and EO -slot are purely transverse and do not contribute to the longitudinal primitive coefficient.

Under the current blockwise working interpretation, the natural primitive longitudinal candidate is the OO block. Therefore the actual extraction formula is

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}.$$

Equivalently,

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.$$

Projection to the low Bloch modes gives

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3.$$

So under the OO -candidate interpretation,

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3.$$

Hence the actual Gate-2-A problem has now been reduced to the single finite extraction:

$$\lambda_{c;OO} \neq 0 ?$$

The only honest caveat is a remaining notation drift across the notes: the algebraic primitive source is unambiguous, but some local formulas were written in a different block-label convention.

CCLXXXII. SOURCE TABLE FOR THE $\Sigma_3 n^\mu$ TRACE IN $\Pi_Q \mathbb{H}_1^\mu \Pi_Q$

CCLXXXIII. STARTING POINT

The exact first-order tangent transport block is

$$\boxed{V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q,} \quad (597)$$

with the additive source split

$$\boxed{\mathbb{H}_1^\mu = \mathbb{H}_{1,\Psi}^\mu + \mathbb{H}_{1,P}^\mu + \mathbb{H}_{1,\sigma}^\mu + \mathbb{H}_{1,X}^\mu + \mathbb{H}_{1,J}^\mu + \mathbb{H}_{1,K}^\mu + \mathbb{H}_{1,EO}^\mu.} \quad (598)$$

The longitudinal coefficient is extracted by

$$\boxed{a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)).} \quad (599)$$

Its branch-sensitive form is

$$\boxed{a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).} \quad (600)$$

CCLXXXIV. SOLVED LOWEST-ORDER CLASS

In the solved lowest-order complete realization, the no-go theorem gives

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (601)$$

Hence the principal longitudinal source is forced onto Branch B:

$$\boxed{a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.}} \quad (602)$$

CCLXXXV. PRIMITIVE SOURCE

The primitive commutator-square seed satisfies

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \Sigma_3,} \quad (603)$$

hence

$$\boxed{V_{\parallel, \text{prim}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3, \quad a_3^{(\text{prim})}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3.} \quad (604)$$

Therefore the primitive commutator-square channel is an explicit principal longitudinal source.

CCLXXXVI. K-SECTOR

The direct tangent-block contribution from the K -sector is

$$\boxed{V_K^\mu = \theta'(\xi) \left[\kappa_E(\xi) E^\mu{}_a \Sigma_a + \kappa_O(\xi) J^\mu{}_a \Sigma_a \right].} \quad (605)$$

Therefore

$$\boxed{a_3^{(K)} = 0.} \quad (606)$$

So the K -sector is purely transverse at principal order.

CCLXXXVII. EO-SLOT

The explicit EO -slot contribution is

$$\boxed{V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].} \quad (607)$$

Therefore

$$\boxed{a_3^{(EO)} = 0.} \quad (608)$$

So the EO -slot is also purely transverse.

CCLXXXVIII. SCALAR/ISOTROPIC SECTORS

If every sector is built from local tangent-isotropic scalar contractions, then

$$\boxed{a_3^{(\Psi)} = a_3^{(P)} = a_3^{(\sigma)} = a_3^{(X)} = a_3^{(EO)} = 0.} \quad (609)$$

Thus these sectors do not generate the principal longitudinal channel at lowest local order.

CCLXXXIX. J-SECTOR STATUS

For the uploaded notes currently inspected, an explicit separate formula

$$a_3^{(J)} = 0$$

was not found as a standalone displayed result.

However, the canonical source table already classifies the principal source content as:

$$\text{primitive commutator-square} \rightarrow n^\mu \sigma_3 \rightarrow \lambda_c,$$

$$\text{transverse even/odd} \rightarrow E^\mu{}_a \sigma_a, \quad J^\mu{}_a \sigma_a,$$

with no additional principal longitudinal source listed.

Therefore the strongest honest current statement is:

$$\boxed{a_3^{(J)} \text{ is not separately extracted in the uploaded notes,}} \quad (610)$$

but

$$\boxed{\text{the principal longitudinal source table assigns the leading}} n^\mu \Sigma_3 \text{ coefficient uniquely to the primitive channel.} \quad (611)$$

CCXC. FINAL SOURCE TABLE

At principal order, the $\Sigma_3 n^\mu$ -trace survival table is:

Source	$\Sigma_3 n^\mu$ -trace at principal order	Status
$\mathbb{H}_{1,\Psi}^\mu$	0	excluded in current local isotropic class
$\mathbb{H}_{1,P}^\mu$	0	excluded in current local isotropic class
$\mathbb{H}_{1,\sigma}^\mu$	0	excluded in current local isotropic class
$\mathbb{H}_{1,X}^\mu$	0	excluded in current local isotropic class
$\mathbb{H}_{1,J}^\mu$	not separately extracted	no explicit principal-source role established
$\mathbb{H}_{1,K}^\mu$	0	purely transverse
$\mathbb{H}_{1,EO}^\mu$	0	purely transverse
primitive commutator-square seed	$\frac{\lambda_c}{2} \theta'(\xi)^3$	unique explicit principal longitudinal source

CCXCI. OO-CANDIDATE INTERPRETATION

Under the current blockwise working interpretation, this principal primitive longitudinal source is assigned to the OO candidate block. Hence the actual extraction reduces to

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B(\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}. \quad (612)$$

If $\lambda_{c;OO} \neq 0$, then

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (613)$$

So the Gate-2-A problem is reduced to the single finite question

$$\lambda_{c;OO} \neq 0 ?$$

CCXCII. FINAL SUMMARY

The first-order tangent transport block is

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q,$$

and the longitudinal coefficient is extracted by

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V^\mu(\xi)).$$

In the solved lowest-order class,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

so the principal longitudinal source is forced to Branch B:

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.}$$

The explicit principal longitudinal source is the primitive commutator-square channel:

$$V_{\parallel, \text{prim}}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3, \quad a_3^{(\text{prim})} = \frac{\lambda_c}{2} \theta'(\xi)^3.$$

At principal order, the following contributions vanish in the $\Sigma_3 n^\mu$ -trace:

$$a_3^{(\Psi)} = a_3^{(P)} = a_3^{(\sigma)} = a_3^{(X)} = a_3^{(EO)} = 0,$$

and also

$$a_3^{(K)} = 0.$$

For the J -sector, the uploaded notes do not yet provide a separate displayed extraction formula for $a_3^{(J)}$. So the strongest honest current statement is not that $a_3^{(J)} = 0$ is proven separately, but that the canonical principal-source table assigns the leading longitudinal coefficient uniquely to the primitive channel.

Under the current blockwise working interpretation, this principal primitive source is assigned to the OO candidate block. Therefore the actual Gate-2-A extraction reduces to

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B(\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}.$$

If $\lambda_{c;OO} \neq 0$, then

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3.$$

Hence the full Gate-2-A problem is reduced to the single finite question

$$\lambda_{c;OO} \neq 0 ?$$

CCXCIII. PRACTICAL SUFFICIENT CONDITIONS FOR $\lambda_{c;OO} \neq 0$

CCXCIV. STARTING POINT

The exact first-order tangent transport block is

$$V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q, \quad (614)$$

and the longitudinal coefficient is extracted by

$$a_3(\xi) = \frac{1}{2 Q_B} (\Sigma_3 n_\mu V^\mu(\xi)). \quad (615)$$

Its branch-sensitive parameterization is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi). \quad (616)$$

In the solved lowest-order complete realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0, \quad (617)$$

so the principal longitudinal source is forced to Branch B:

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 \quad \text{at principal order.} \quad (618)$$

CCXCV. PRIMITIVE PRINCIPAL SOURCE

The primitive commutator-square channel is defined by

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (619)$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \Sigma_3, \quad (620)$$

hence the primitive transport source is

$$V_{\parallel, \text{prim}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \Sigma_3. \quad (621)$$

Therefore

$$a_3^{(\text{prim})}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3. \quad (622)$$

CCXCVI. SUBTERM ELIMINATION AT PRINCIPAL ORDER

At principal order, the following source sectors do *not* contribute to the $\Sigma_3 n^\mu$ -trace:

$$\boxed{a_3^{(K)} = 0, \quad a_3^{(EO)} = 0,} \quad (623)$$

and, inside the current local tangent-isotropic scalar class,

$$\boxed{a_3^{(\Psi)} = a_3^{(P)} = a_3^{(\sigma)} = a_3^{(X)} = 0.} \quad (624)$$

Thus the strongest honest principal-order source table is:

$$\boxed{\text{primitive commutator-square source} \implies n^\mu \Sigma_3 \implies \lambda_c.} \quad (625)$$

This means that, within the current solved LO library, the principal longitudinal coefficient is unique.

CCXCVII. OO-CANDIDATE EXTRACTION

Under the current blockwise working interpretation, the primitive longitudinal source is assigned to the *OO* candidate block:

$$\boxed{V_{OO}^\mu(\xi) := \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q.} \quad (626)$$

Define

$$\boxed{a_{3;OO}(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu V_{OO}^\mu(\xi)).} \quad (627)$$

Then the actual primitive extraction is

$$\boxed{\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q)}{\theta'(\xi)^3}.} \quad (628)$$

Equivalently,

$$\boxed{\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.} \quad (629)$$

CCXCVIII. TRANSPORT-LEVEL SUFFICIENT CONDITION

A clean sufficient condition for

$$\lambda_{c;OO} \neq 0$$

is:

$$\boxed{\exists U \subset \mathbb{R}_\xi \text{ of positive measure such that } \theta'(\xi) \neq 0 \text{ on } U,} \quad (630)$$

and

$$\boxed{\frac{1}{2_{Q_B}} (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_{1,OO}^\mu(\xi) \Pi_Q) \text{ has the same nonzero sign as } \theta'(\xi)^3 \text{ on } U.} \quad (631)$$

Then

$$\boxed{\lambda_{c;OO} \neq 0.} \quad (632)$$

Because the principal-order source is unique in the current solved library, (631) is not merely sufficient in practice: it is the effective principal-order nonvanishing test.

CCXCIX. EQUIVALENT SYMMETRY/PARITY/SUPPORT FORMULATION

The same content can be written as four practical conditions:

a. (i) *Nontrivial defect support*

$$\boxed{\theta'(\xi) \neq 0.} \quad (633)$$

b. (ii) *Allowed parity* The source must be odd overall at first order, so that $\theta'(\xi)^3 \Sigma_3$ is not parity-forbidden:

$$\boxed{\text{the projected source is reflection-odd and compatible with } n^\mu \Sigma_3.} \quad (634)$$

c. (iii) *Nontrivial internal trace*

$$\boxed{Q_B(\Sigma_3 n_\mu \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q) \neq 0.} \quad (635)$$

d. (iv) *No principal-order cancellation* Since the principal longitudinal source is unique in the current library, there is no competing principal-order term to cancel it. Hence once (635) holds with nonzero θ'^3 -coefficient, $\lambda_{c;OO}$ survives.

CCC. FULL GATE-2 SUFFICIENT CONDITION

Transport-level survival is not yet the final physical longitudinal coefficient. Projecting to the low Bloch modes gives

$$\boxed{\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3.} \quad (636)$$

Therefore a sufficient condition for the full Gate 2 longitudinal survival is:

$$\boxed{\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_c \neq 0.} \quad (637)$$

If these hold, then

$$\boxed{\lambda_{\parallel} \neq 0.} \quad (638)$$

CCCI. HONEST CAVEAT

There remains a notation drift across the uploaded notes: some primitive formulas were written in a local block-labeling convention that resembles a *TT*-style notation, while the blockwise audit summary assigns the primitive longitudinal carrier to the *OO* candidate.

Hence the fully honest statement is:

$$\boxed{\text{the algebraic principal source is unambiguously } \theta'(\xi)^3 n^\mu \Sigma_3,} \quad (639)$$

and

$$\boxed{\text{the current blockwise working interpretation assigns it to the } OO \text{ candidate.}} \quad (640)$$

CCCII. FINAL SUMMARY

The actual longitudinal transport coefficient is extracted from

$$\boxed{a_3(\xi) = \frac{1}{2 Q_B} (\Sigma_3 n_\mu V^\mu(\xi)), \quad V^\mu(\xi) = \Pi_Q \mathbb{H}_1^\mu(\xi) \Pi_Q.}$$

Its branch-sensitive form is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi).$$

In the solved lowest-order class,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

so the principal longitudinal source is uniquely the primitive commutator-square channel:

$$V_{\parallel, \text{prim}}^{\mu} = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \Sigma_3.$$

At principal order:

$$a_3^{(K)} = 0, \quad a_3^{(EO)} = 0,$$

and inside the current local tangent-isotropic scalar class,

$$a_3^{(\Psi)} = a_3^{(P)} = a_3^{(\sigma)} = a_3^{(X)} = 0.$$

Therefore the strongest honest principal-order source table leaves a unique explicit longitudinal source. Under the current *OO*-candidate interpretation, the primitive coefficient is extracted by

$$\lambda_{c;OO} = \lim_{\theta \rightarrow 0} \frac{Q_B(\Sigma_3 n_{\mu} \Pi_Q \mathbb{H}_{1,OO}^{\mu}(\xi) \Pi_Q)}{\theta'(\xi)^3}.$$

A clean transport-level sufficient condition is:

$$\theta'(\xi) \neq 0, \quad Q_B(\Sigma_3 n_{\mu} \Pi_Q \mathbb{H}_{1,OO}^{\mu}(\xi) \Pi_Q) \text{ has a nonzero } \theta'(\xi)^3 \text{ coefficient.}$$

In the current solved library, this is effectively the principal-order yes/no test for

$$\lambda_{c;OO} \neq 0.$$

For the full Gate 2 longitudinal survival, a sufficient condition is

$$\lambda_{c;OO} \neq 0, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \neq 0, \quad C_c \neq 0.$$

Then

$$\lambda_{\parallel} \neq 0.$$

The only honest caveat is a remaining notation drift across the uploaded notes: the algebraic primitive source is unambiguous, but some local formulas use a different block-label convention than the current *OO*-candidate interpretation.

CCCI. FINAL GATE-2 VERDICT WITH TRANSPORT/CALIBRATION SPLIT

CCCIV. SHORT EXPLANATION

The longitudinal carrier problem is now reduced to a branch-sensitive finite audit. The exact branch decision tree is:

$$\text{first test } \Lambda_{\parallel}^{(0)}; \quad \text{if it vanishes, test } \lambda_c.$$

In the solved lowest-order complete realization,

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.$$

Therefore the practical Gate-2 route is forced onto Branch B.

CCCV. 1. BRANCH-B PRIMITIVE SEED

Define

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}.$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (641)$$

Hence the primitive longitudinal transport channel is

$$V_{(\parallel), \text{tr}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (642)$$

Projecting to the shell-localized low modes gives

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \quad I_{\theta'^3} \neq 0. \quad (643)$$

Therefore the direct Branch-B implication is

$$\lambda_c \neq 0 \implies \lambda_\parallel \neq 0. \quad (644)$$

CCCVI. 2. CANONICAL SOURCE CLASSIFICATION

The principal first-order source table is

scalar/isotropic $\rightarrow$ spectator,

primitive commutator-square $\rightarrow n^\mu \sigma_3 \rightarrow \lambda_c$,

transverse even $\rightarrow E^\mu{}_a \sigma_a \rightarrow \Lambda_E$,

transverse odd $\rightarrow J^\mu{}_a \sigma_a \rightarrow \Lambda_O$,

irreducible/forbidden $\rightarrow$ projected out.

Thus, at principal order, the longitudinal source is unique in the current solved library.

CCCVII. 3. OO-CANDIDATE BLOCK INTERPRETATION

Under the current blockwise working interpretation, the primitive longitudinal carrier is assigned to the OO candidate block. Then

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}, \quad (645)$$

and

$$\lambda_\parallel = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (646)$$

So the transport-level Gate-2 question is reduced to

$$\lambda_{c;OO} \neq 0 ? \quad (647)$$

CCCVIII. 4. ONE-MEDIATOR BRANCH-B SHELL PROJECTION

In the one-mediator Branch-B truncation, the shell-projected longitudinal coefficient is written as

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3. \quad (648)$$

This is the same transport/calibration structure in a different notation:

$$\text{transport amplitude} \times \text{Bloch overlap} \times \text{defect-profile overlap}.$$

Hence a clean practical sufficient condition is

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0. \quad (649)$$

Under the identification of $C_{\parallel}$ with the longitudinal Bloch normalization constant C_c , this is equivalent to requiring

$$C_c \neq 0$$

together with the nonzero transport branch amplitude and nonzero defect overlap.

CCCIX. 5. FINAL PRACTICAL VERDICT

Combining the Branch-B direct test and the shell-projected overlap form, the strongest practical Gate-2 statement is:

$$\boxed{\text{Gate 2 passes if the current microscopic branch realizes Branch B with } \lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0.} \quad (650)$$

Equivalently, in the one-mediator notation,

$$\boxed{\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0.} \quad (651)$$

CCCX. 6. HONEST CAVEATS

Two honest caveats remain:

a. (i) Block-label drift Some local primitive formulas are written in a notation that resembles a TT -style block label, while the blockwise audit summary assigns the primitive longitudinal carrier to the OO candidate.

Therefore the fully honest statement is:

$$\boxed{\text{the algebraic primitive source is unambiguous, but the local block labeling is not yet fully cleaned up.}}$$

b. (ii) Final Bloch overlap evaluation The criterion

$$C_{\parallel} \neq 0 \quad \text{or equivalently} \quad C_c \neq 0$$

is exact, but its actual refined locked Bloch evaluation remains to be inserted explicitly.

CCCXI. 7. FINAL GATE-2 CLOSURE STATEMENT

Therefore the strongest honest current closure statement is:

$$\boxed{\text{Transport level: Gate 2 is reduced to the single finite extraction } \lambda_{c;OO} \neq 0.} \quad (652)$$

$$\boxed{\text{Full physical level: Gate 2 passes if } \lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0.} \quad (653)$$

CCCXII. FINAL SUMMARY

The longitudinal gate is now reduced to a branch-sensitive transport/calibration split.
The exact branch decision tree is

$$\boxed{\text{first test } \Lambda_{\parallel}^{(0)}; \quad \text{if it vanishes, test } \lambda_c.}$$

In the solved lowest-order complete realization,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,}$$

so the practical Gate-2 route is forced onto Branch B.

The primitive Branch-B transport seed is

$$\boxed{V_{(\parallel),\text{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^{\mu} \sigma_3,}$$

and therefore

$$\boxed{\lambda_c \neq 0 \implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3} \neq 0.}$$

Under the current *OO*-candidate interpretation, the transport-level Gate-2 problem reduces to the single finite extraction

$$\boxed{\lambda_{c;OO} \neq 0 ?}$$

At the shell-projected one-mediator Branch-B level,

$$\boxed{\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3.}$$

Hence the strongest practical full Gate-2 sufficient condition is

$$\boxed{\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0.}$$

Equivalently, in the one-mediator notation,

$$\boxed{\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad I_3 \neq 0 \implies \lambda_{\parallel} \neq 0.}$$

Thus the strongest honest current closure statement is:

$$\boxed{\text{Transport level: Gate 2 is reduced to the single finite extraction } \lambda_{c;OO} \neq 0;}$$

$$\boxed{\text{Full physical level: Gate 2 passes if } \lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0.}$$

The remaining caveats are only:

$$\boxed{\text{block-label drift in some local notes,} \quad \text{and the still-uninserted explicit Bloch evaluation of } C_{\parallel}.}$$

CCCXIII. FINAL GATE-3 VERDICT WITH TRANSPORT/CALIBRATION SPLIT

CCCXIV. SHORT EXPLANATION

The transverse carrier problem is now reduced to a transport/calibration split, completely parallel to the longitudinal Gate 2.

The transport-side question is whether the shell-adapted transverse witness pair survives:

$$(c_E, c_O) \neq (0, 0).$$

The physical-side question is whether the surviving transport witness is not killed by the Bloch-frame overlap:

$$(C_E, C_O) \neq (0, 0).$$

CCCXV. 1. FIXED REDUCED WITNESS DICTIONARY

The reduced shell coefficients are already fixed by

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}, \quad (654)$$

with

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0.$$

Therefore the transport-side transverse pass condition is exactly

$$(c_E, c_O) \neq (0, 0) \iff (\alpha, \beta) \neq (0, 0) \text{ before Bloch calibration.} \quad (655)$$

CCCXVI. 2. WHY THE NAIVE LOCAL JK SHORTCUT IS FALSE

Take the shell-adapted leading forms

$$J \sim j \Sigma_3, \quad K \sim k_E \Sigma_1 + k_O \Sigma_2.$$

Then the simplest local real Hermitian JK bilinear reduces to

$$\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = k_E \{\Sigma_3, \Sigma_1\} + k_O \{\Sigma_3, \Sigma_2\} = 0. \quad (656)$$

Therefore

$$\boxed{\text{the naive local real Hermitian } JK \text{ term does not suffice.}} \quad (657)$$

Hence the honest microscopic frontier is not JK without momentum structure, but the derivative-resolved shell audit

$$\boxed{\partial_{p_{\nu}} \widehat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_{\nu}} \widehat{\Sigma}_{KK}(G_*; p)|_0.} \quad (658)$$

CCCXVII. 3. EXPLICIT CLASS II COMPRESSION

Inside the explicit folded-Bragg Class II branch, the derivative bookkeeping compresses to the single Fourier convolution

$$\widehat{T}_{(II)}^{\alpha}(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q'), \quad \alpha = 1, 2. \quad (659)$$

In the minimal first-shell-dominant reduction, this further reduces to

$$\widehat{T}_{(II)}^{\alpha}(G_*) \propto i A_{G_*}(C_{\alpha} \cdot G_*), \quad \alpha = 1, 2. \quad (660)$$

Therefore the strongest practical explicit branch-level sufficient test is

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.} \quad (661)$$

Under (661),

$$\boxed{(\alpha, \beta) \neq (0, 0)} \quad (662)$$

is realized in the current explicit Class II branch.

CCCXVIII. 4. ROLE OF THE K -SECTOR

At the explicit shell-adapted leading level, the intervalley K -sector harmonic has

$$\mathcal{K}_{\parallel}^a(G_*; 0) = 0,$$

hence

$$\boxed{c_{\parallel}^{(K)} = 0}, \quad (663)$$

while

$$\boxed{c_E^{(K)} \neq 0 \text{ in general,} \quad c_O^{(K)} \neq 0 \text{ in general.}} \quad (664)$$

So the K -sector is a direct transverse carrier, while pure axial JJ cannot generate the transverse pair by itself. Hence the transverse witness must come from genuinely nontrivial JK and/or KK realization.

CCCXIX. 5. BLOCH CALIBRATION LAYER

Even if the transport-side transverse witness survives, the physical coefficients still require the Bloch overlap layer:

$$\boxed{\alpha = C_E g_E, \quad \beta = C_O g_O.} \quad (665)$$

Thus the final physical transverse pass condition is

$$\boxed{(C_E, C_O) \neq (0, 0).} \quad (666)$$

Equivalently, if

$$(C_E, C_O) = (0, 0),$$

then the transverse witness is killed in the actual Bloch frame even if the transport-side branch realizes a nonzero (c_E, c_O) .

CCCXX. 6. FINAL PRACTICAL VERDICT

Combining transport and calibration, the strongest practical Gate-3 statement is:

$$\boxed{\text{Gate 3 passes if the current microscopic branch realizes a nonzero derivative-resolved transverse source,}} \quad (667)$$

in practice by the explicit Class II branch test

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0,} \quad (668)$$

together with the Bloch calibration requirement

$$\boxed{(C_E, C_O) \neq (0, 0).} \quad (669)$$

Equivalently, in reduced witness language,

$$\boxed{(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0).} \quad (670)$$

CCCXXI. 7. HONEST CAVEAT

The final honest caveat is:

$$\boxed{\text{As a working-branch acceptance statement, Gate 3 is reduced to finite explicit overlap tests.}} \quad (671)$$

But

$$\boxed{\text{As a fully unconditional microscopic theorem, it still rests on the derivative-resolved } JK/KK \text{ shell audit.}} \quad (672)$$

So the correct wording is:

finite computation left on the explicit branch, with a structural caveat for unconditional closure.

CCCXXII. FINAL SUMMARY

The transverse gate is now reduced to a transport/calibration split exactly parallel to Gate 2.
The reduced witness dictionary is

$$\boxed{\alpha = c_E I_\theta, \quad \beta = c_O I_\theta,}$$

so the transport-side condition is

$$\boxed{(c_E, c_O) \neq (0, 0).}$$

However, the naive local real Hermitian JK term vanishes:

$$\boxed{\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = 0.}$$

Therefore the honest microscopic frontier is

$$\boxed{\partial_{p_\nu} \hat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_\nu} \hat{\Sigma}_{KK}(G_*; p)|_0.}$$

Inside the explicit folded-Bragg Class II branch, this derivative bookkeeping compresses to

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \hat{J}_i^A(Q) \hat{K}_{i,\alpha}^A(Q'),}$$

and in the minimal first-shell-dominant reduction to the practical sufficient test

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0.}$$

At the shell-adapted leading level, the K -sector is a direct transverse carrier:

$$\boxed{c_{\parallel}^{(K)} = 0, \quad c_E^{(K)} \neq 0 \text{ in general,} \quad c_O^{(K)} \neq 0 \text{ in general.}}$$

After Bloch calibration, the physical coefficients are

$$\boxed{\alpha = C_E g_E, \quad \beta = C_O g_O,}$$

so the final physical transverse pass condition is

$$\boxed{(C_E, C_O) \neq (0, 0).}$$

Therefore the strongest practical Gate-3 verdict is

$$\boxed{\text{Gate 3 passes if } A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0).}$$

Equivalently,

$$\boxed{(c_E, c_O) \neq (0, 0), \quad (C_E, C_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0).}$$

The honest caveat is:

$$\boxed{\text{as a working-branch acceptance statement, Gate 3 is reduced to finite explicit overlap tests,}}$$

but

$$\boxed{\text{as a fully unconditional microscopic theorem, it still rests on the derivative-resolved } JK/KK \text{ shell audit.}}$$

CCCXXIII. CARRIER-LEVEL ACCEPTANCE THEOREM BEFORE MASS PROTECTION

CCCXXIV. GOAL

We now combine Gate 2 and Gate 3 into a single carrier-level acceptance statement. This theorem should decide whether the current refined locked + Class II branch already realizes a nondegenerate massless Dirac *carrier candidate*, before the independent mass-protection audit of Gate 4.

CCCXXV. FIXED REDUCED WITNESS DICTIONARY

Inside the shell seed basis, the reduced coefficients are exactly

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta},} \quad (673)$$

with

$$I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0.$$

Therefore the carrier-level nondegeneracy condition is precisely

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (674)$$

CCCXXVI. GATE 2: LONGITUDINAL BRANCH-B VERDICT

The branch-sensitive longitudinal decomposition is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi). \quad (675)$$

In the solved lowest-order complete realization,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (676)$$

Hence the practical longitudinal route is forced onto Branch B.
Under the current *OO*-candidate interpretation,

$$\boxed{\lambda_{c;OO} \neq 0} \quad (677)$$

is the transport-level longitudinal condition, and the reduced coefficient is

$$\lambda_{\parallel} = \frac{\lambda_{c;OO}}{2} I_3, \quad I_3 := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (678)$$

After Bloch calibration, this requires

$$C_{\parallel} \neq 0 \quad (\text{equivalently } C_c \neq 0). \quad (679)$$

Thus the practical Gate-2 verdict is

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (680)$$

CCCXXVII. GATE 3: TRANSVERSE VERDICT

The transverse reduced coefficients are

$$\alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}. \quad (681)$$

So the transport-side transverse condition is

$$(c_E, c_O) \neq (0, 0). \quad (682)$$

However, the naive local real Hermitian JK shortcut fails because

$$\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = 0. \quad (683)$$

Therefore the honest microscopic frontier is the derivative-resolved shell audit

$$\partial_{p_{\nu}} \widehat{\Sigma}_{JK}(G_*; p)|_0, \quad \partial_{p_{\nu}} \widehat{\Sigma}_{KK}(G_*; p)|_0. \quad (684)$$

Inside the explicit folded-Bragg Class II working branch, this compresses to the practical sufficient test

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0. \quad (685)$$

After Bloch calibration, the physical transverse condition is

$$(C_E, C_O) \neq (0, 0). \quad (686)$$

Hence the practical Gate-3 verdict is

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0) \implies (\alpha, \beta) \neq (0, 0). \quad (687)$$

CCCXXVIII. INTEGRATED CARRIER-LEVEL THEOREM

Combining (680) and (687), we obtain the strongest practical carrier-level acceptance theorem before mass protection:

$$\lambda_{c;OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0, \quad (688)$$

and

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (689)$$

If both blocks hold, then

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad (690)$$

so the current refined locked + Class II branch is accepted as a viable microscopic *carrier* for the anisotropic massless Dirac block.

CCCXXIX. EQUIVALENT COMPACT FORM

In the one-mediator Branch-B truncation, the practical closure can be summarized as

$$\boxed{\alpha_X C_{\parallel} \neq 0, \quad (\gamma_E C_E, \gamma_O C_O) \neq (0, 0),} \quad (691)$$

assuming the nonzero overlap data

$$I_3 \neq 0, \quad J_1 \neq 0, \quad \beta_X \neq 0.$$

CCCXXX. HONEST CAVEATS

Two caveats must still be stated honestly.

a. (i) Longitudinal block-label drift. The algebraic primitive source is unambiguous, namely the

$$\theta'(\xi)^3 n^\mu \Sigma_3$$

channel, but some local notes still carry a different block-label convention than the current *OO*-candidate interpretation.

b. (ii) Transverse unconditional closure. As a working-branch statement, Gate 3 is reduced to finite explicit overlap tests. As a fully unconditional microscopic theorem, it still rests on the derivative-resolved *JK/KK* shell audit.

CCCXXXI. WHAT REMAINS AFTER THIS THEOREM

Once the carrier-level theorem is accepted, the only logically independent remaining gate is mass protection:

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.} \quad (692)$$

This is Gate 4.

CCCXXXII. FINAL SUMMARY

The carrier-level acceptance problem before mass protection is now reduced to two practical blocks.

a. Longitudinal block. In the solved lowest-order class,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0,}$$

so the practical Gate-2 route is forced to Branch B. Under the current *OO*-candidate interpretation, Gate 2 passes if

$$\boxed{\lambda_{c;OO} \neq 0, \quad I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 \neq 0, \quad C_{\parallel} (= C_c) \neq 0.}$$

Then

$$\boxed{\lambda_{\parallel} \neq 0.}$$

b. Transverse block. The naive local real Hermitian *JK* shortcut is false, so the honest frontier is the derivative-resolved *JK/KK* shell audit. However, inside the explicit folded-Bragg Class II branch, Gate 3 is practically reduced to

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0).}$$

Then

$$\boxed{(\alpha, \beta) \neq (0, 0).}$$

c. Integrated carrier-level theorem. Therefore the strongest practical carrier-level acceptance statement before mass protection is:

$$\boxed{\lambda_{c,OO} \neq 0, \quad I_3 \neq 0, \quad C_{\parallel}(=C_c) \neq 0,}$$

and

$$\boxed{A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0, \quad (C_E, C_O) \neq (0, 0).}$$

If these hold, then

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),}$$

so the current refined locked + Class II branch is accepted as a viable microscopic *carrier* for the anisotropic massless Dirac block.

d. Compact one-mediator form. Assuming

$$I_3 \neq 0, \quad J_1 \neq 0, \quad \beta_X \neq 0,$$

the same closure is summarized by

$$\boxed{\alpha_X C_{\parallel} \neq 0, \quad (\gamma_E C_E, \gamma_O C_O) \neq (0, 0).}$$

e. What remains. After this carrier-level theorem, the only logically independent remaining gate is mass protection:

$$\boxed{A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$